Weekly Report

Dean Reiter

March 23, 2026

Detector Hits in Cosmic and Beam Data

Figures

1) Trigger hit time distributions
   - All hits, in-time hits, out-of-time hits
2) Multiplicity histograms
   - GEM, Trigger, GEM vs Trigger
3) Detector Hits Present in Events
4) Hit Position Maps
5) Hit Position Correlations
   - Top v Bottom GEMs, Trigger vs GEMs, Trigger vs GEM Projections

Dataset Information

- Readout configs: Left+Trigger or Right+Trigger
- Dipole settings: elastic line on one side, off on other side
- 1M events each
- GEMs at 3950V

*Many dead strips in left bottom GEM

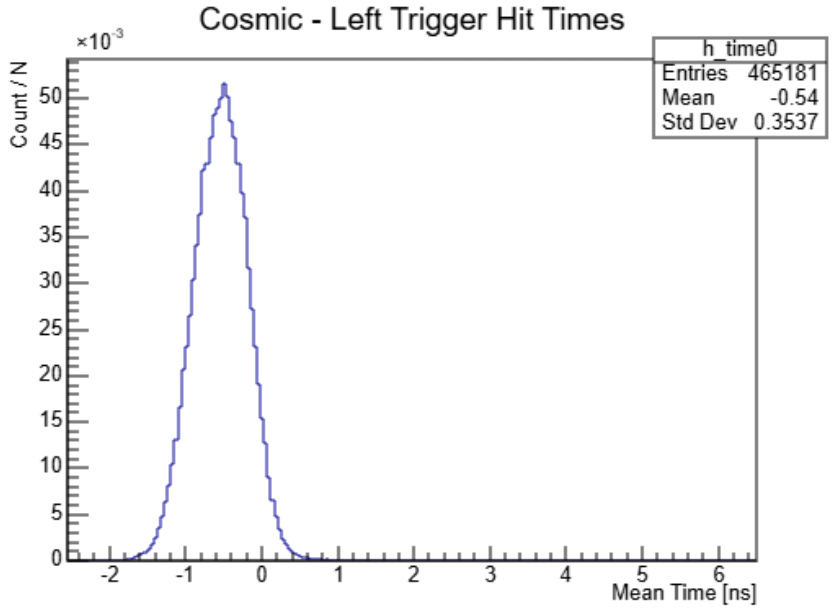
Run	Date	Setup	Readout	Dipole	Beam
3622*	4 Jan 2026	Cosmic	L+Trig		
3932	15 Jan 2026	Cosmic	R+Trig		
4001	21 Jan 2026	e-e-	R+Trig	0/95	15 MeV, 5uA, 13us, 1.5%, 1100Hz
4010	21 Jan 2026	e-e-	L+Trig	95/0	‘’

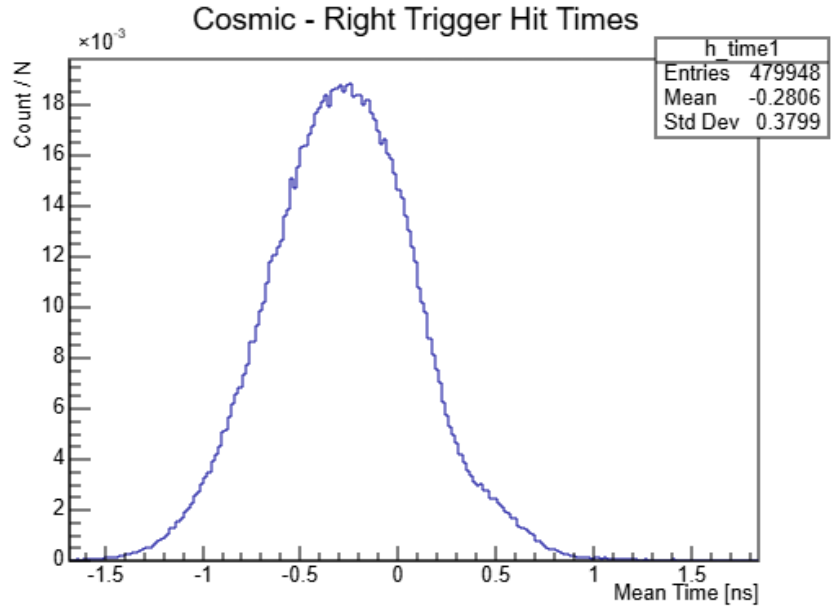
Trigger Hit Times

- "In-Time" = $\left|t_{paddle,\,L}\right| < 5$ ns AND $\left|t_{paddle,\,R}\right| < 5$ ns

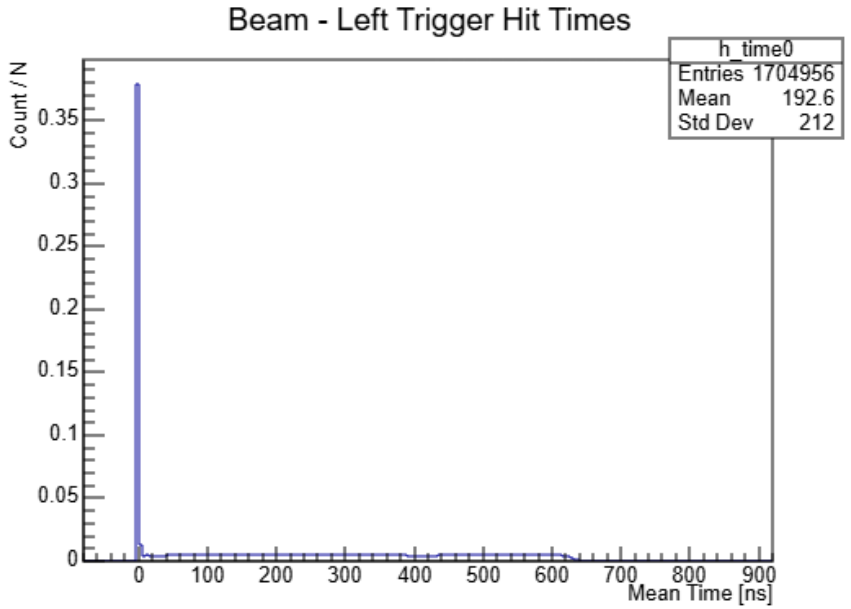
All Trigger Hits

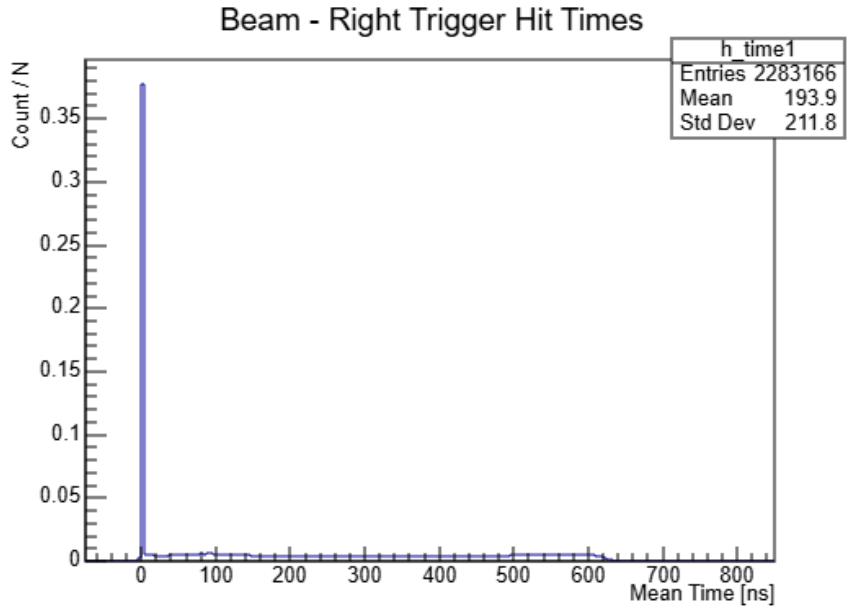
Cosmic





Beam

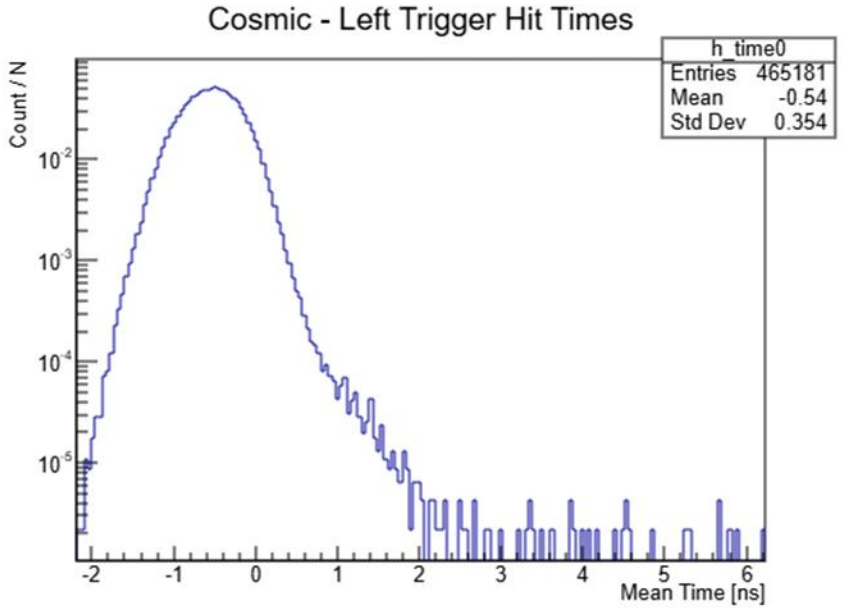


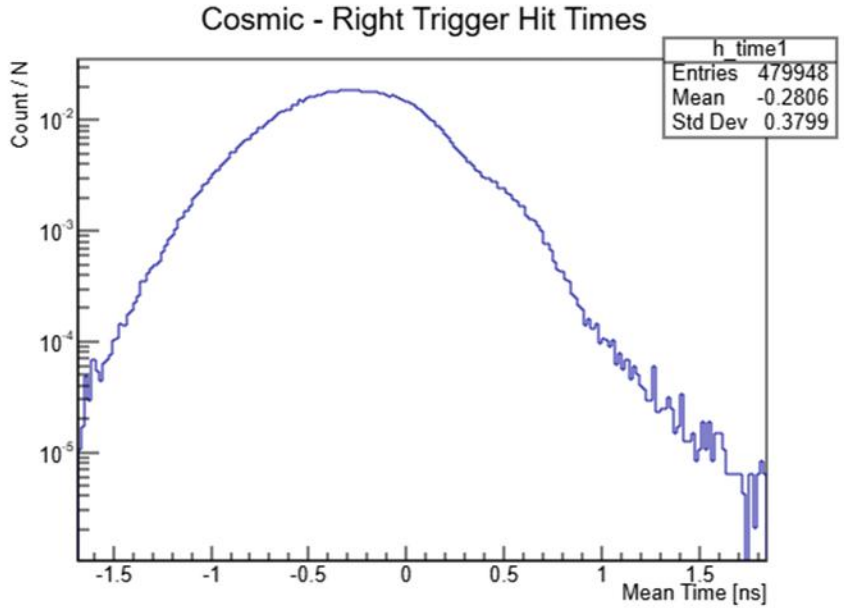


All Trigger Hits

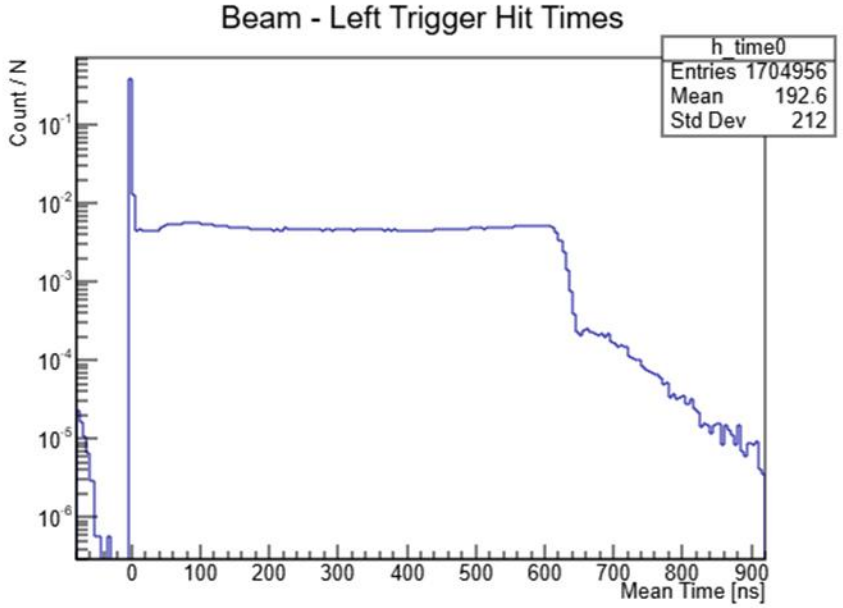
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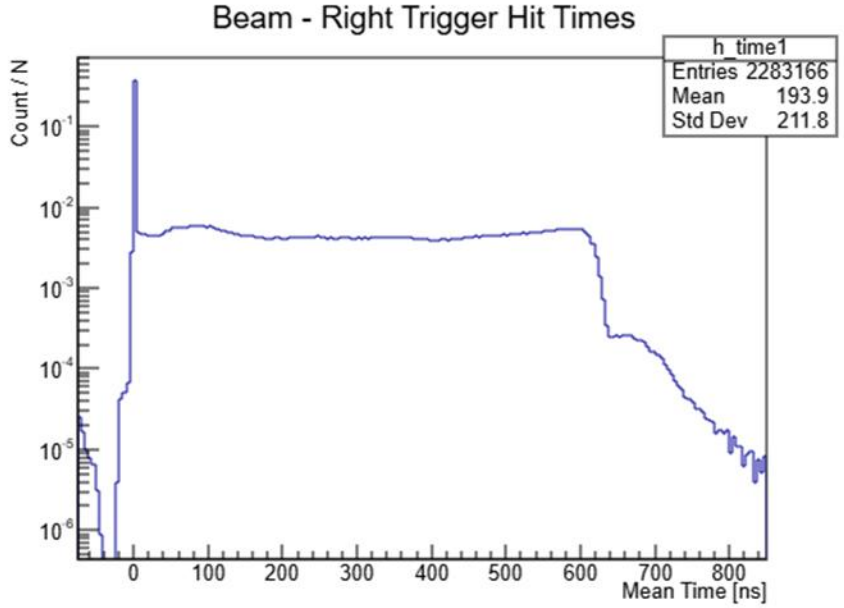
Cosmic





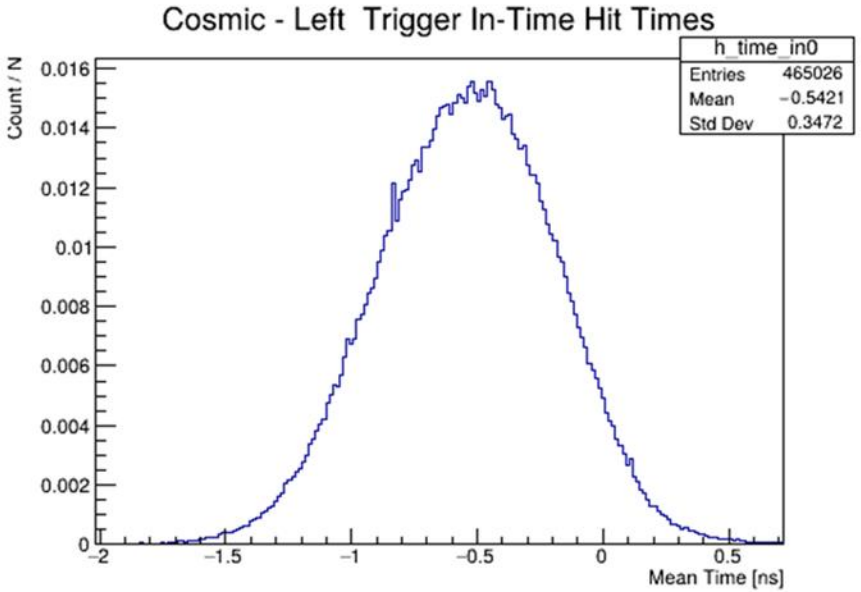
Beam

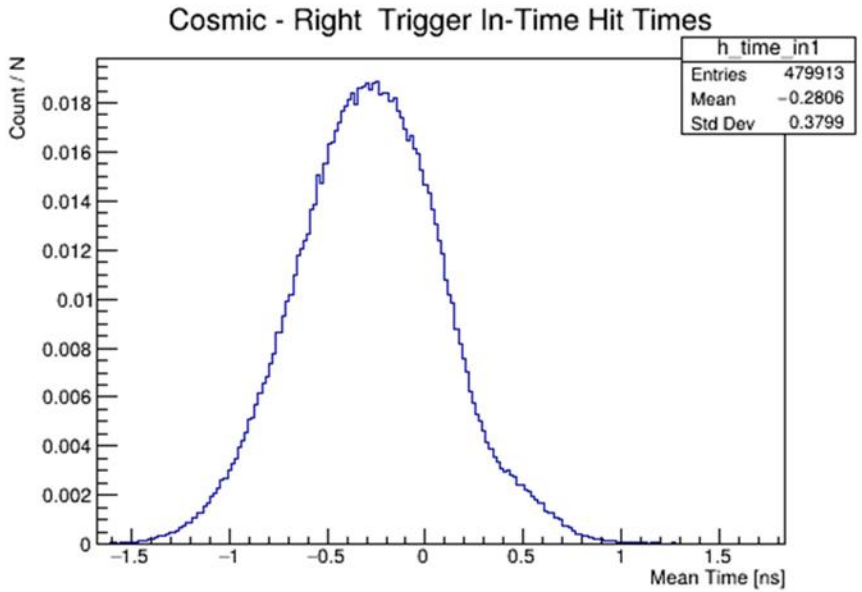




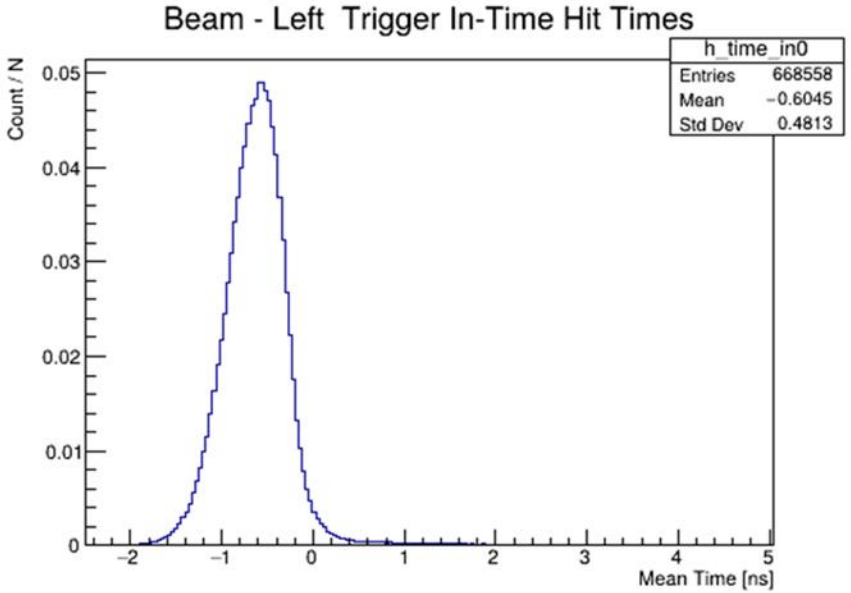
In Time Hits

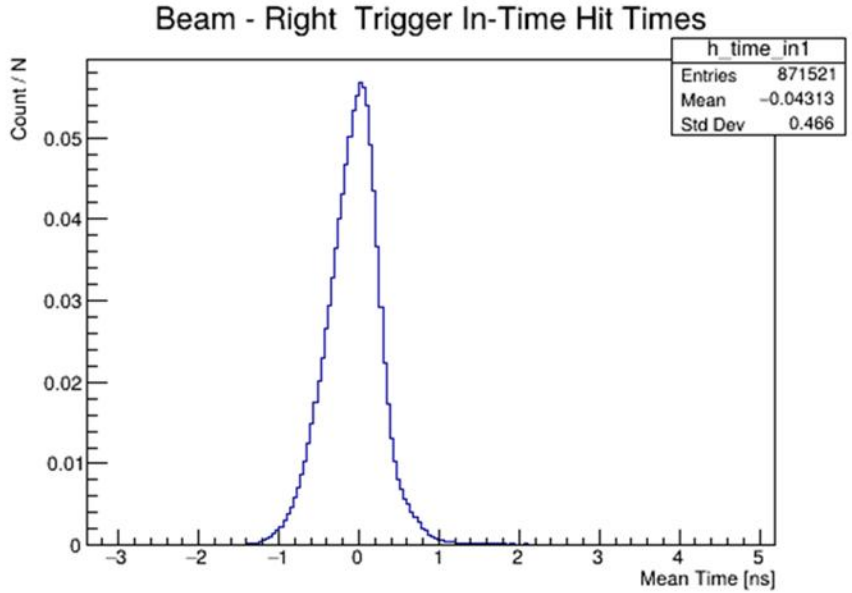
Cosmic





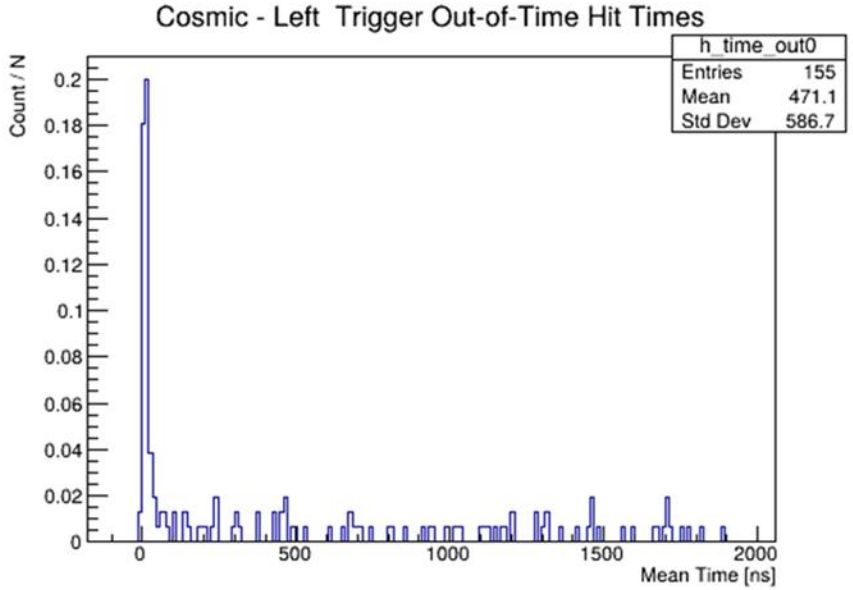
Beam

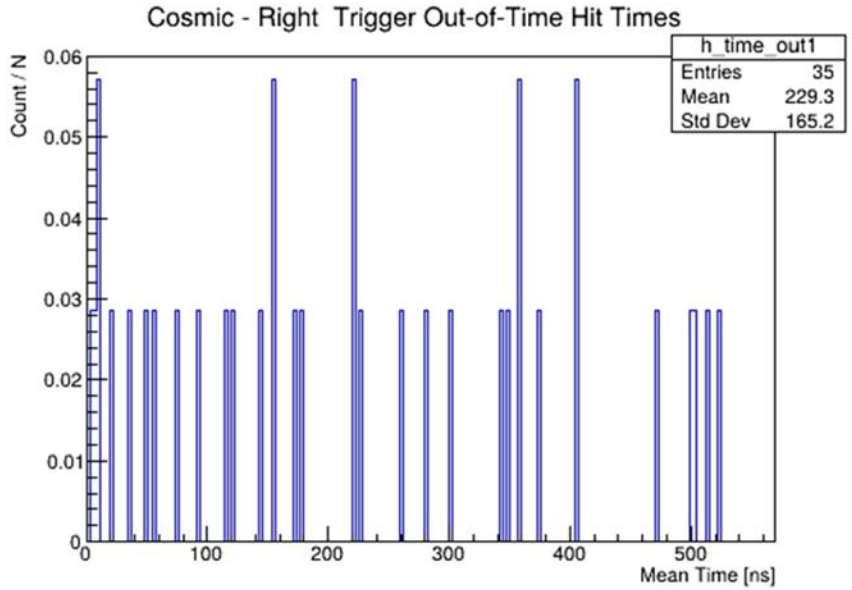




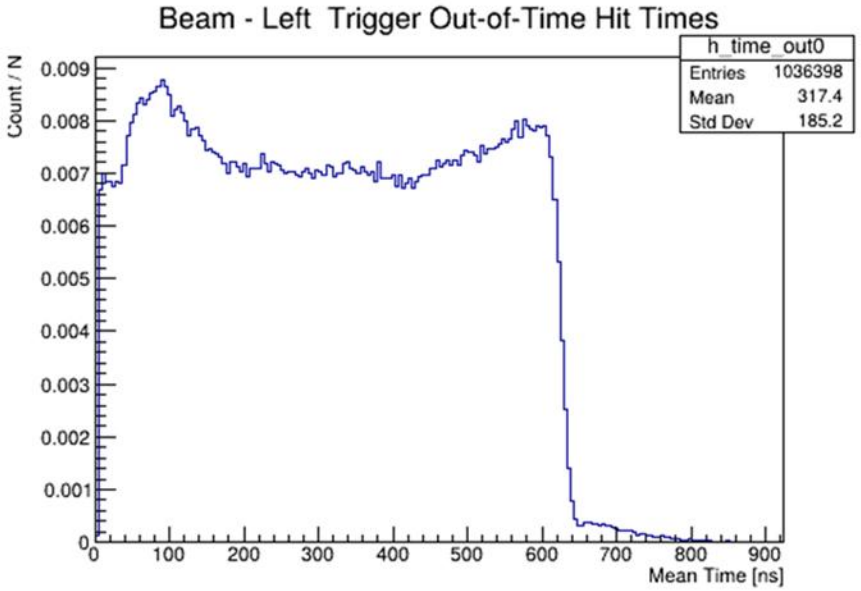
Out of Time Hits

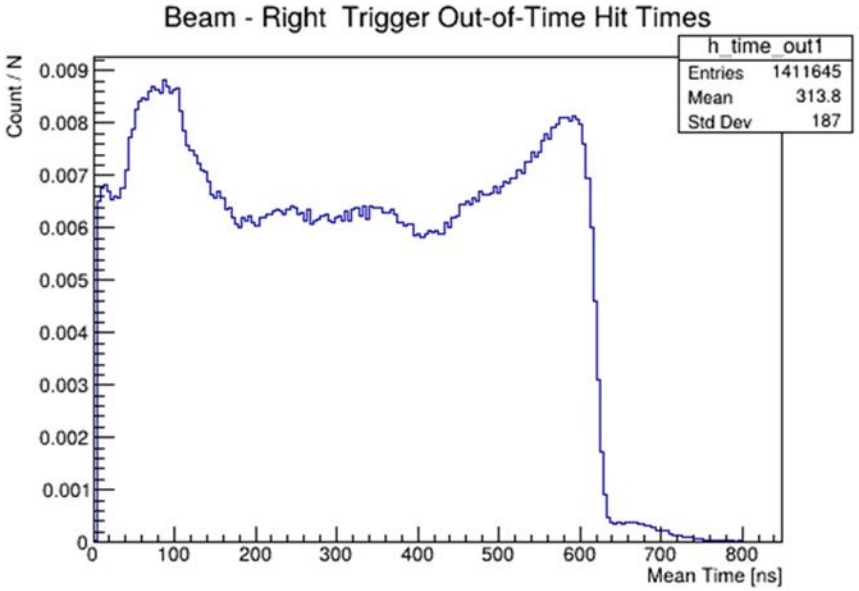
Cosmic





Beam

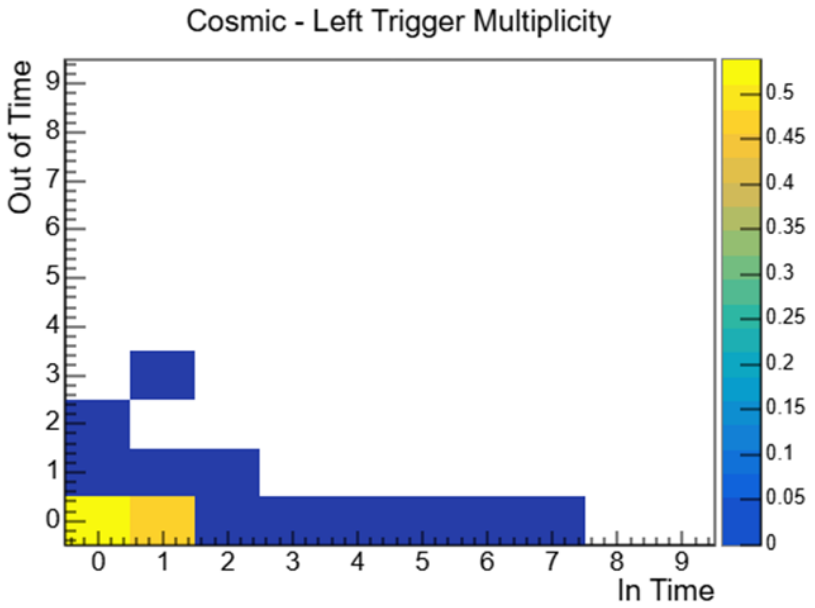


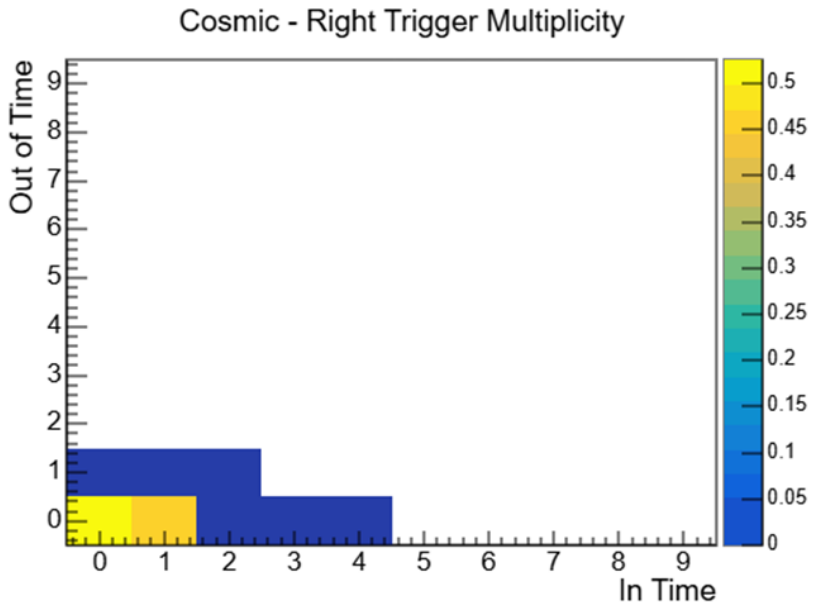


Detector Hit Multiplicity

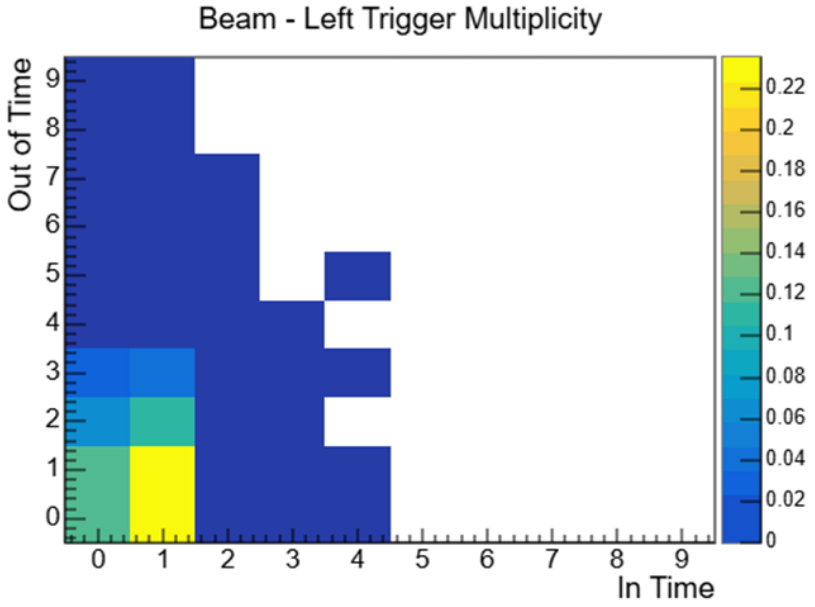
Out of Time vs In Time Trigger

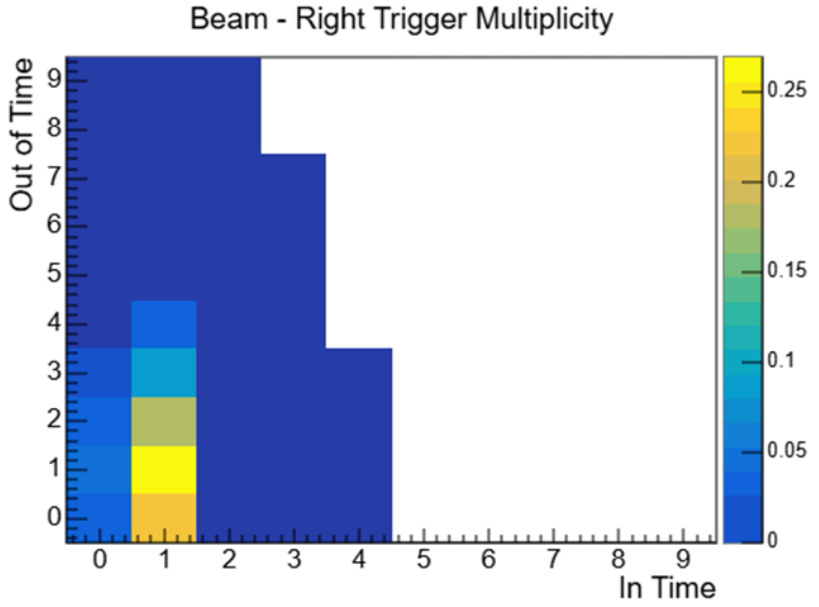
Cosmic





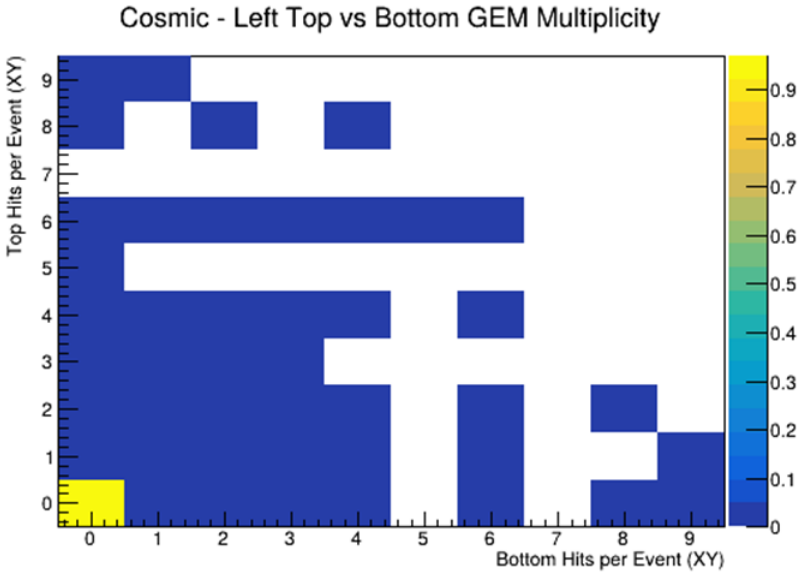
Beam

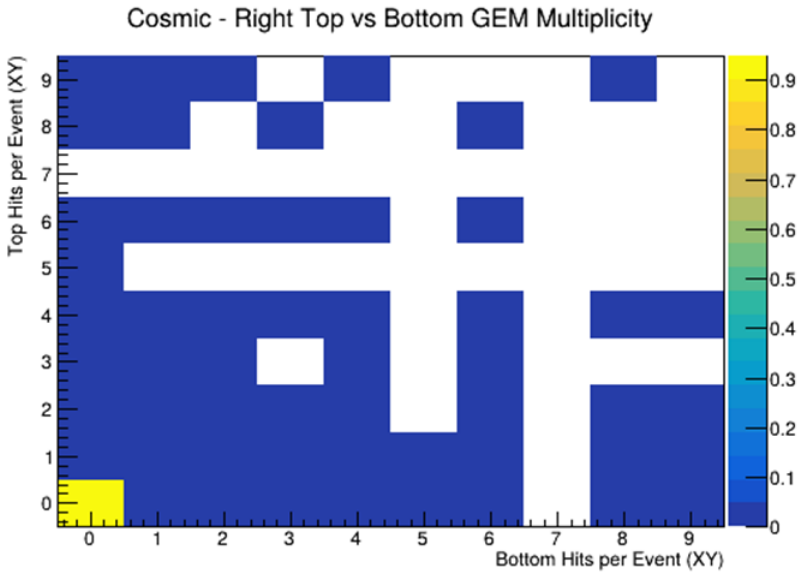




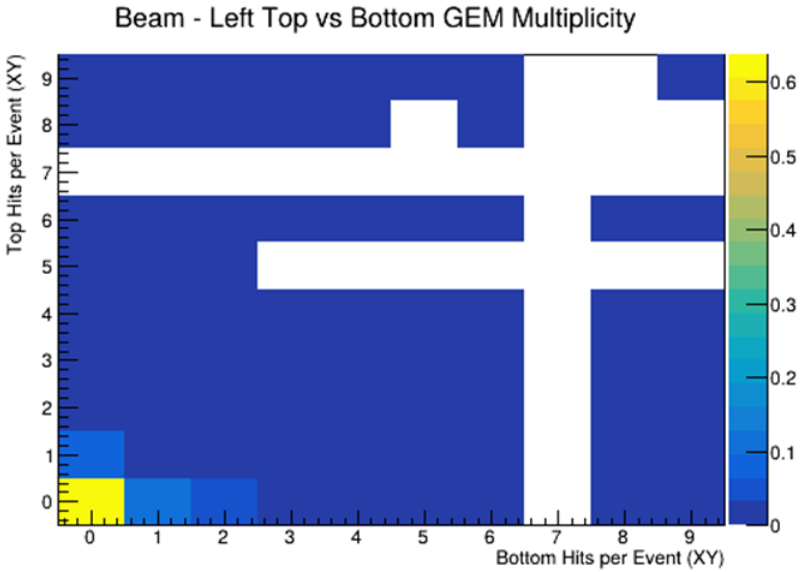
Top vs Bottom GEM

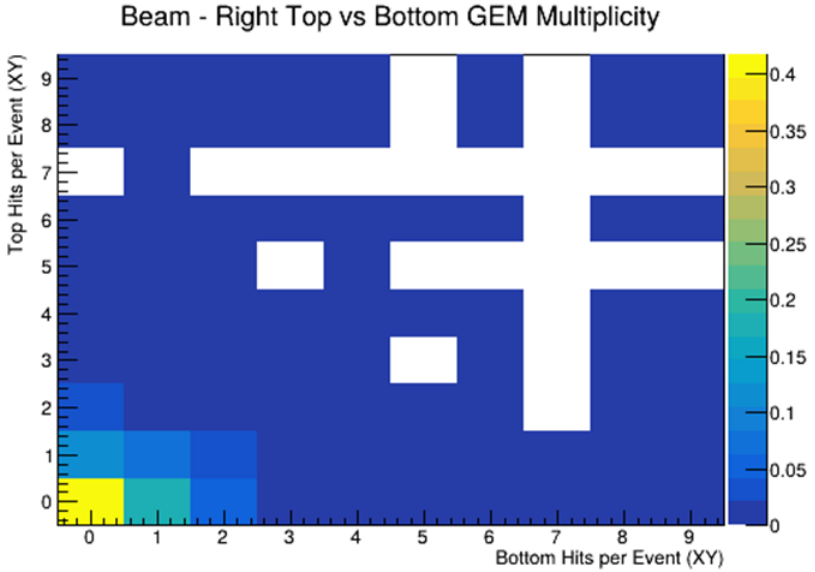
Cosmic





Beam

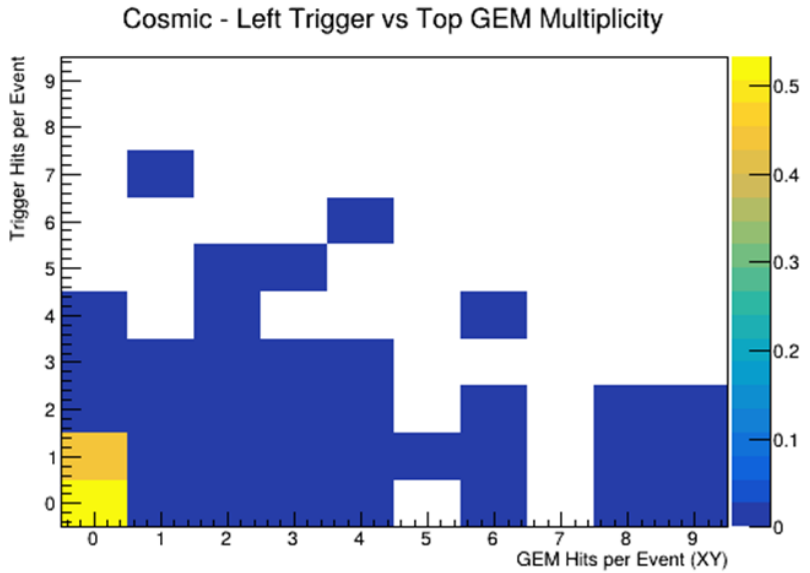


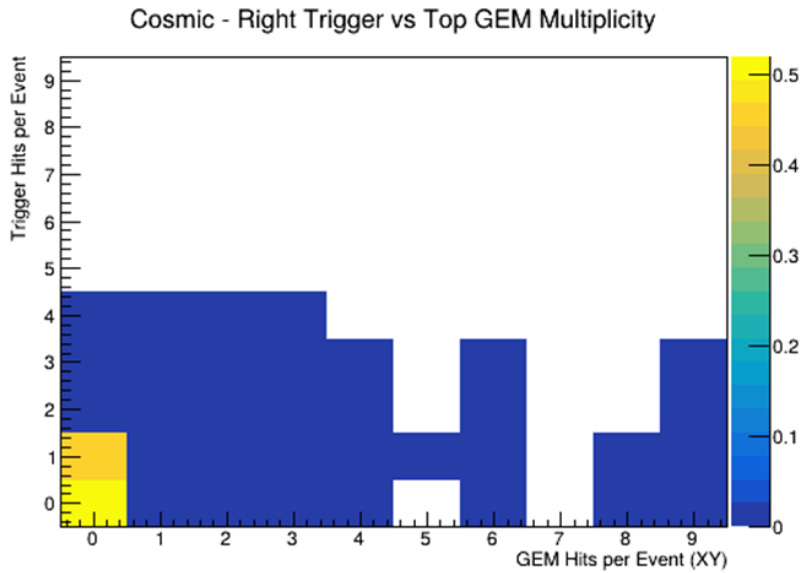


Trigger vs Top GEM Hit Multiplicity

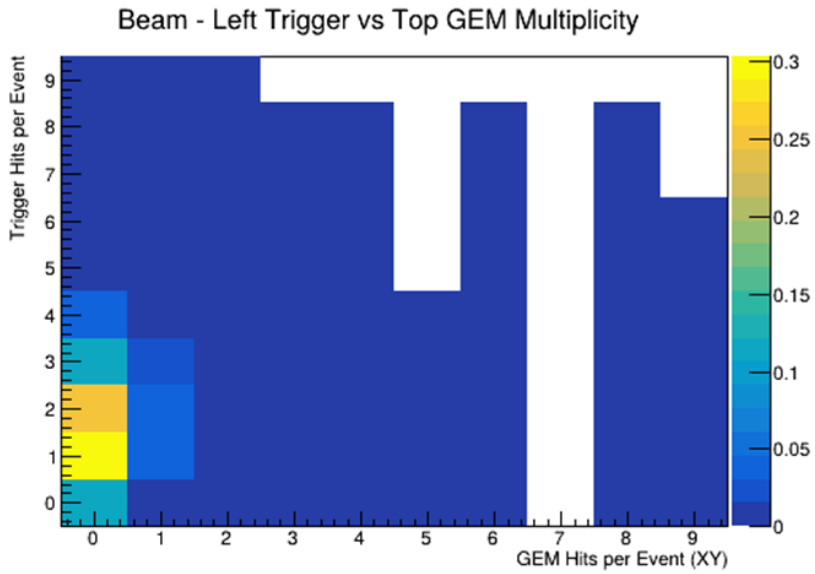
Trigger vs Top GEM

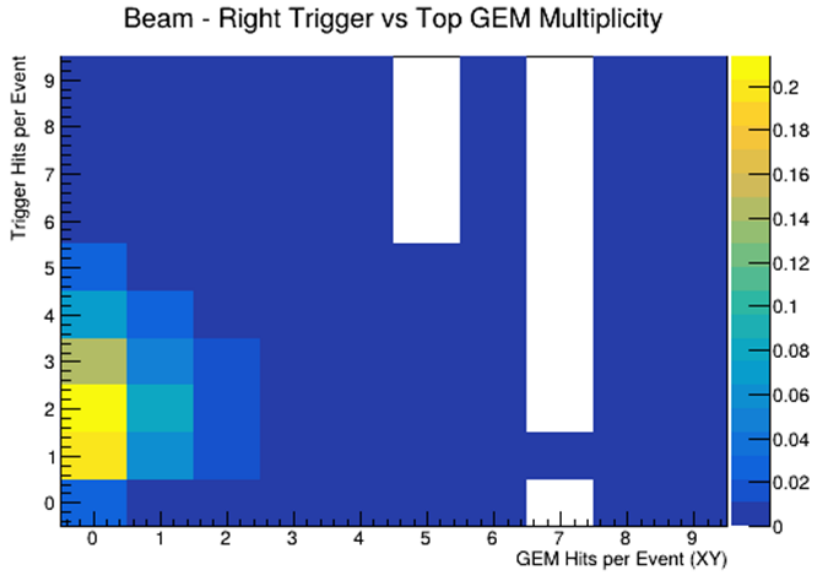
Cosmic





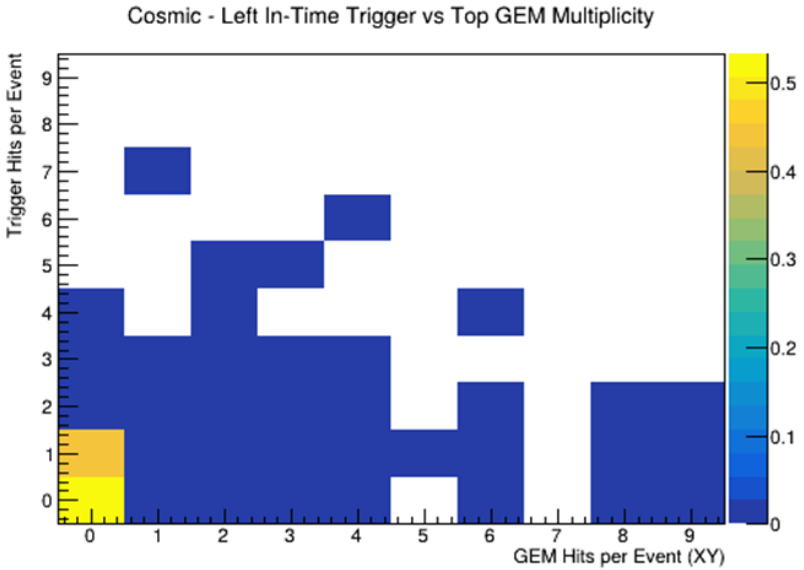
Beam

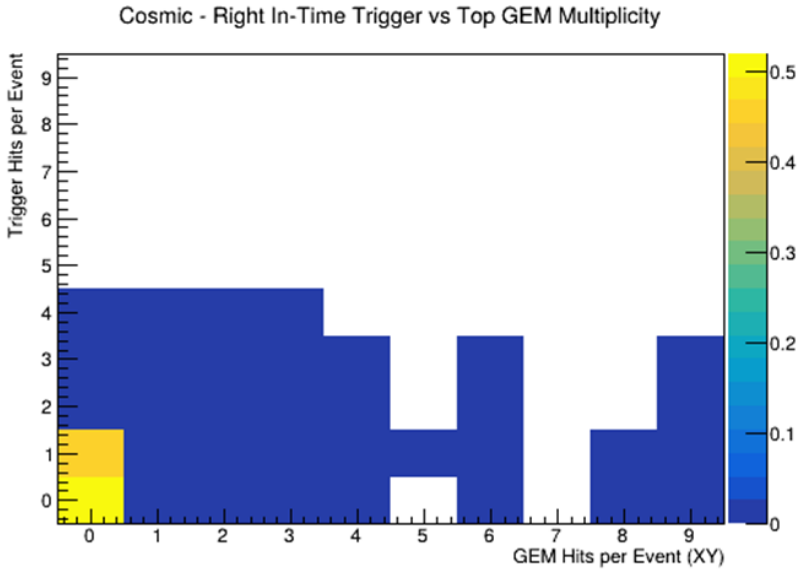




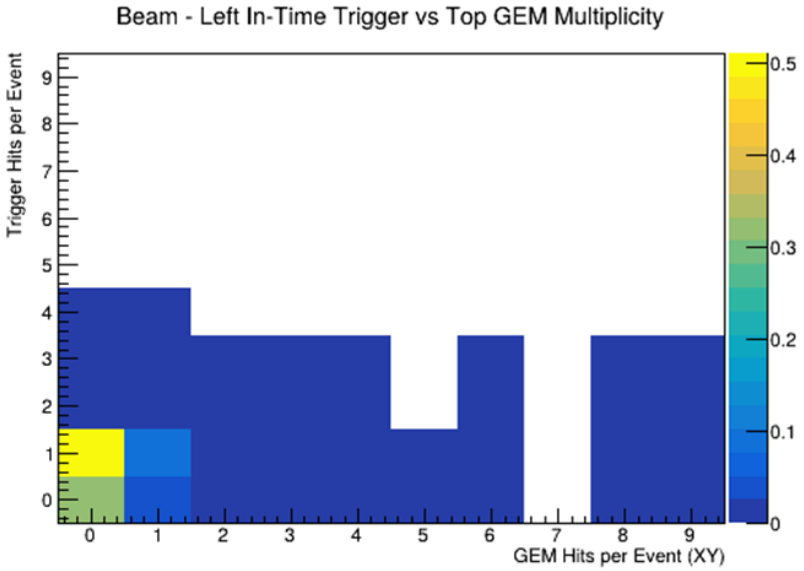
Trigger vs Top GEM In Time Hits

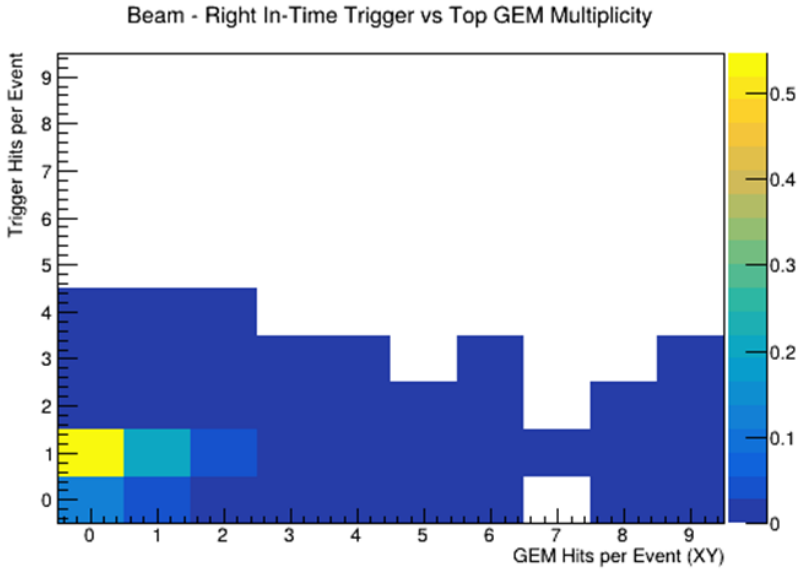
Cosmic





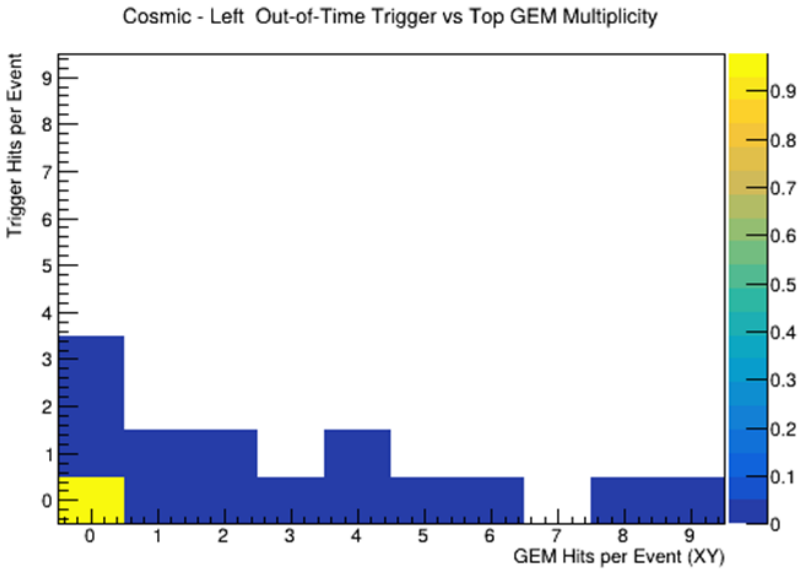
Beam

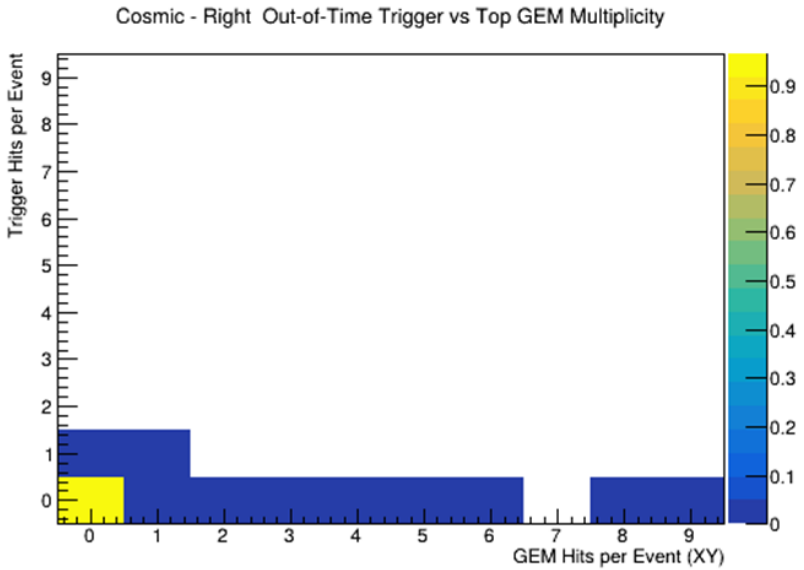




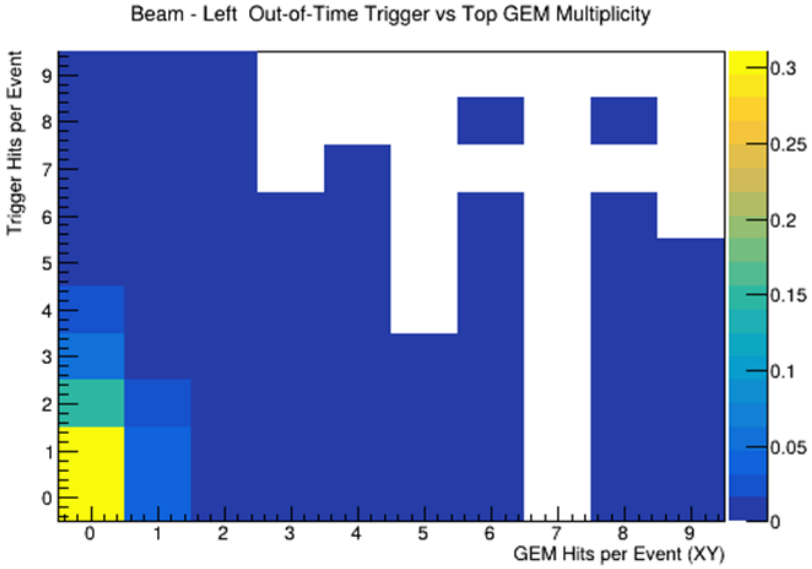
Trigger vs Top GEM Out of Time Hits

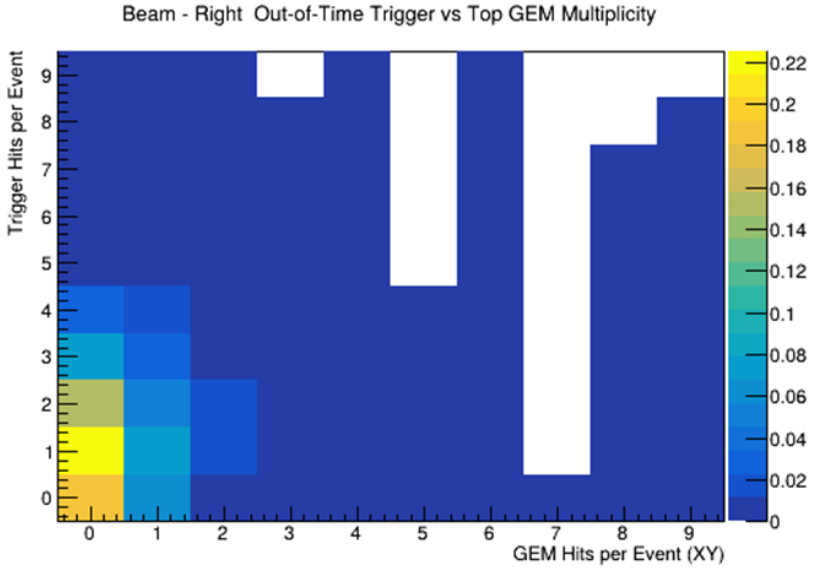
Cosmic





Beam

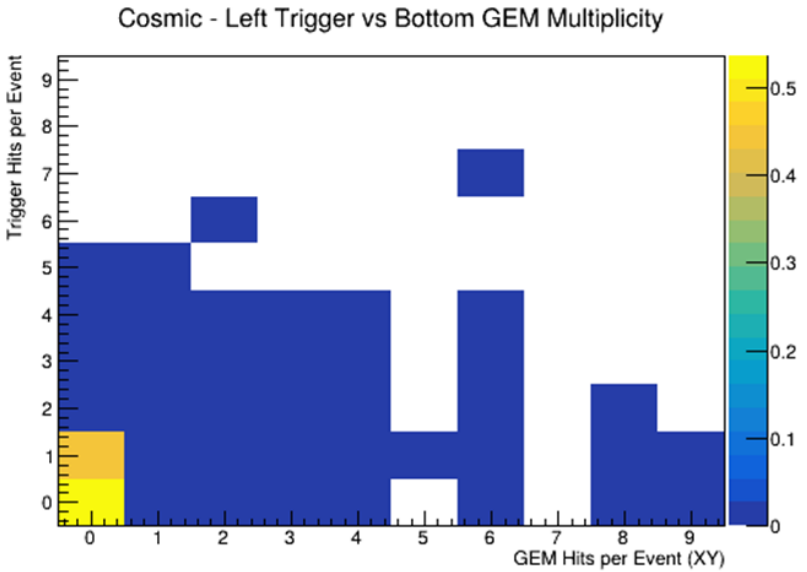


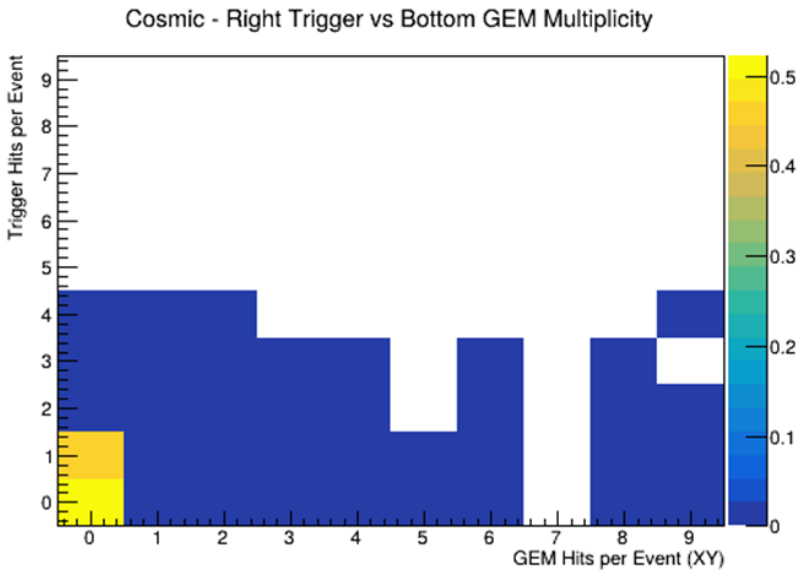


Trigger vs Bottom GEM Multiplicity

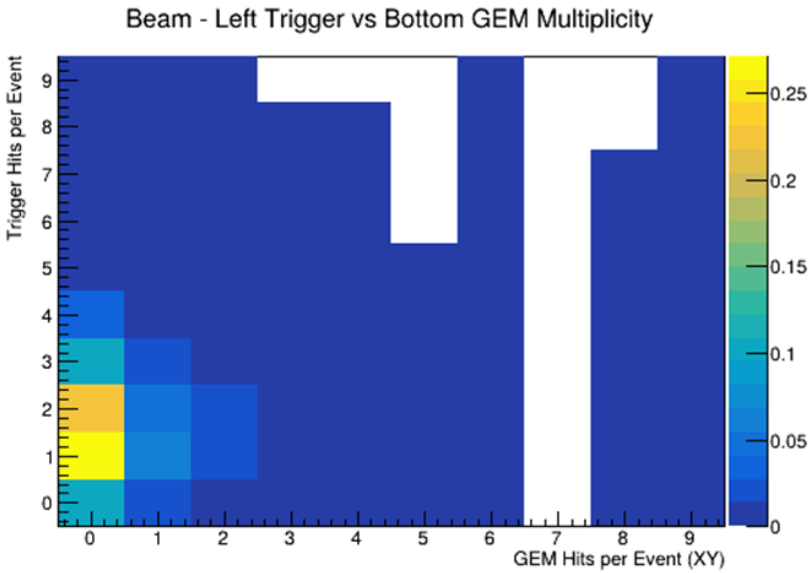
Trigger vs Bottom GEM

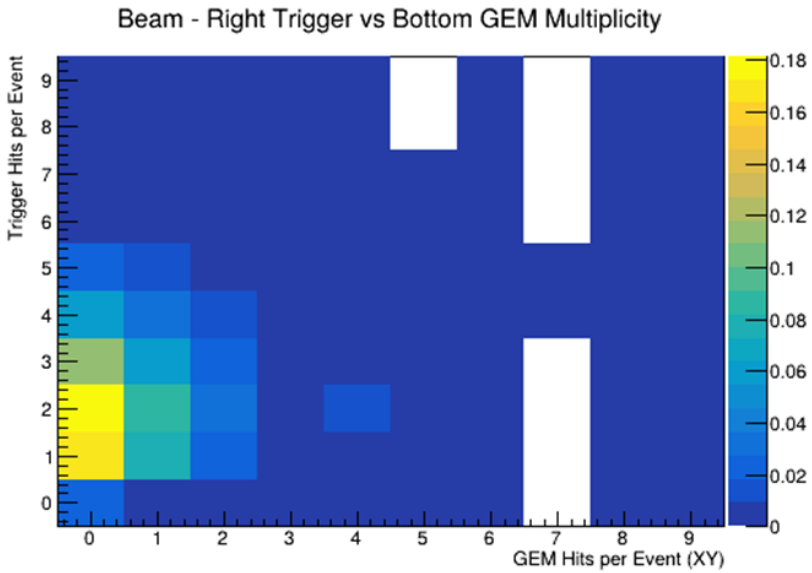
Cosmic





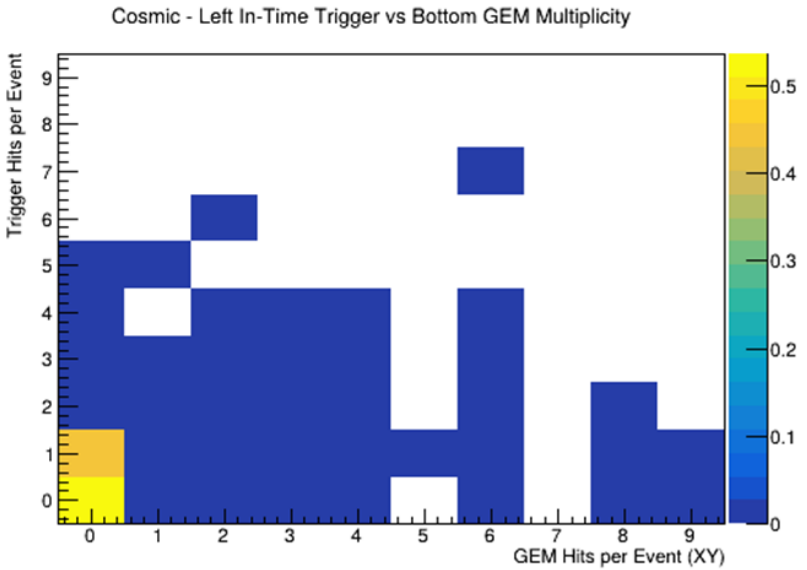
Beam

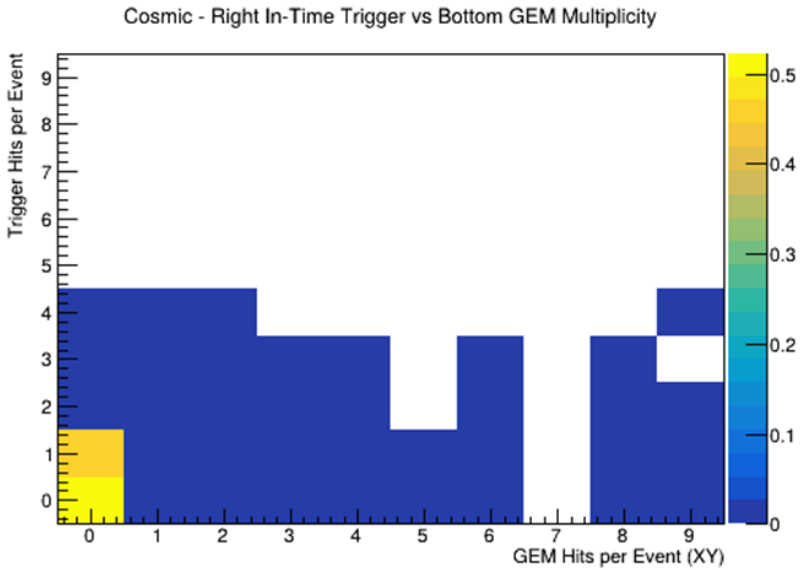




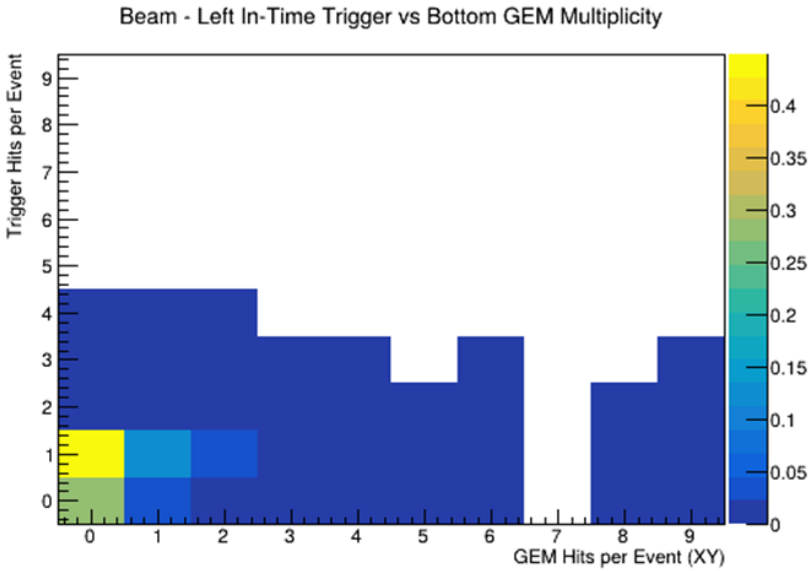
Trigger vs Bottom GEM In Time Hits

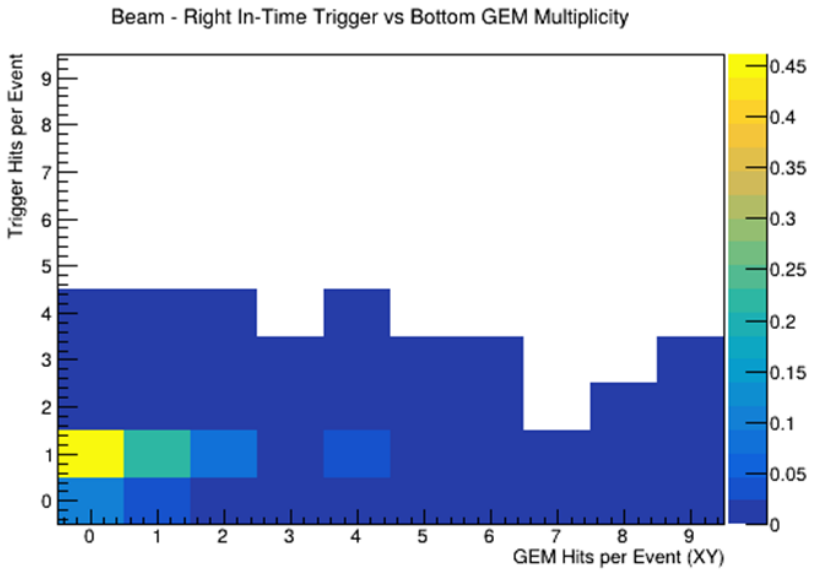
Cosmic





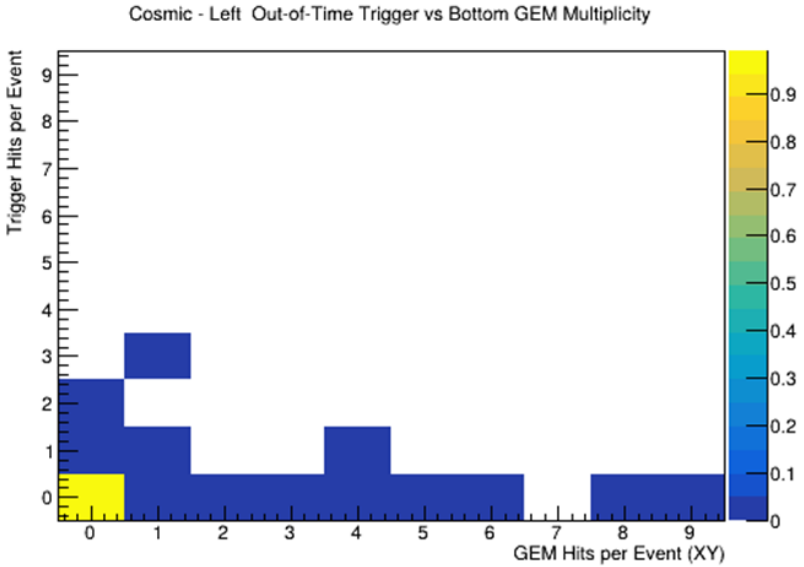
Beam

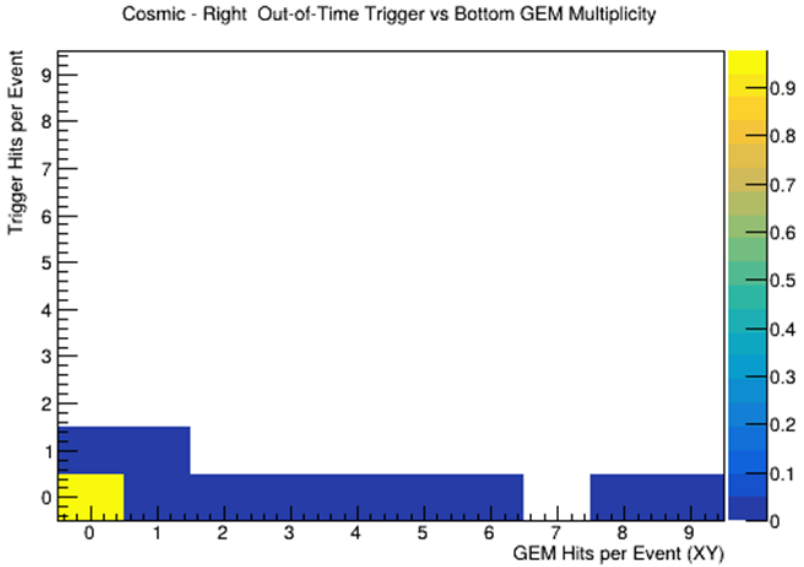




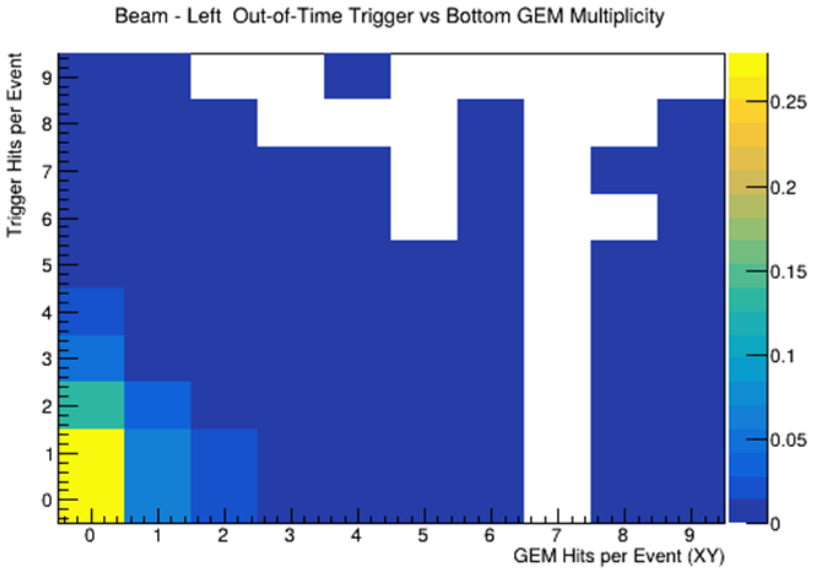
Trigger vs Bottom GEM Out of Time Hits

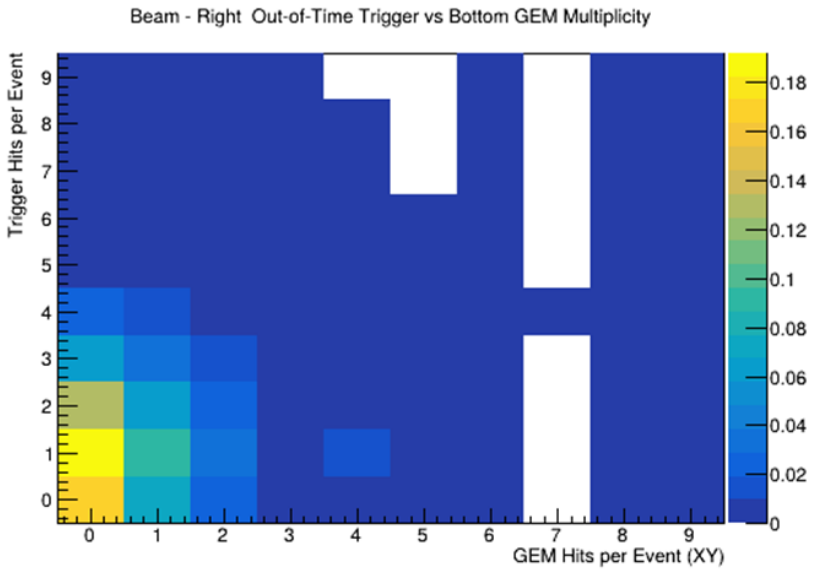
Cosmic





Beam

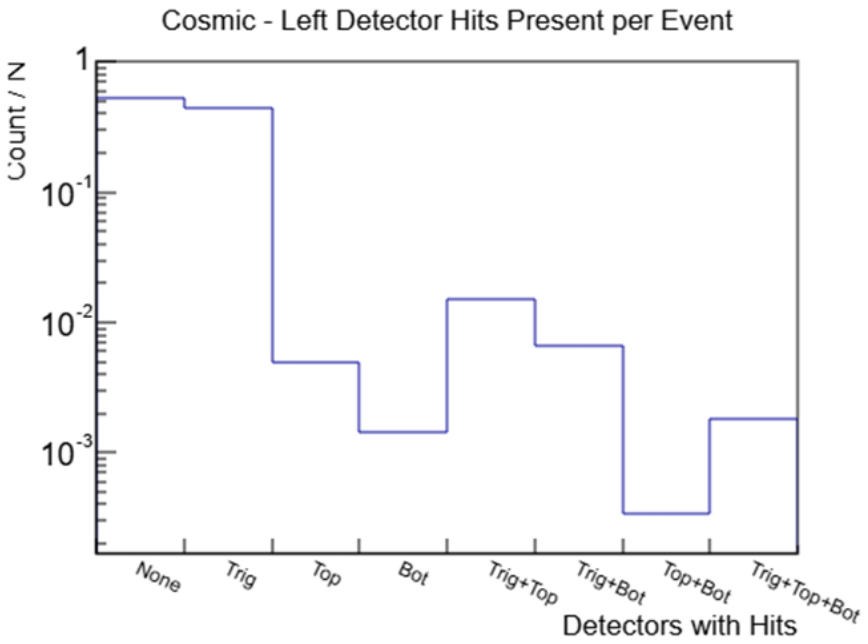


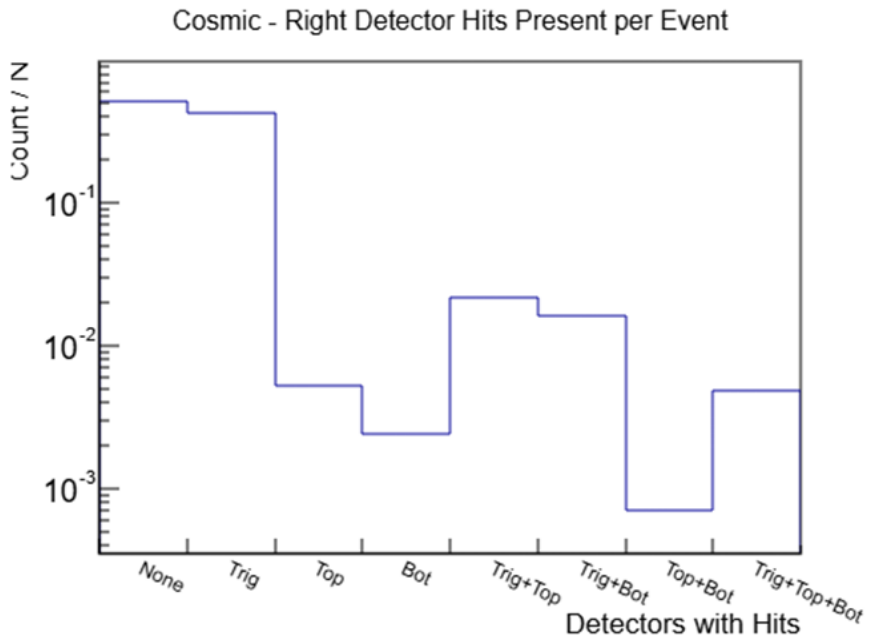


Detectors with Hits per Event

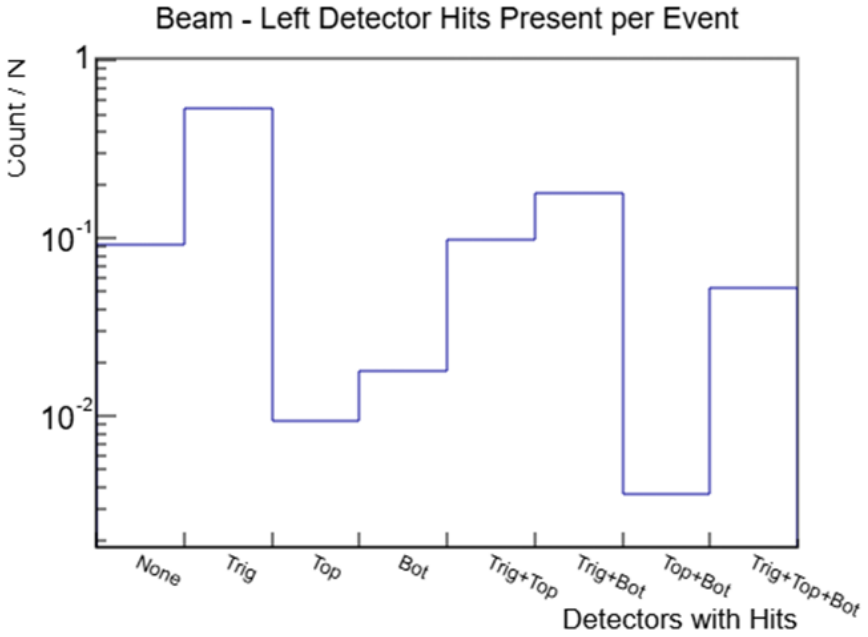
Events with hits present in only these detectors

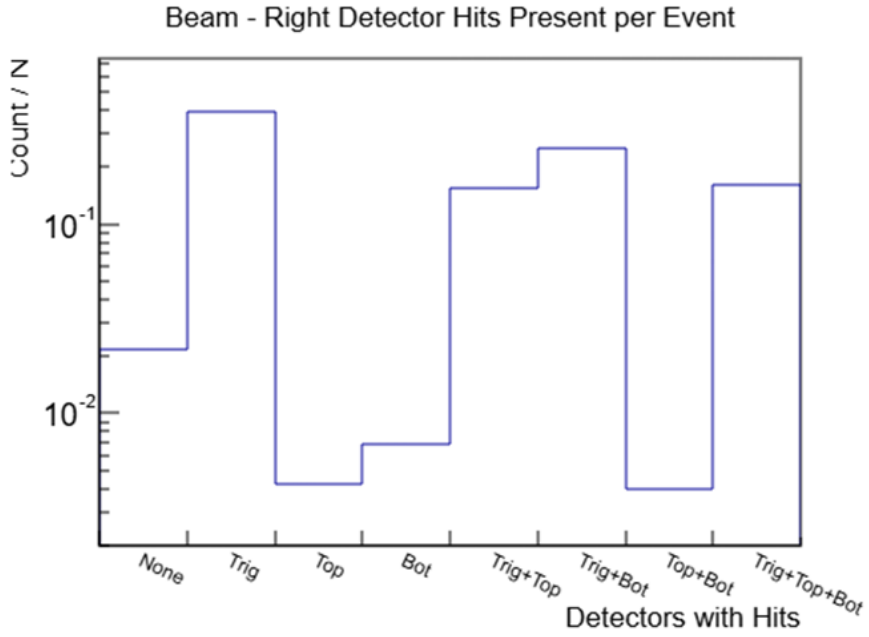
Cosmic





Beam

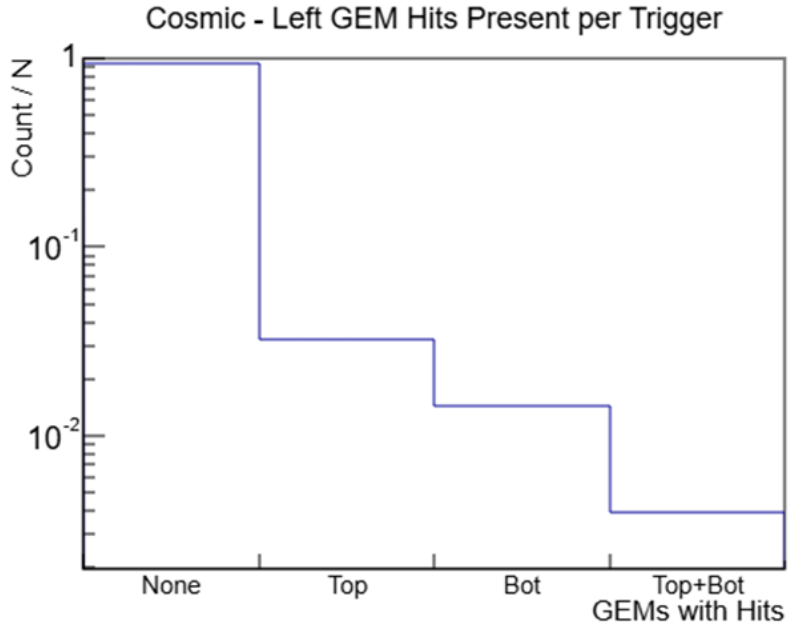


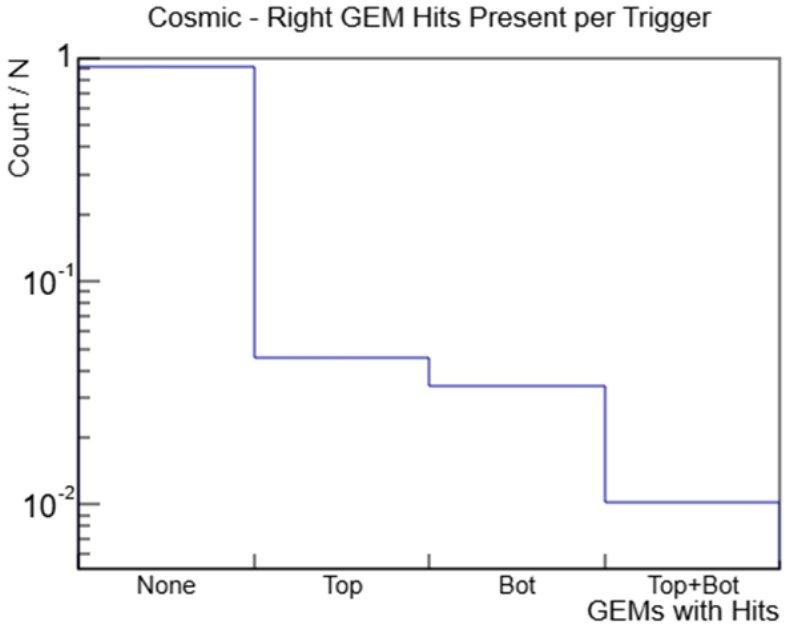


GEMs with Hits per Events with Trigger Hits

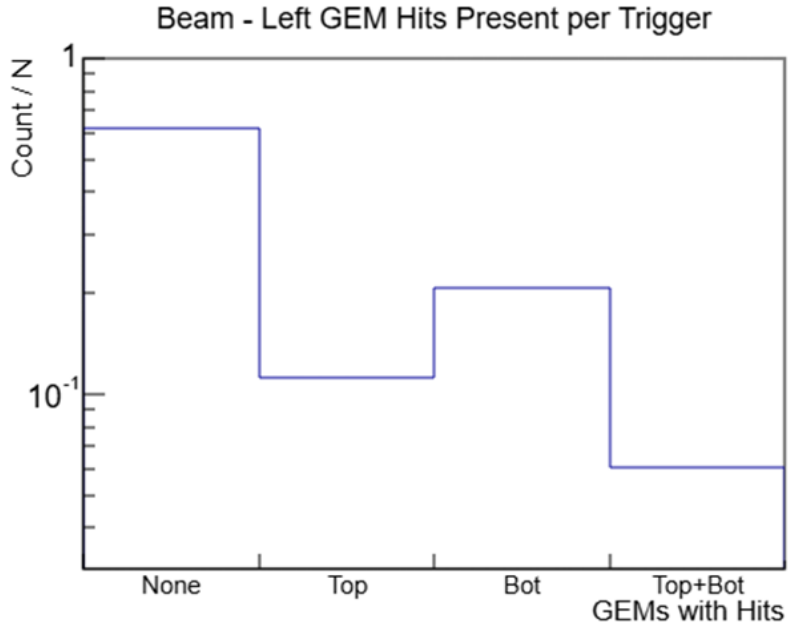
Events with trigger hits with hits present in only these GEM detectors

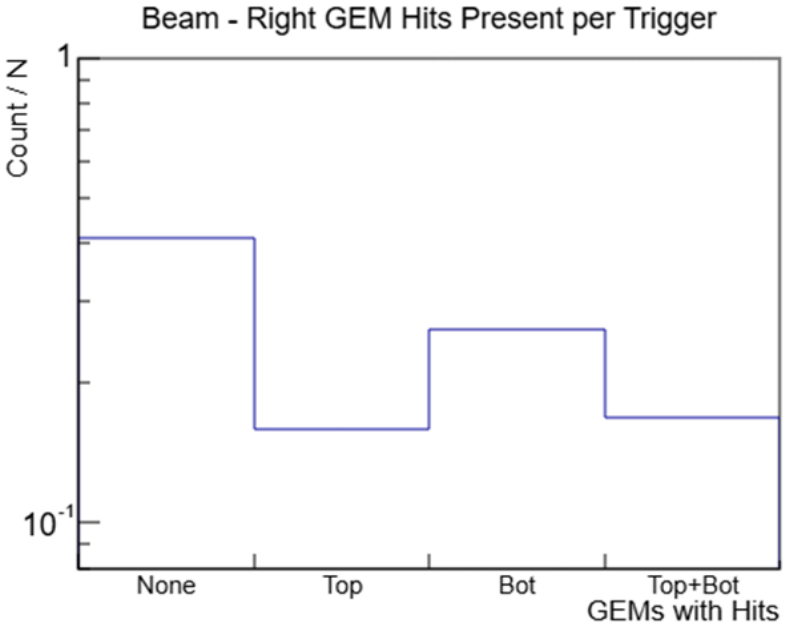
Cosmic





Beam





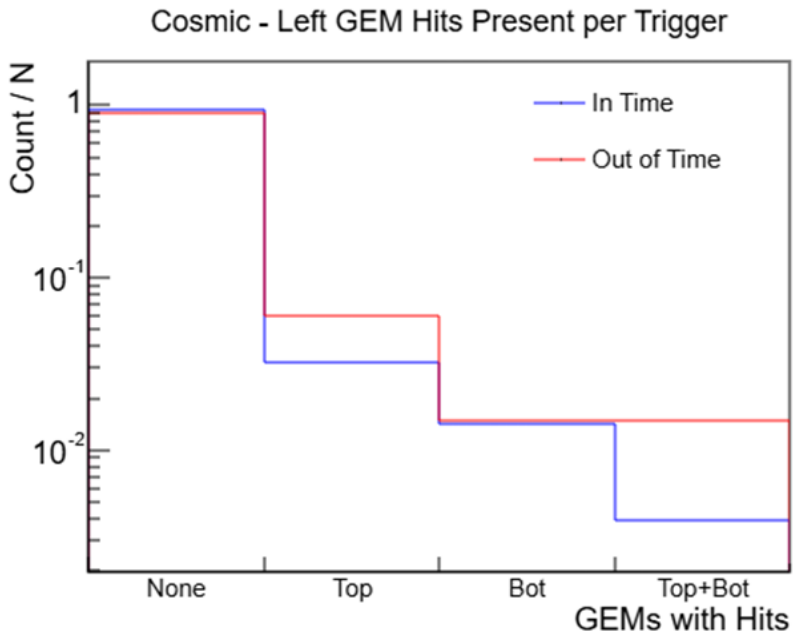
Events with trigger hits with hits present in only these GEM detectors

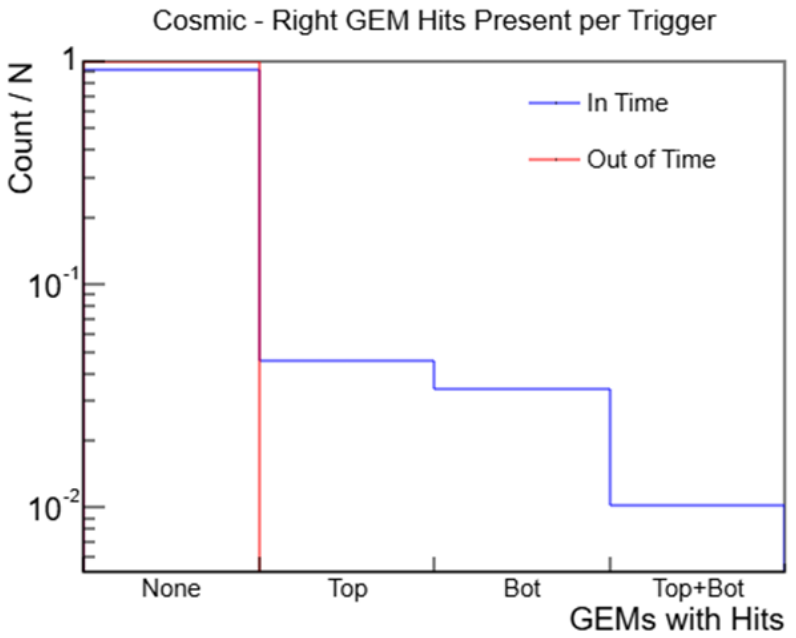
Cosmic

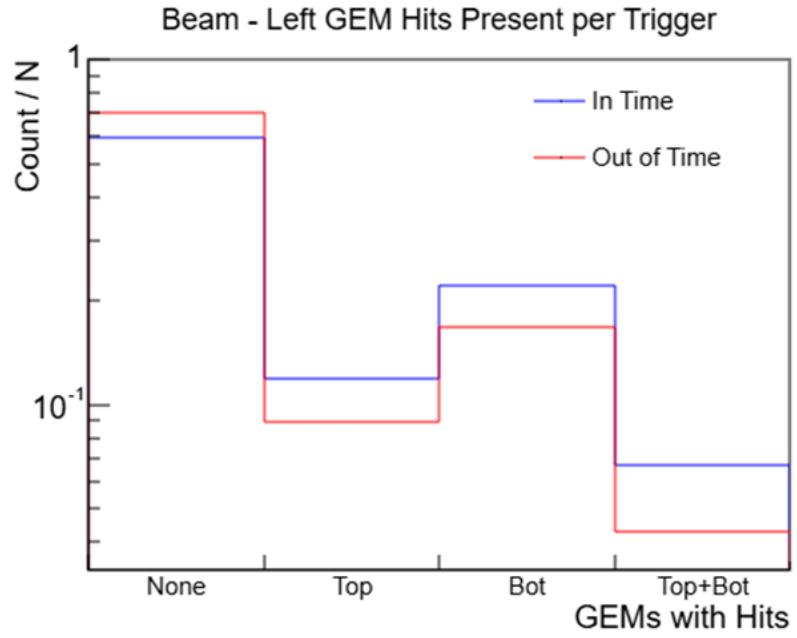
Events with at least one in-time trigger hit

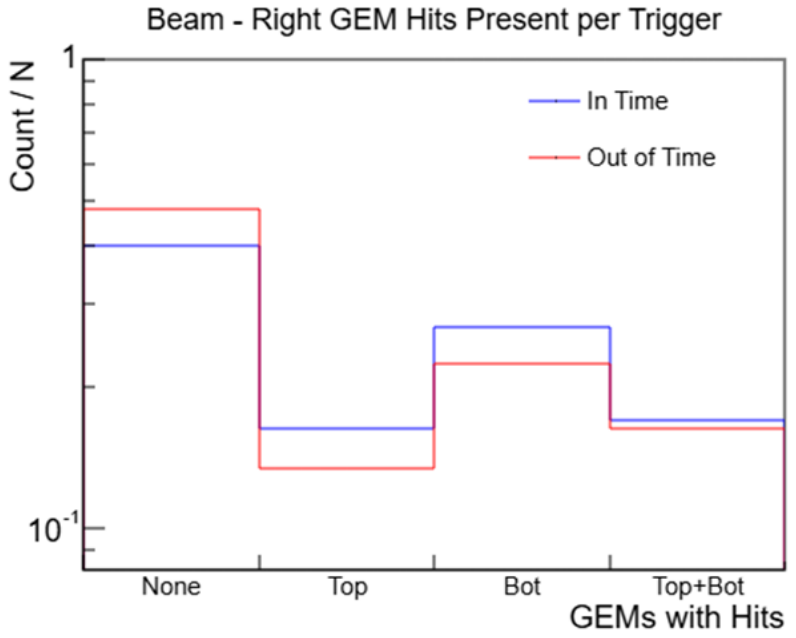
Events with only out-of-time trigger hits

Beam



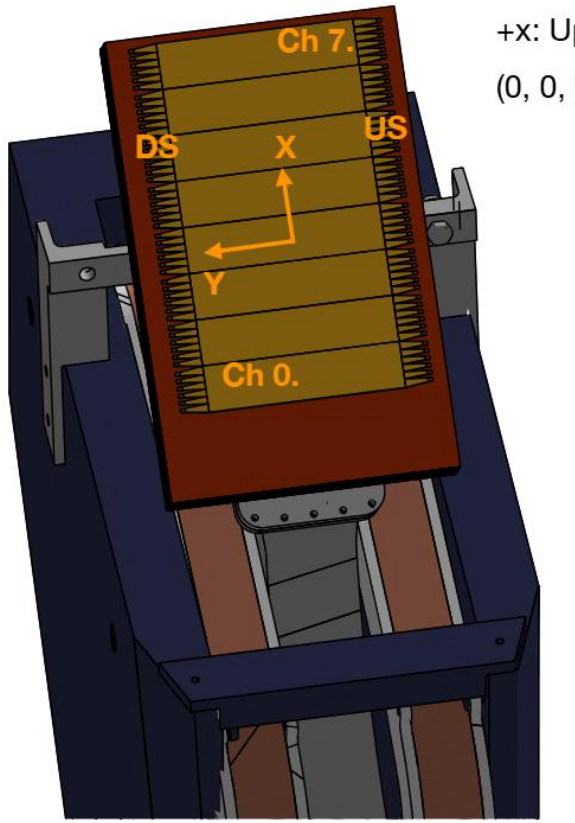






Local Coordinate System

Trigger paddle local coordinate:



+z: out of the trigger plane away from magnets

+y: left along the long side of paddle

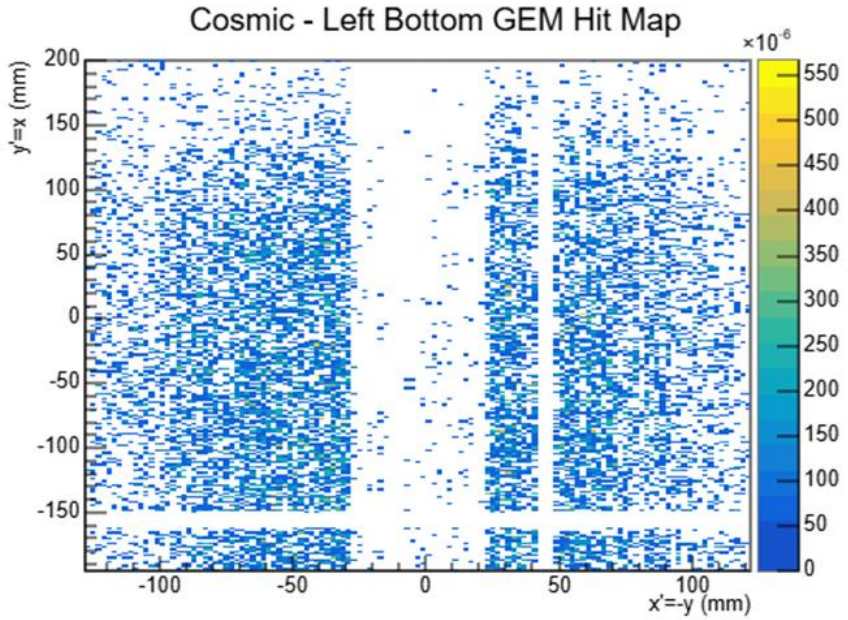
+x: Up along the short side of paddle

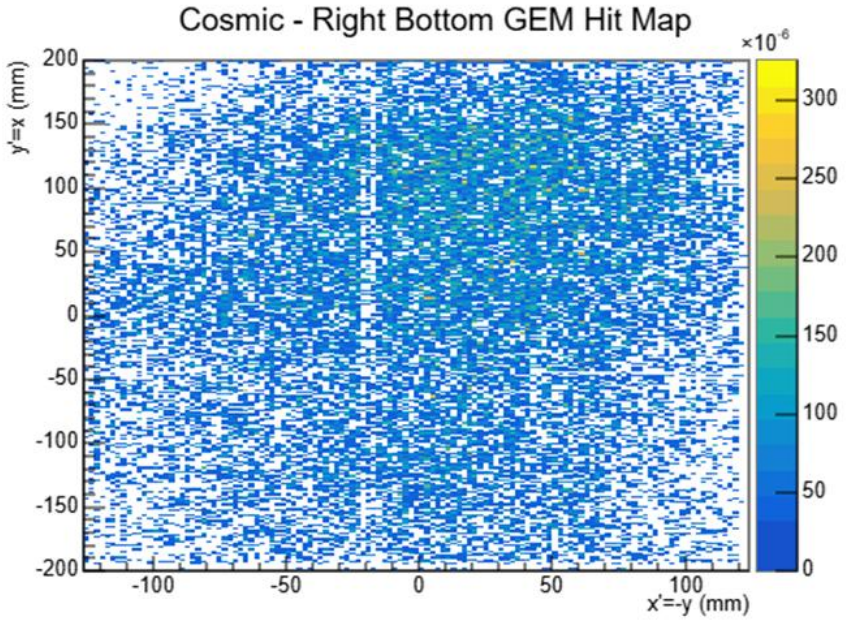
(0, 0, 0): center of the paddle

GEM Hit Positions

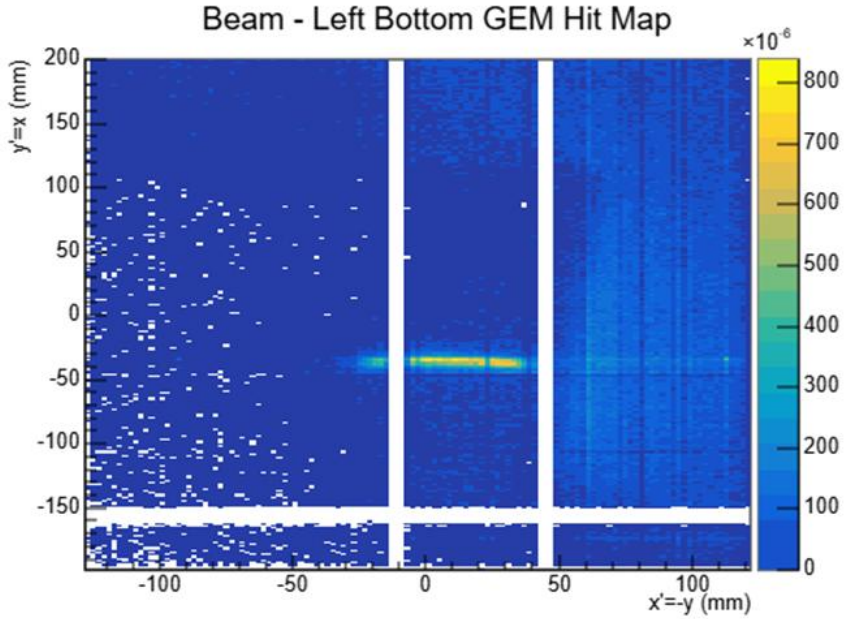
Bottom GEMs

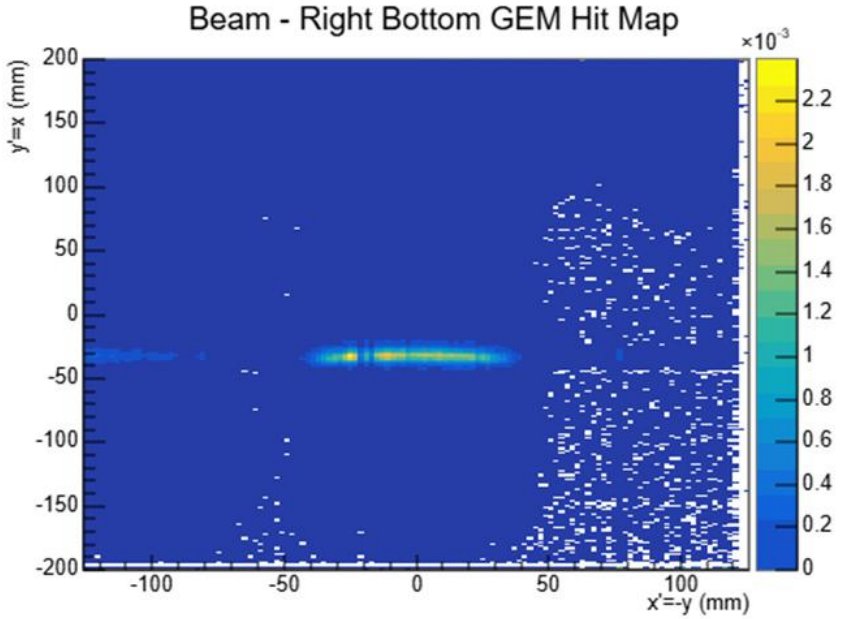
Cosmic





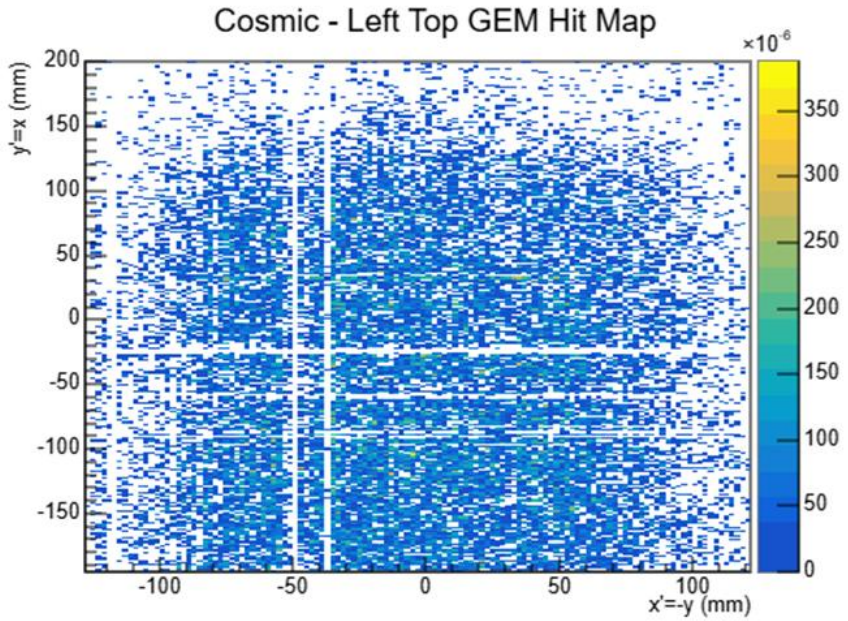
Beam

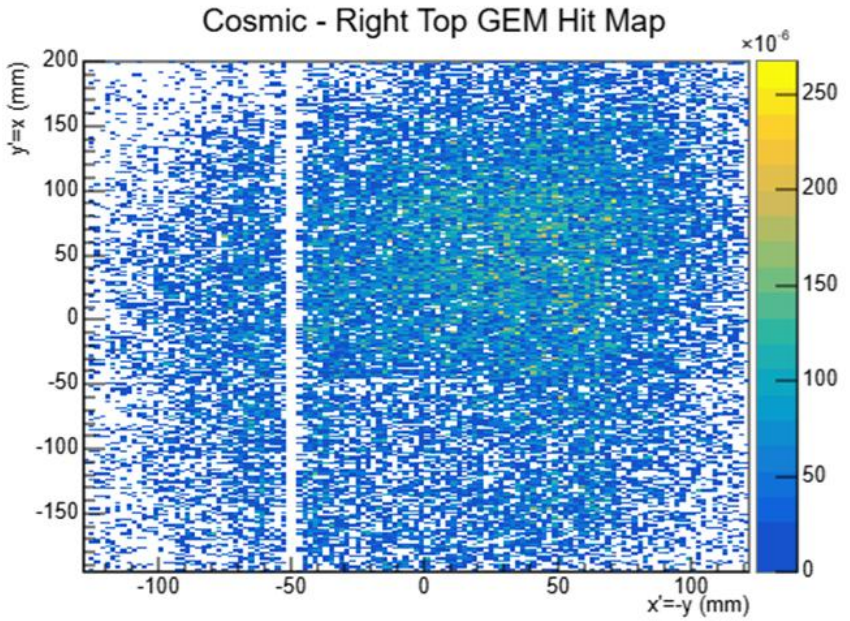




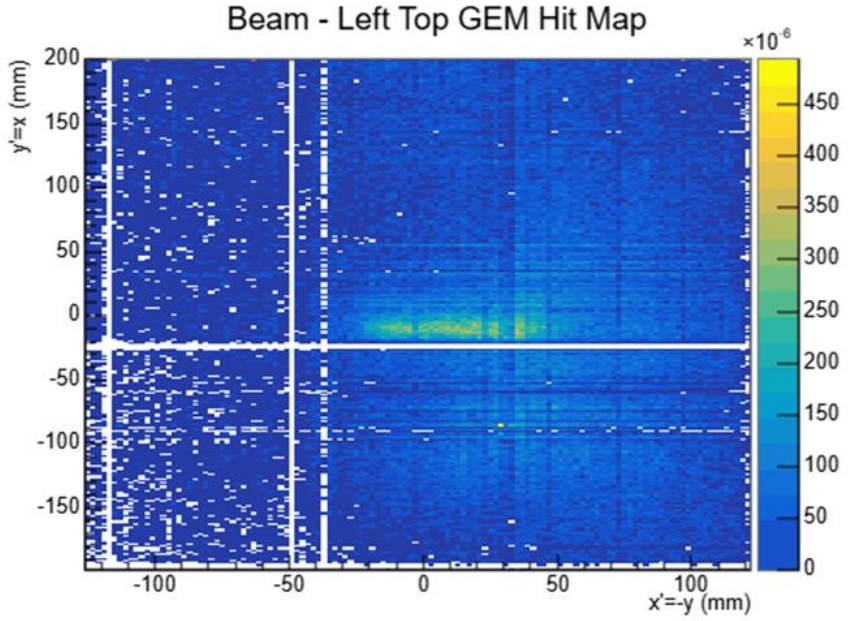
Top GEMs

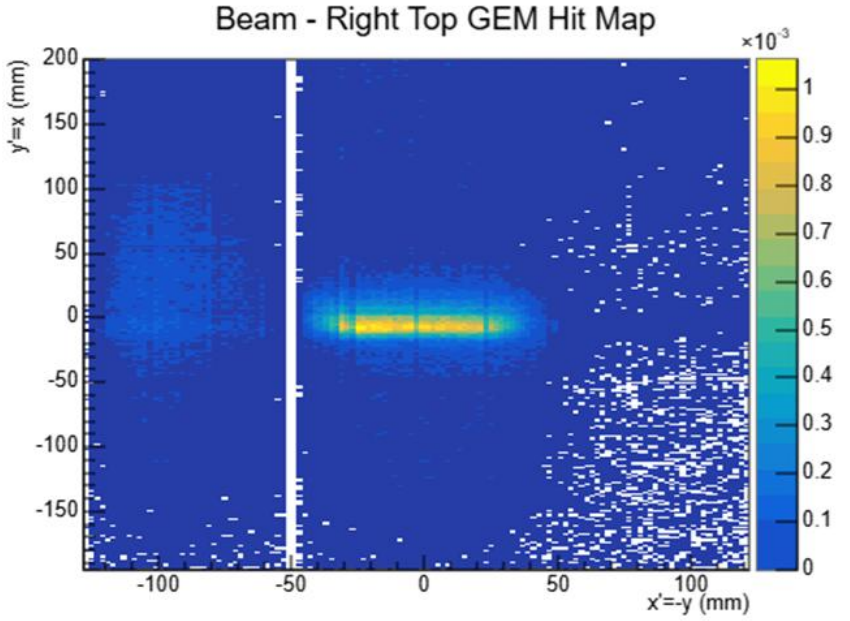
Cosmic





Beam

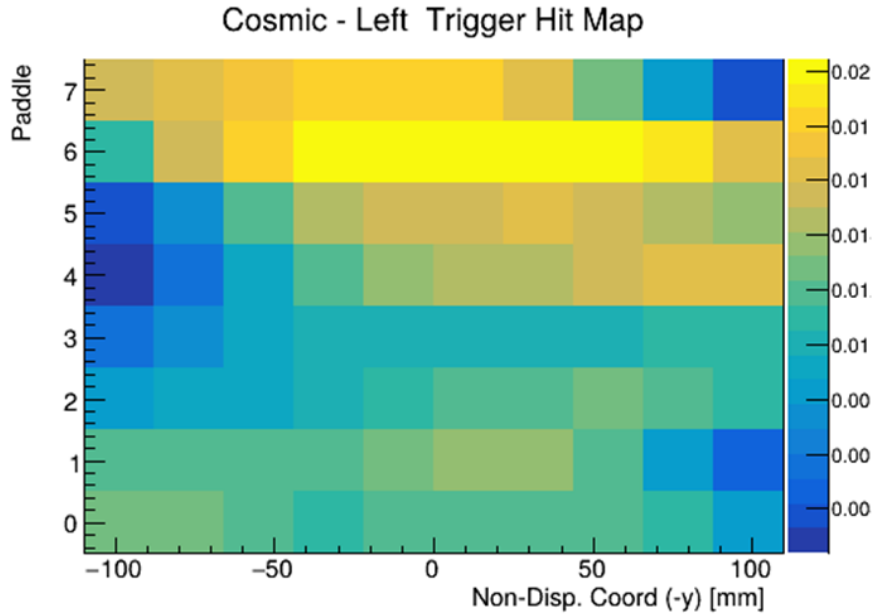


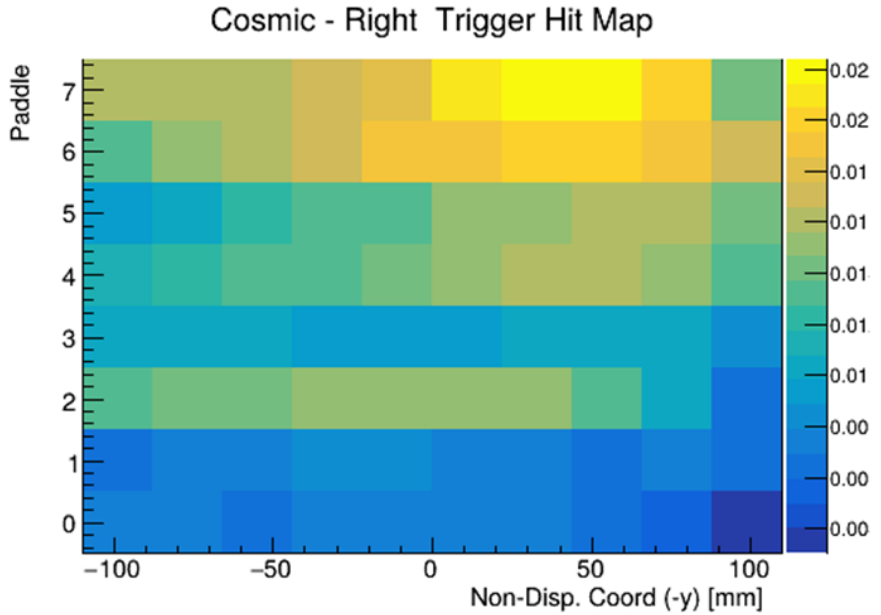


Trigger Hit Positions

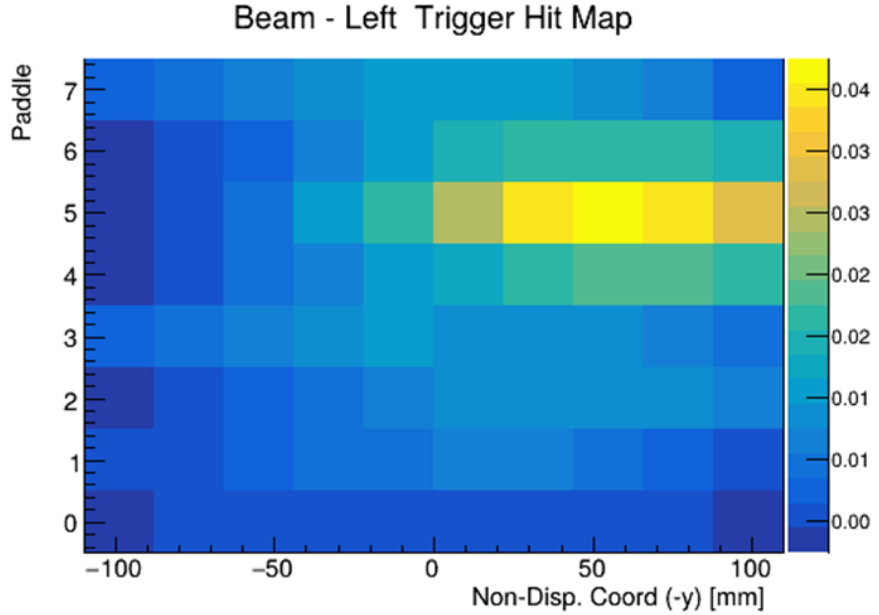
All Trigger Hits

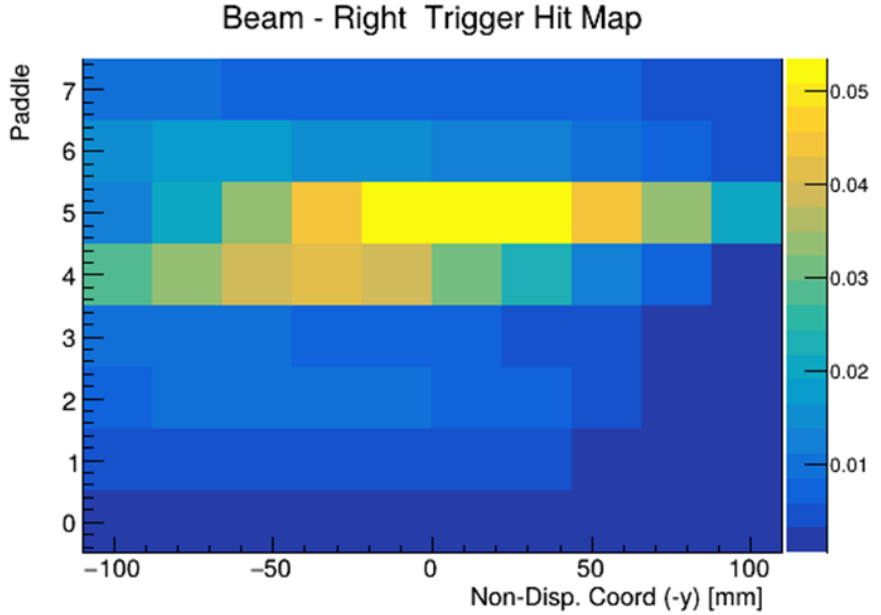
Cosmic





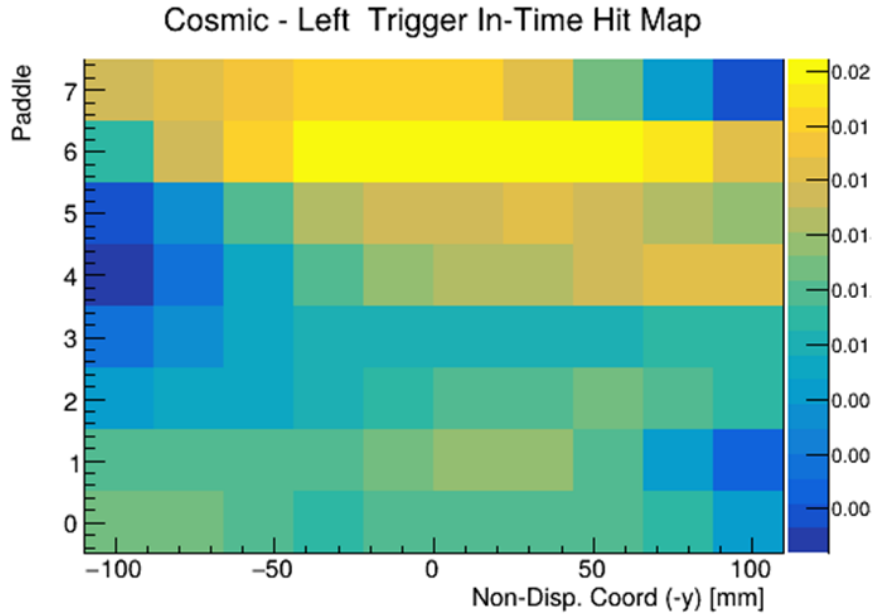
Beam

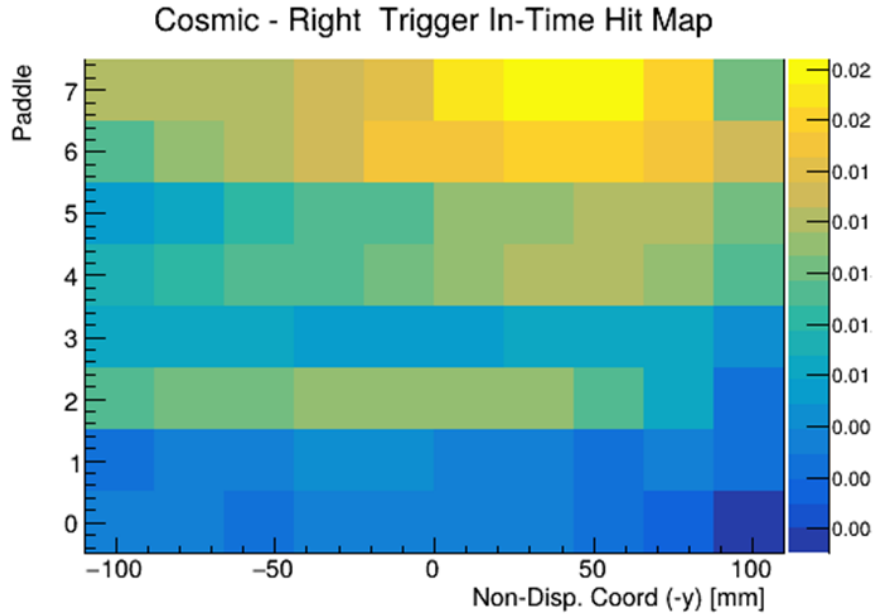




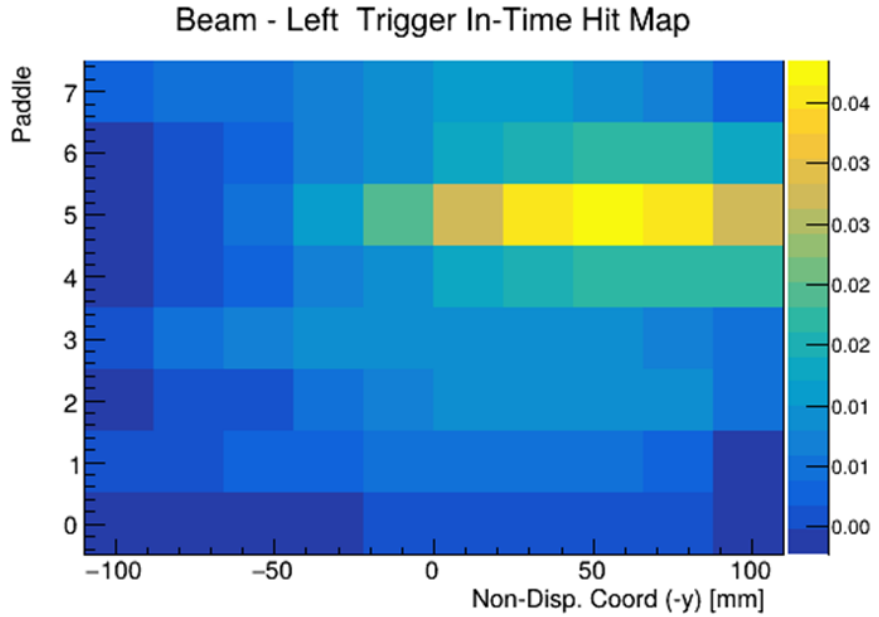
In Time Hits

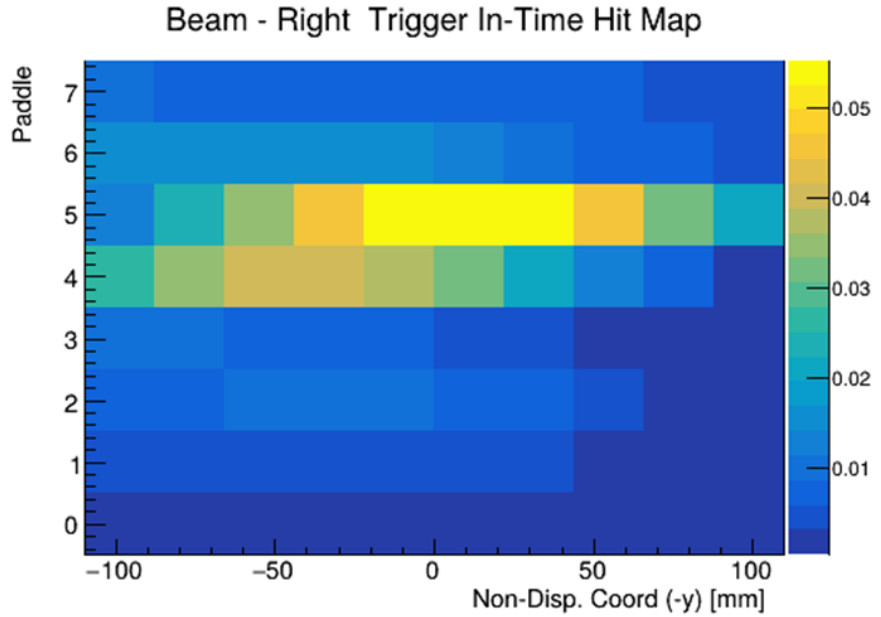
Cosmic





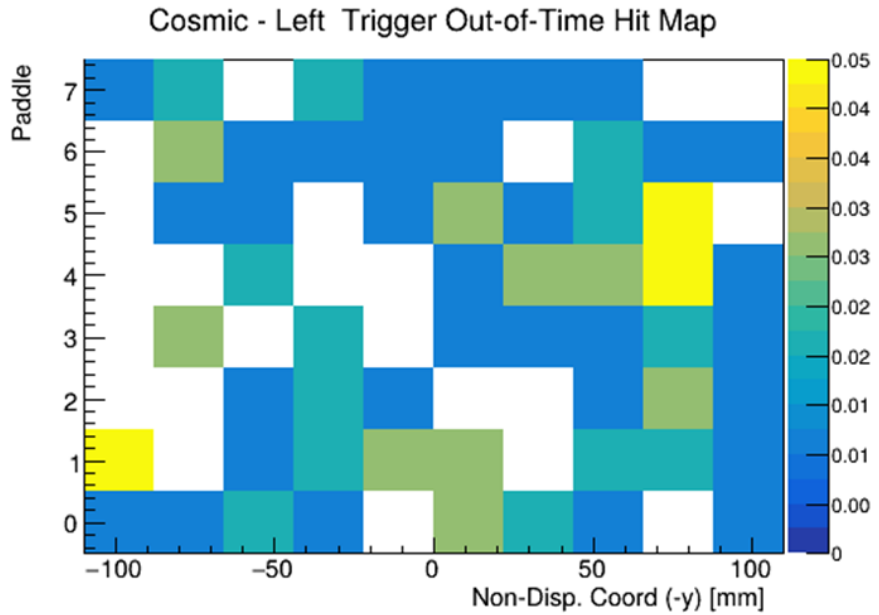
Beam

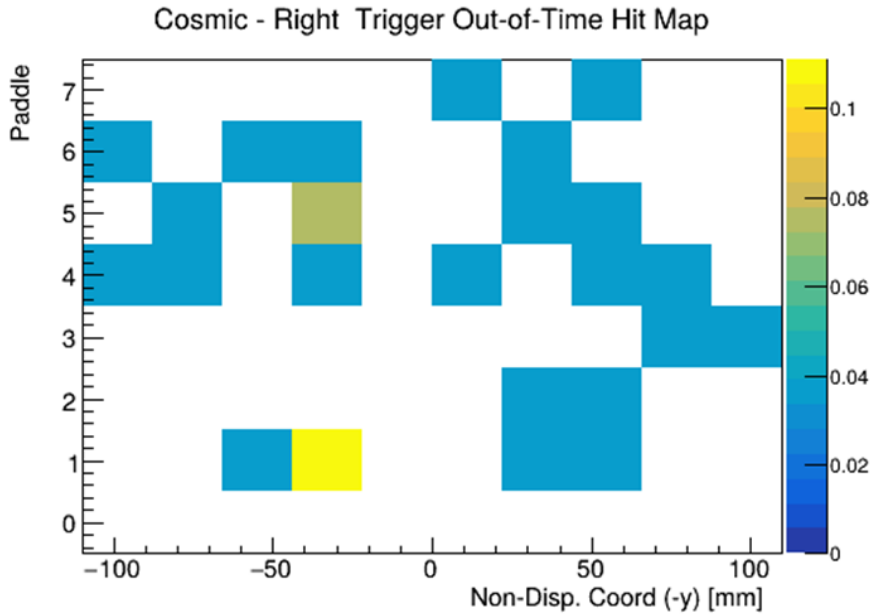




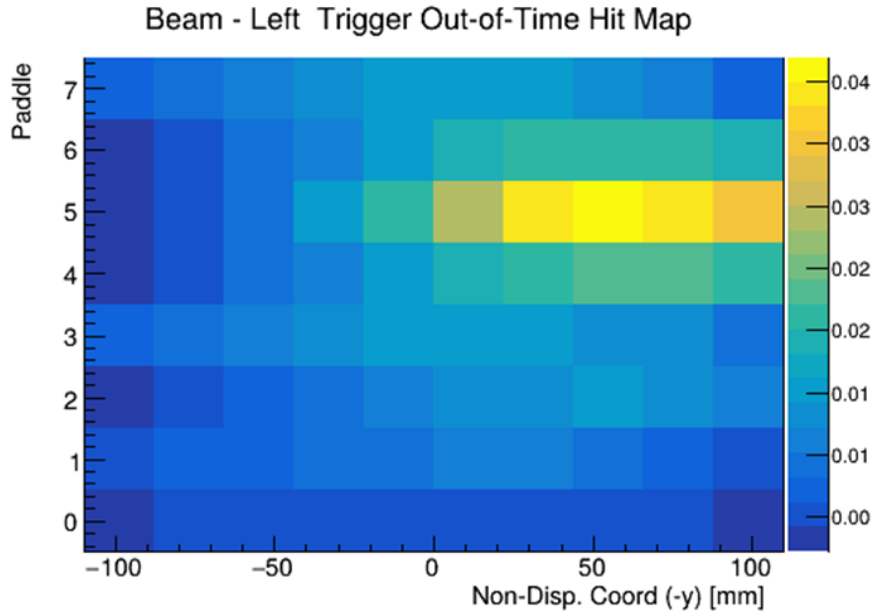
Out of Time Hits

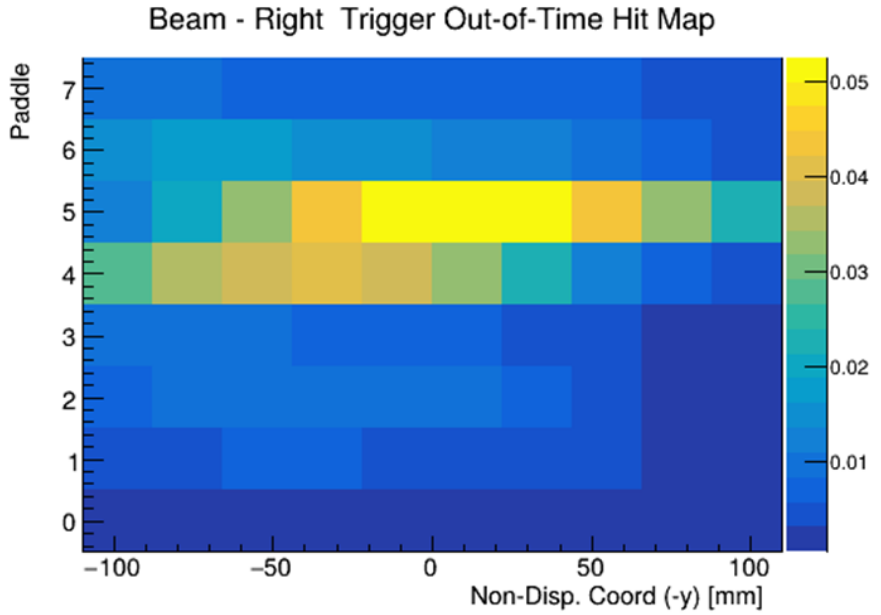
Cosmic





Beam

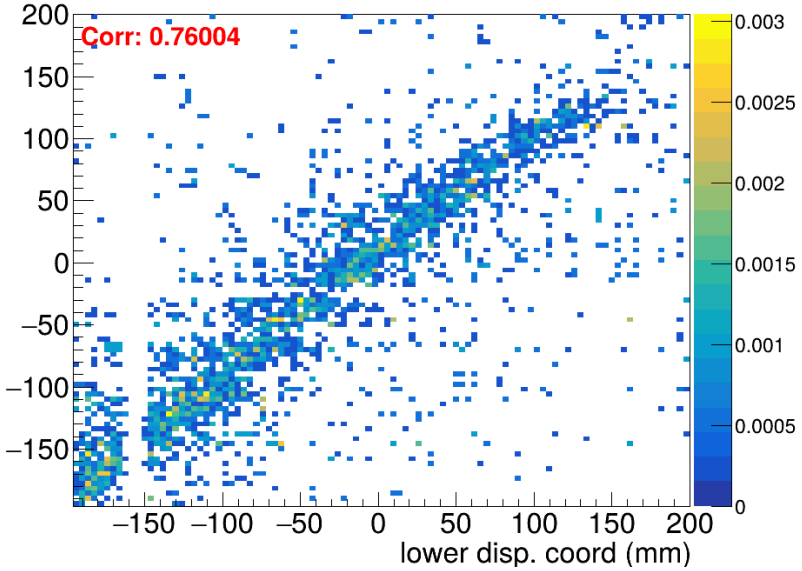




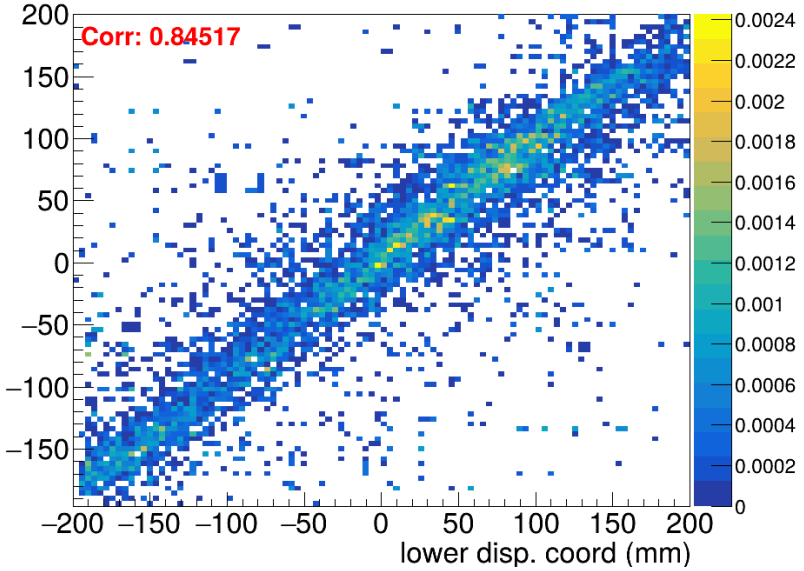
Top vs Bottom GEM Hit Position Correlations

	Avg. Top vs Bottom GEM Cluster Position Correlation
Cosmic	0.80
Beam	0.24

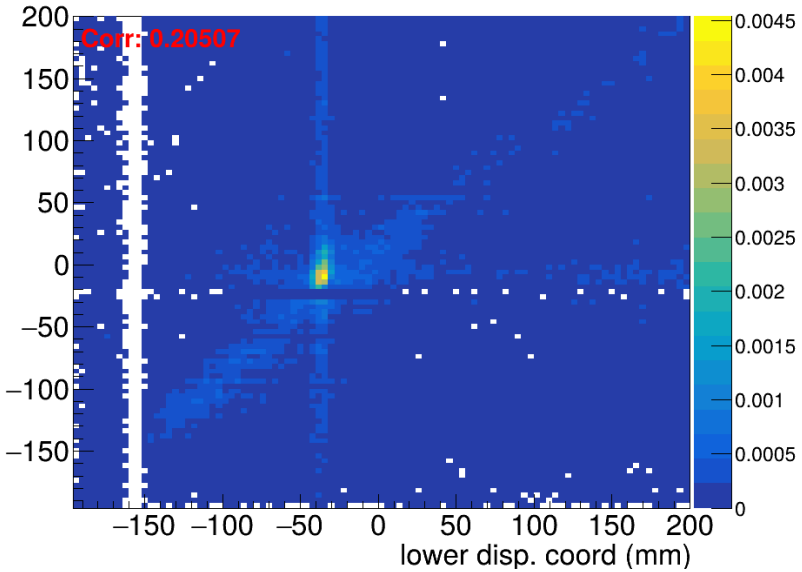
Cosmic Left: Top GEM X vs Bottom GEM X



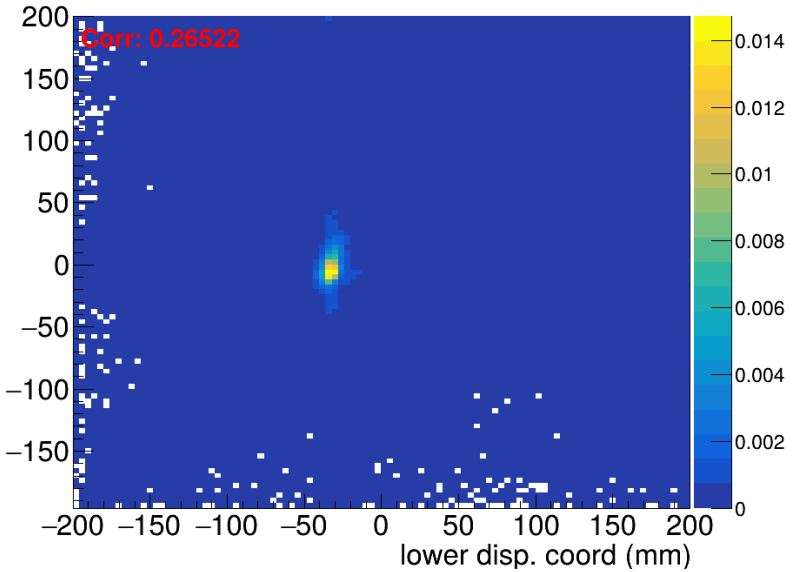
Cosmic Right: Top GEM X vs Bottom GEM X



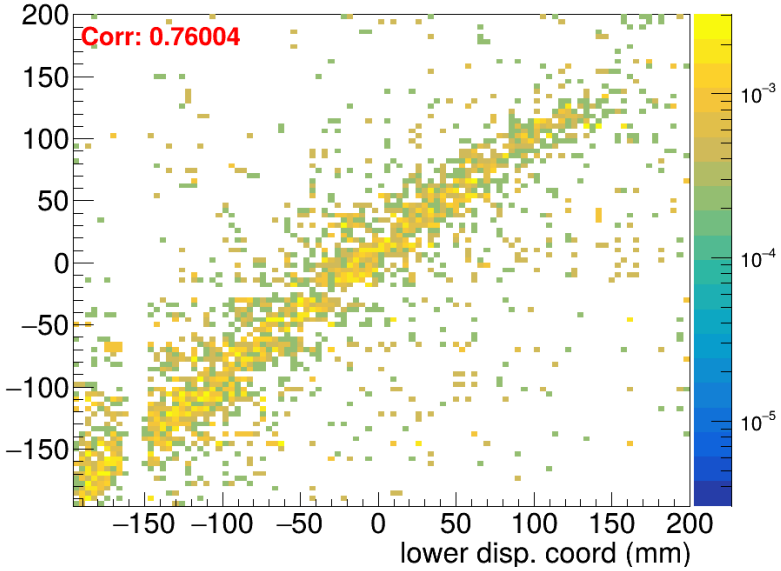
Beam Left: Top GEM X vs Bottom GEM X



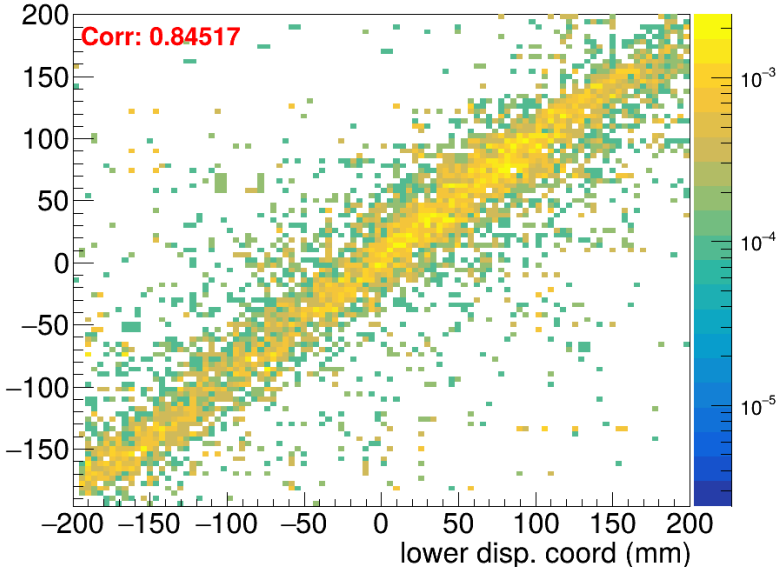
Beam Right: Top GEM X vs Bottom GEM X



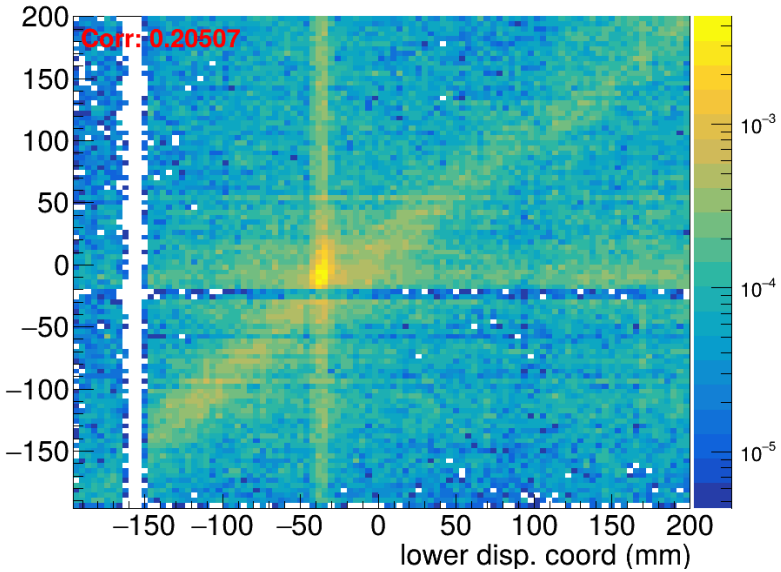
Cosmic Left: Top GEM X vs Bottom GEM X



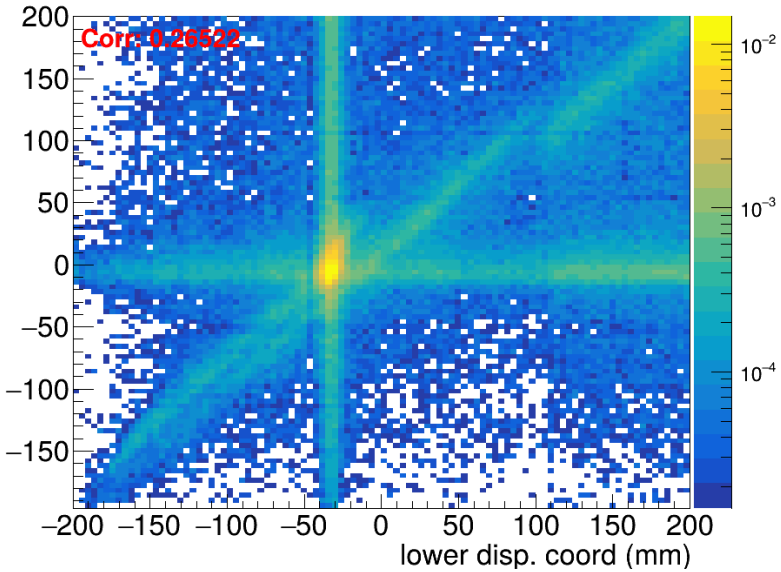
Cosmic Right: Top GEM X vs Bottom GEM X



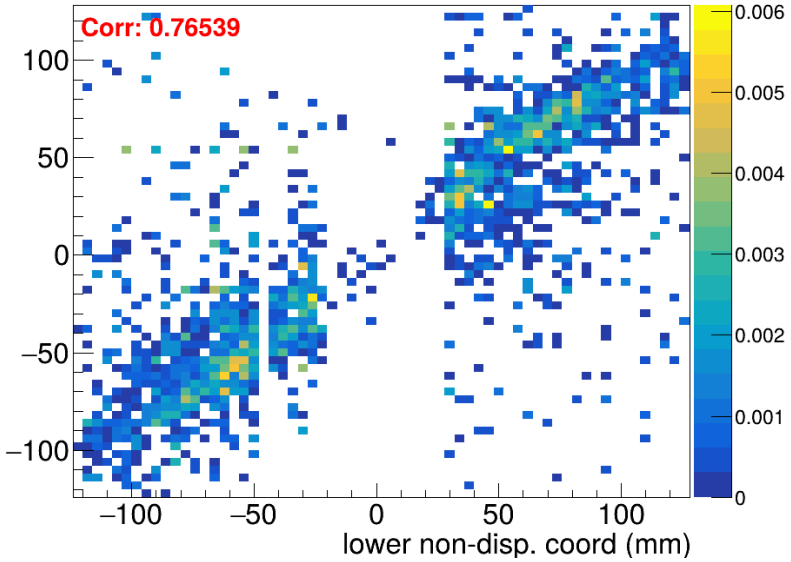
Beam Left: Top GEM X vs Bottom GEM X



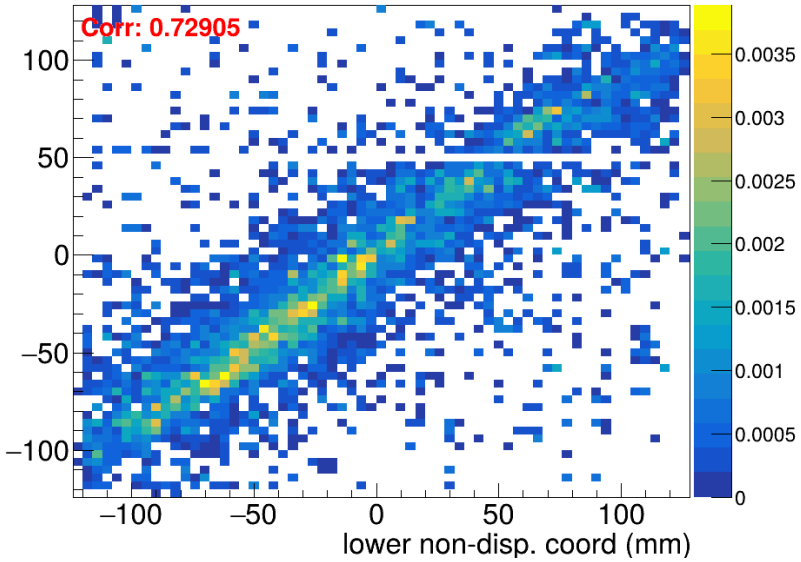
Beam Right: Top GEM X vs Bottom GEM X



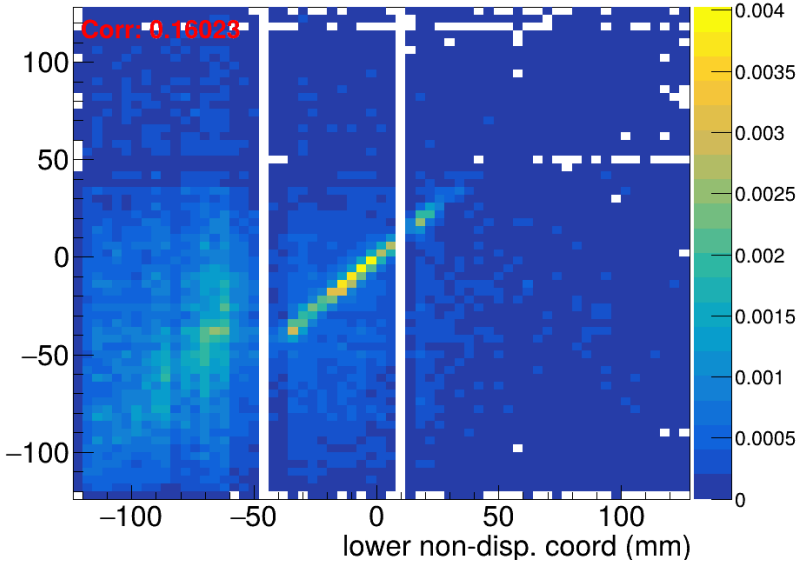
Cosmic Left: Top GEM Y vs Bottom GEM Y



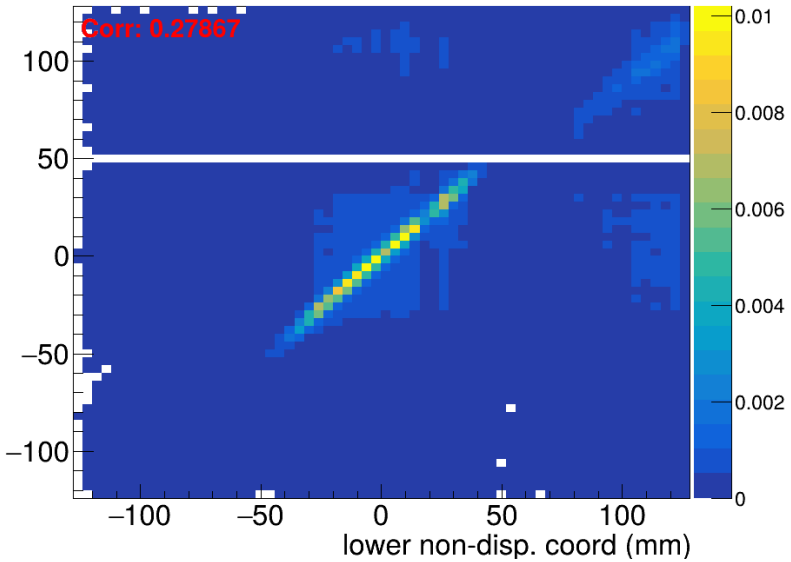
Cosmic Right: Top GEM Y vs Bottom GEM Y



Beam Left: Top GEM Y vs Bottom GEM Y



Beam Right: Top GEM Y vs Bottom GEM Y



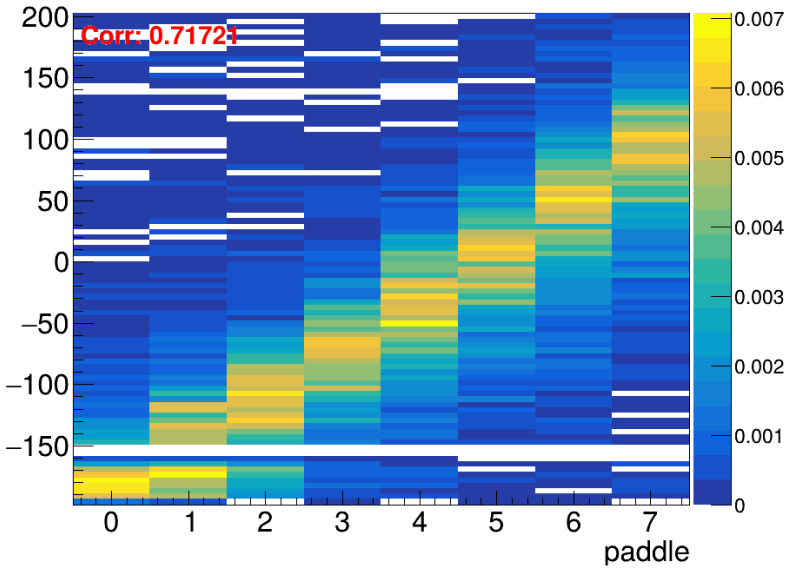
GEM Cluster vs Trigger Hit Position Correlations

- Average GEM cluster vs trigger hit position correlations
  - Average across all 4 GEMs

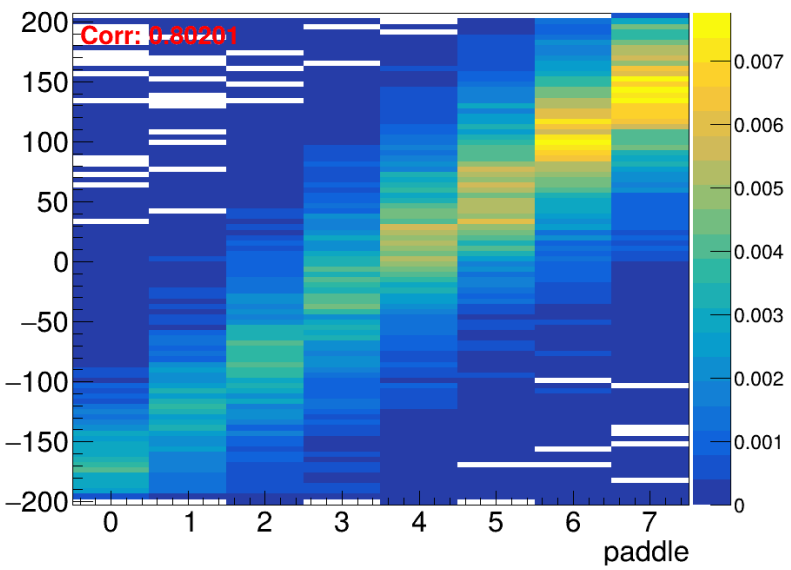
	In Time	Out of Time
Cosmic	0.73	0.14
Beam	0.22	0.02

Bottom GEM vs Trigger Hit X Correlations

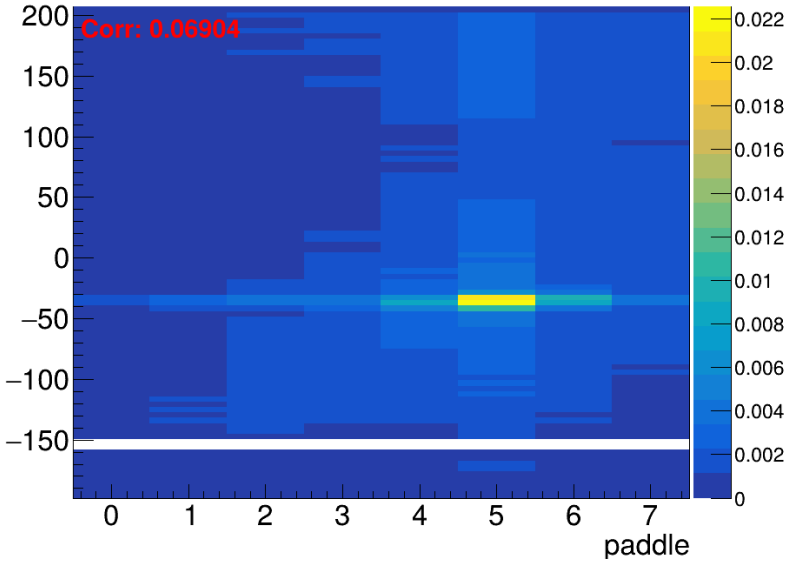
Cosmic Left: Bottom GEM X vs Paddle #



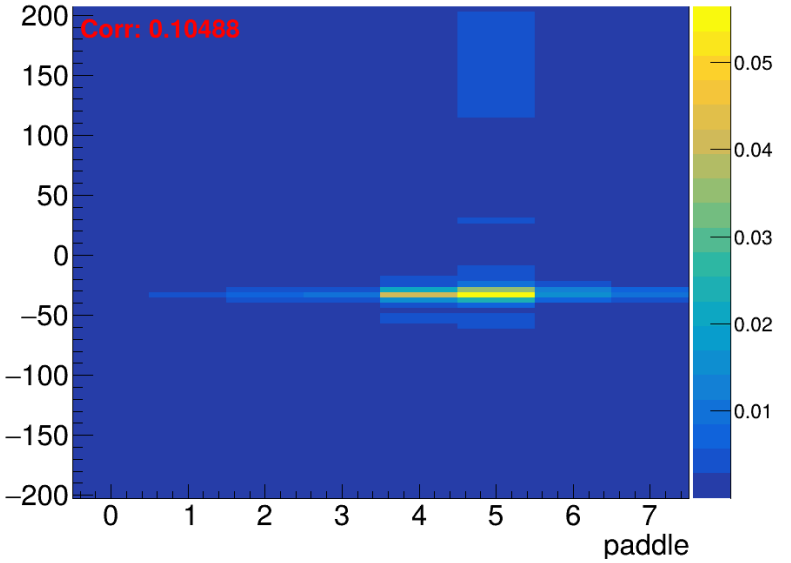
Cosmic Right: Bottom GEM X vs Paddle #



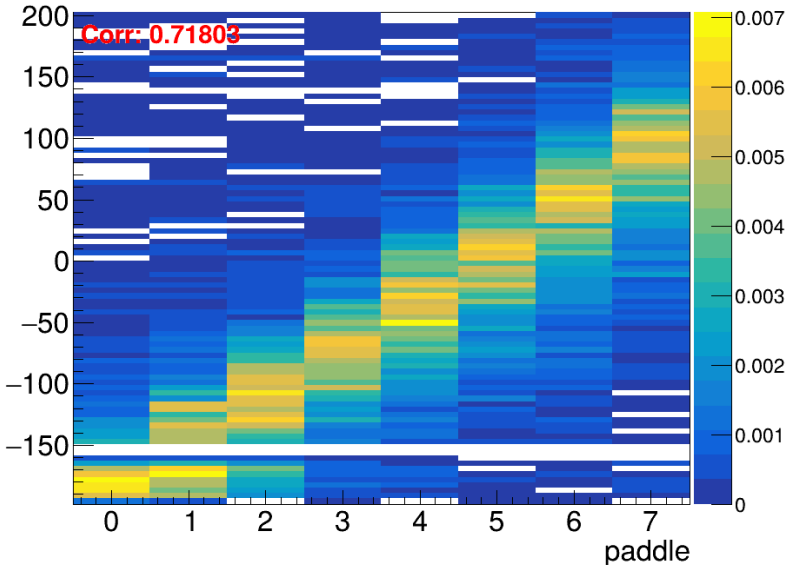
Beam Left: Bottom GEM X vs Paddle #



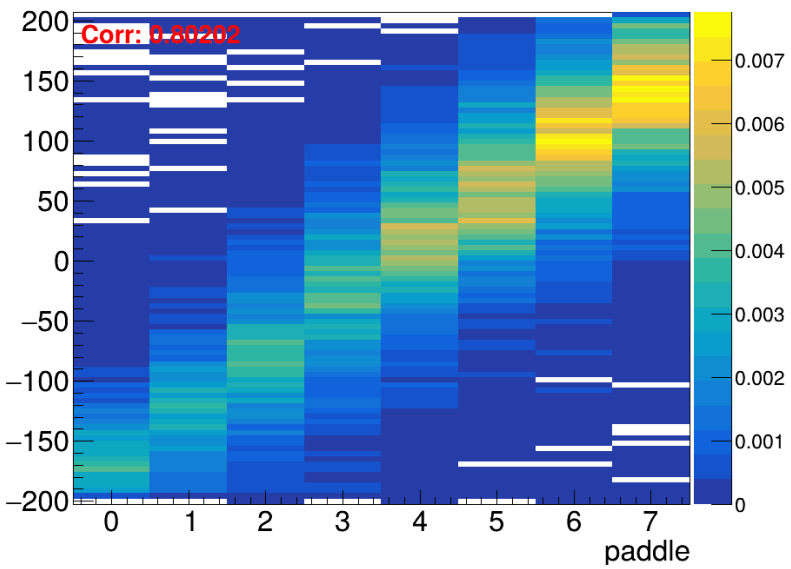
Beam Right: Bottom GEM X vs Paddle #



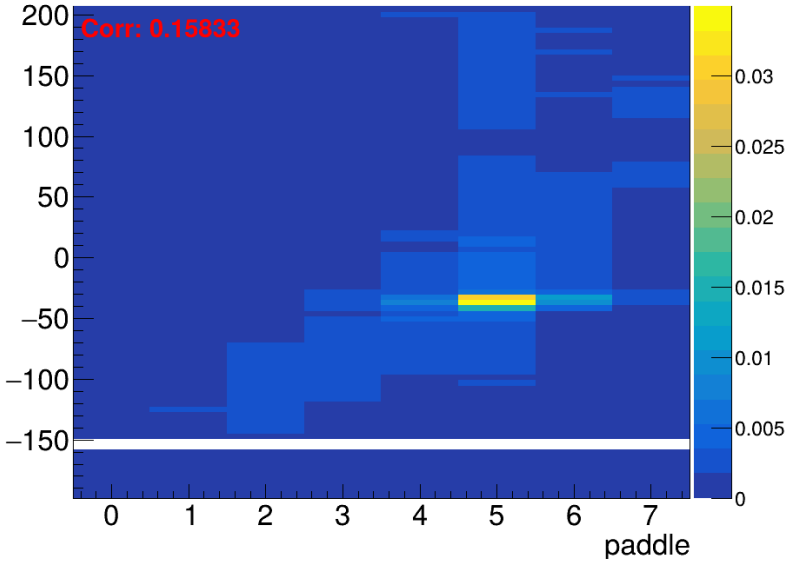
Cosmic Left: Bottom GEM X vs In-Time Paddle #



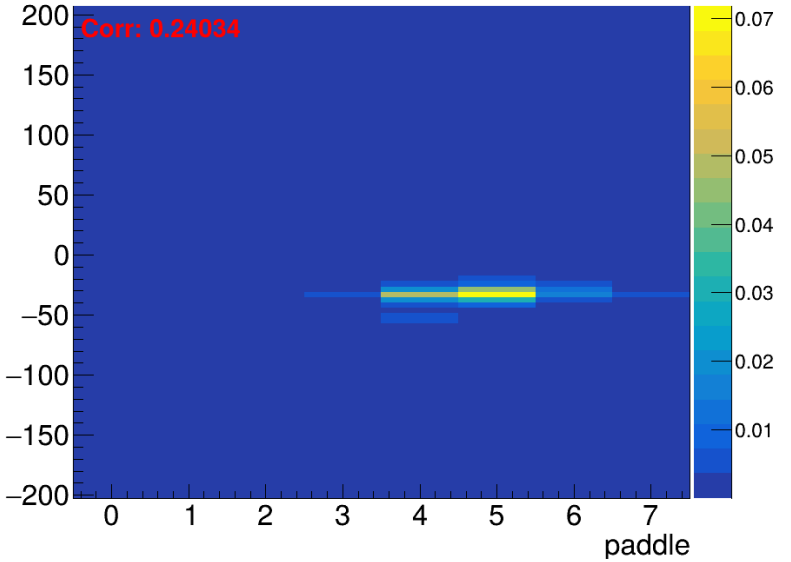
Cosmic Right: Bottom GEM X vs In-Time Paddle #



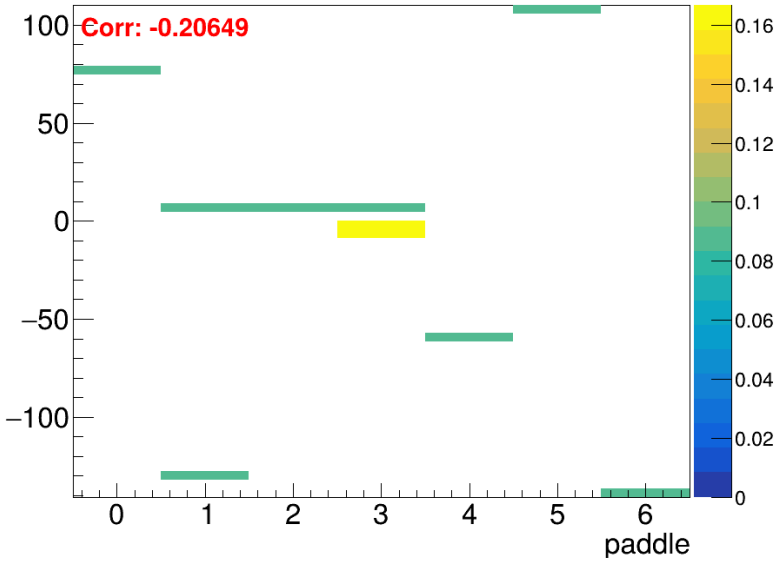
Beam Left: Bottom GEM X vs In-Time Paddle #



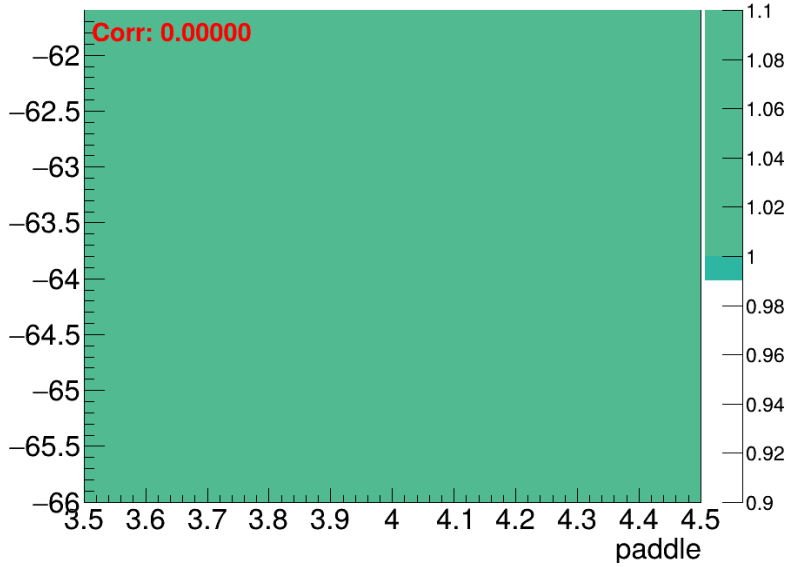
Beam Right: Bottom GEM X vs In-Time Paddle #



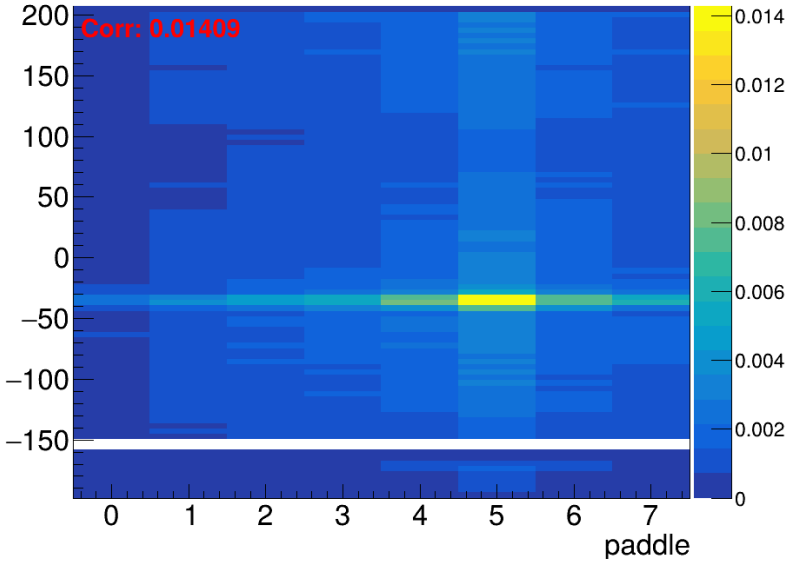
Cosmic Left: Bottom GEM X vs Out-of-Time Paddle #



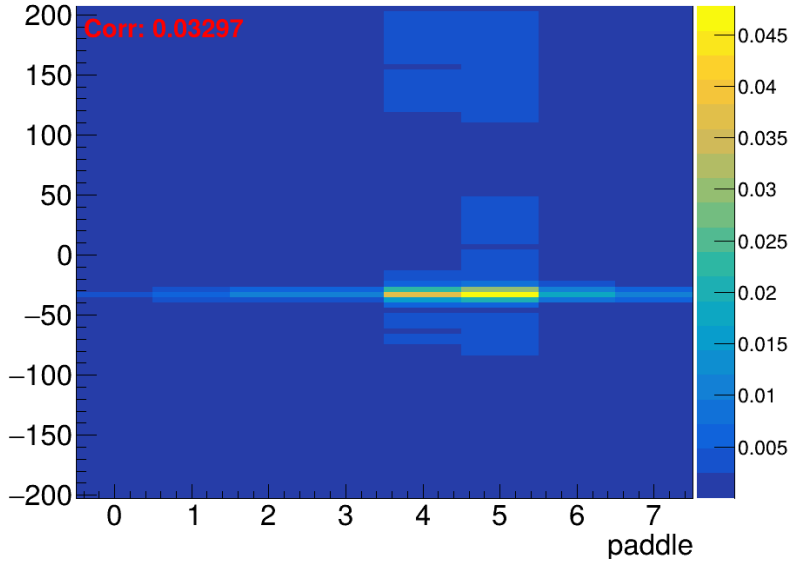
Cosmic Right: Bottom GEM X vs Out-of-Time Paddle #



Beam Left: Bottom GEM X vs Out-of-Time Paddle #

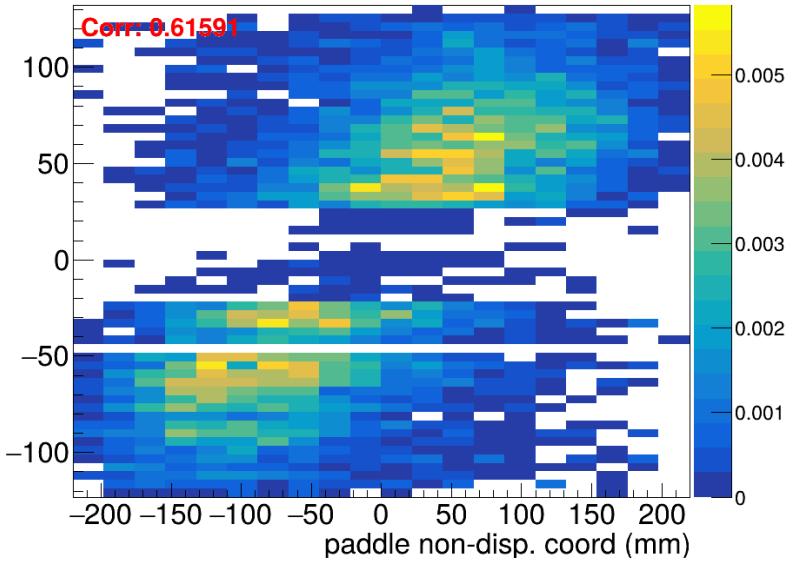


Beam Right: Bottom GEM X vs Out-of-Time Paddle #

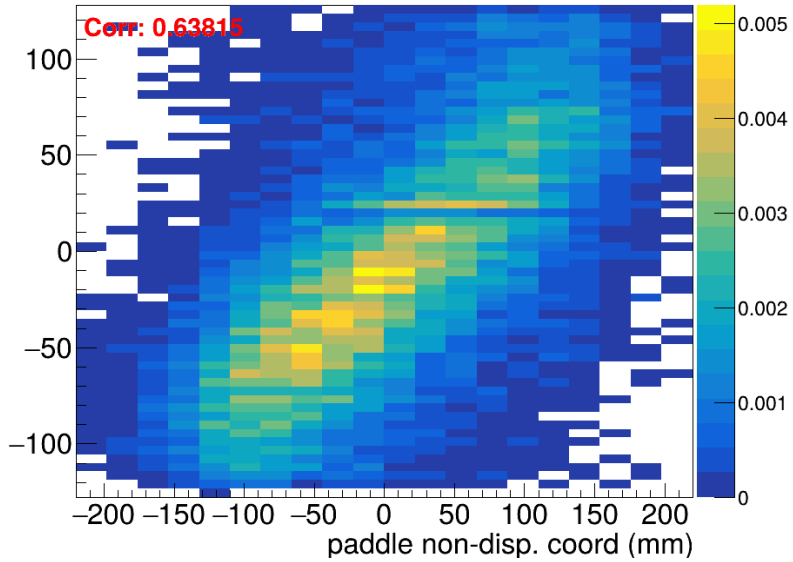


Bottom GEM vs Trigger Hit Y Correlations

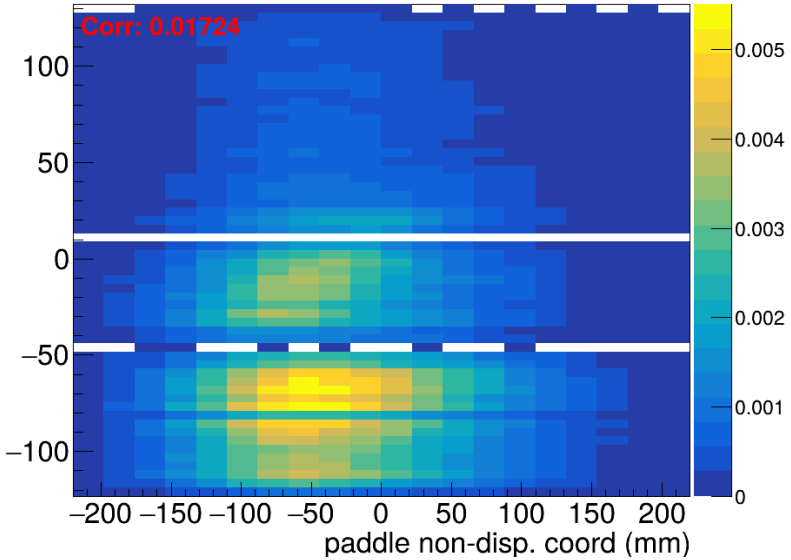
Cosmic Left: Bottom GEM Y vs Trigger Y



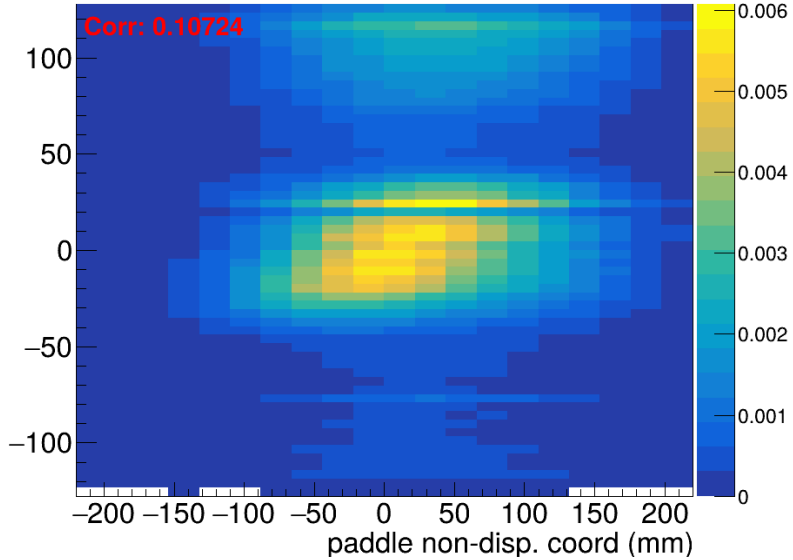
Cosmic Right: Bottom GEM Y vs Trigger Y



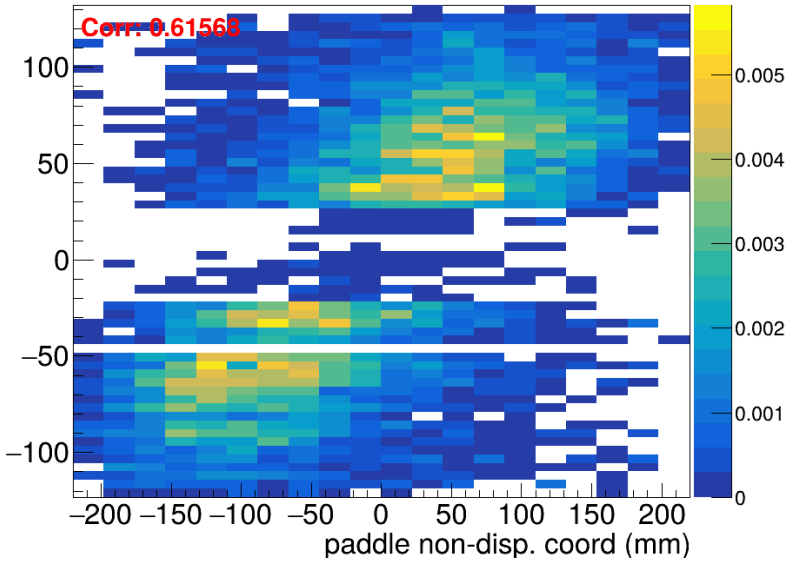
Beam Left: Bottom GEM Y vs Trigger Y



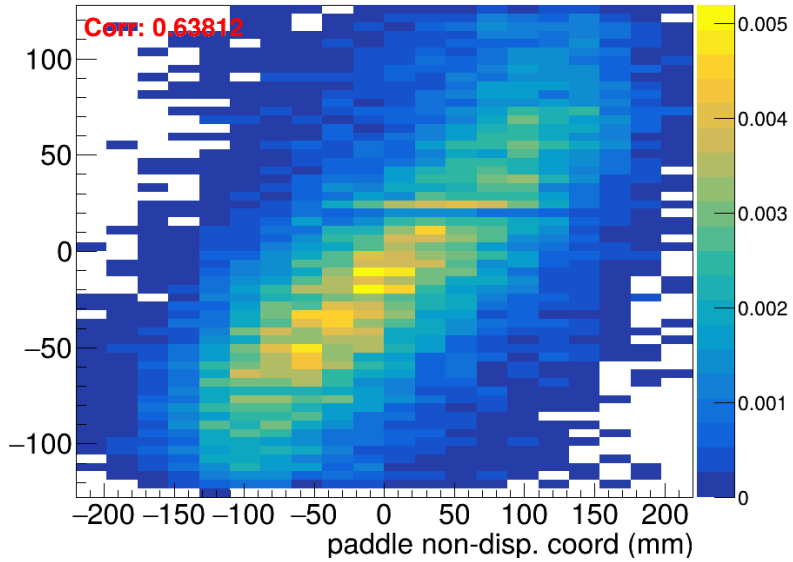
Beam Right: Bottom GEM Y vs Trigger Y



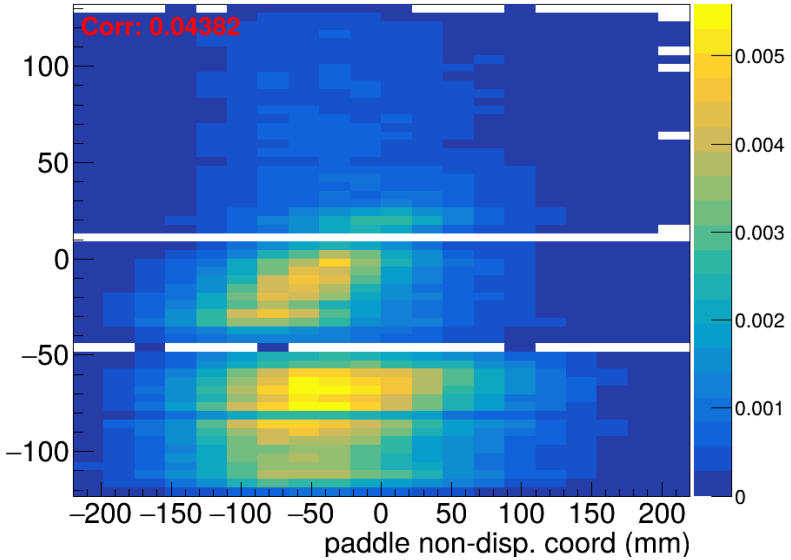
Cosmic Left: Bottom GEM Y vs In-Time Trigger Y



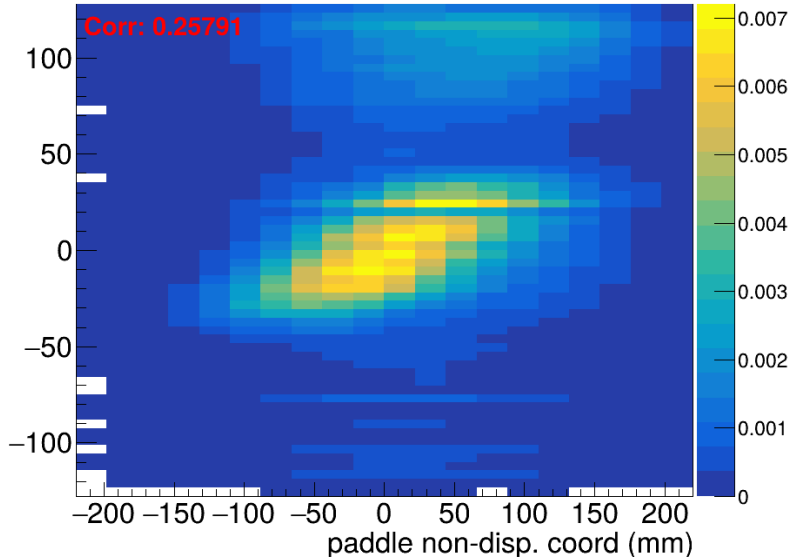
Cosmic Right: Bottom GEM Y vs In-Time Trigger Y



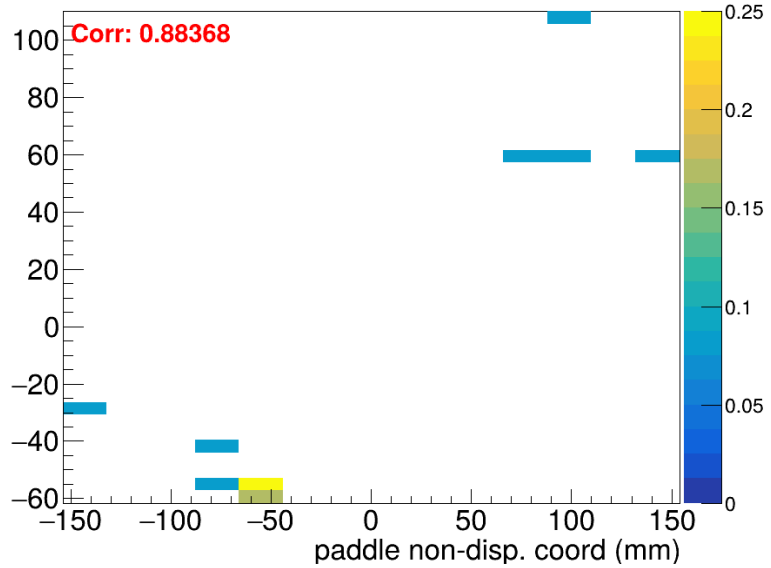
Beam Left: Bottom GEM Y vs In-Time Trigger Y



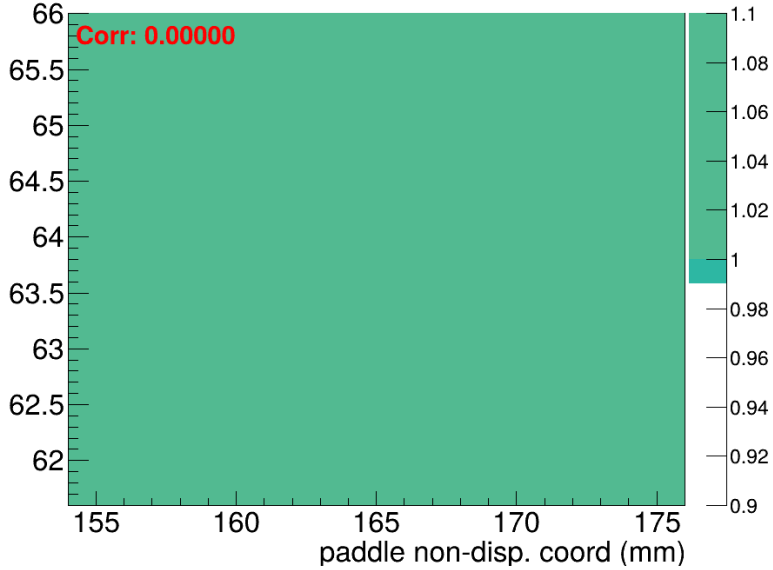
Beam Right: Bottom GEM Y vs In-Time Trigger Y



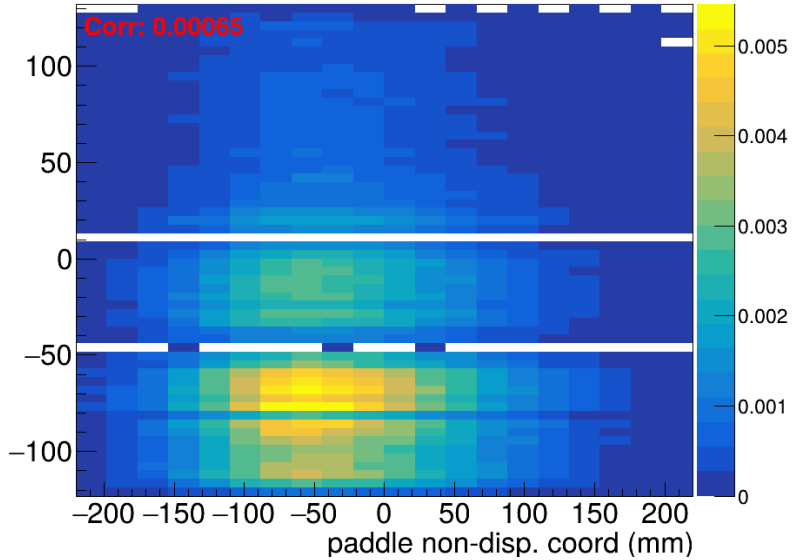
Cosmic Left: Bottom GEM Y vs Out-of-Time Trigger Y



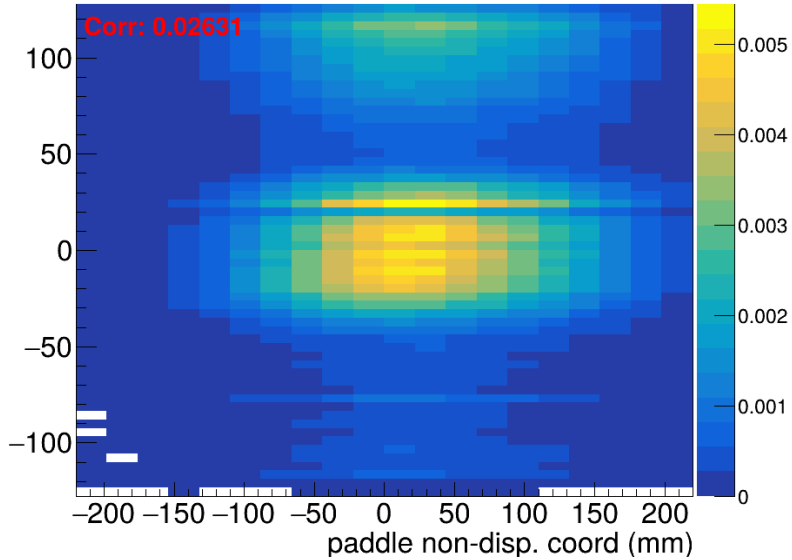
Cosmic Right: Bottom GEM Y vs Out-of-Time Trigger Y



Beam Left: Bottom GEM Y vs Out-of-Time Trigger Y

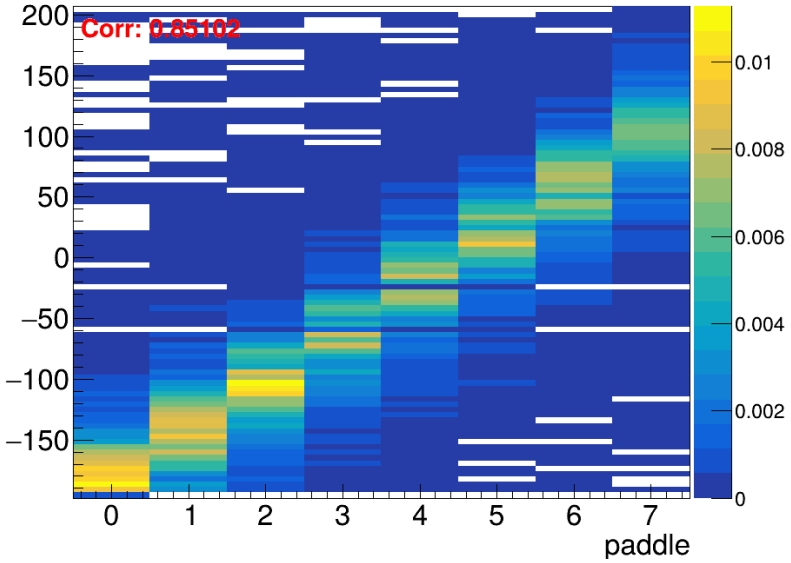


Beam Right: Bottom GEM Y vs Out-of-Time Trigger Y

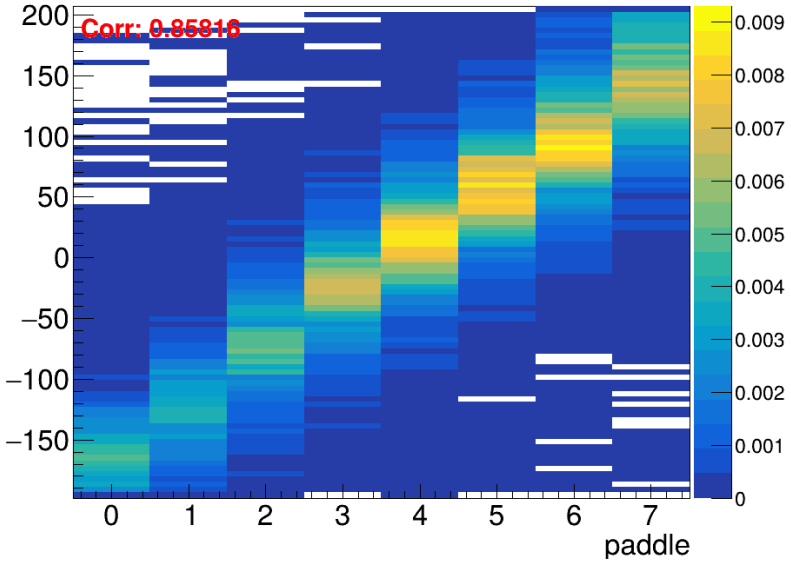


Top GEM vs Trigger Hit X Correlations

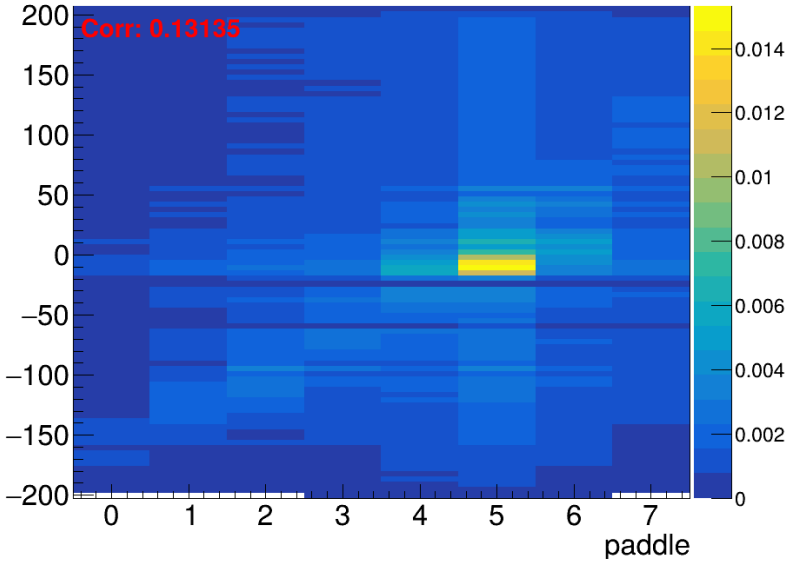
Cosmic Left: Top GEM X vs Paddle #



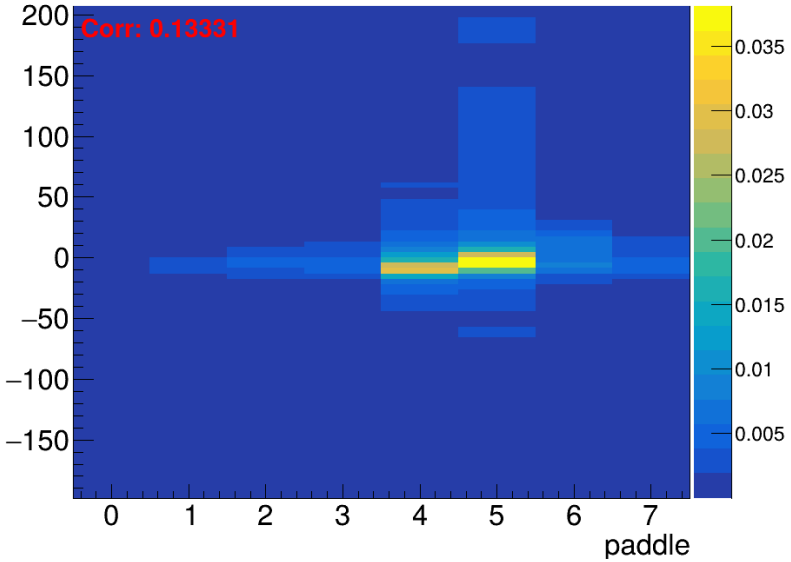
Cosmic Right: Top GEM X vs Paddle #



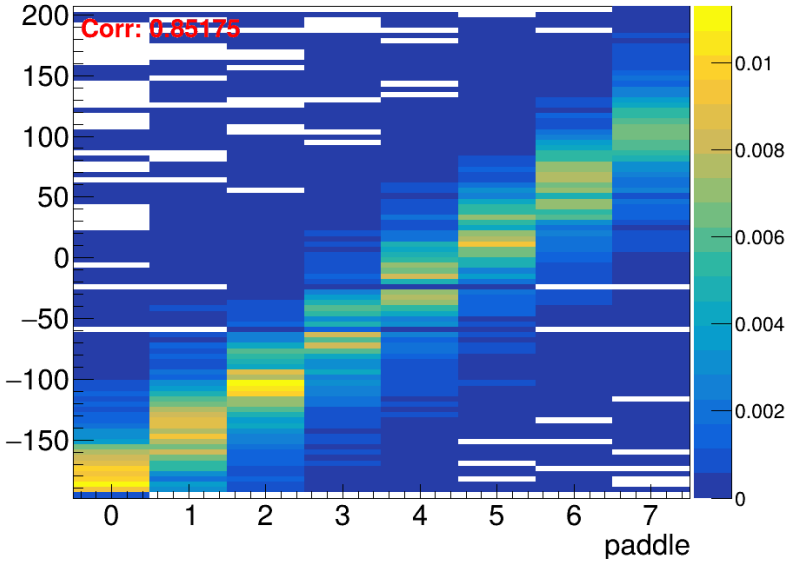
Beam Left: Top GEM X vs Paddle #



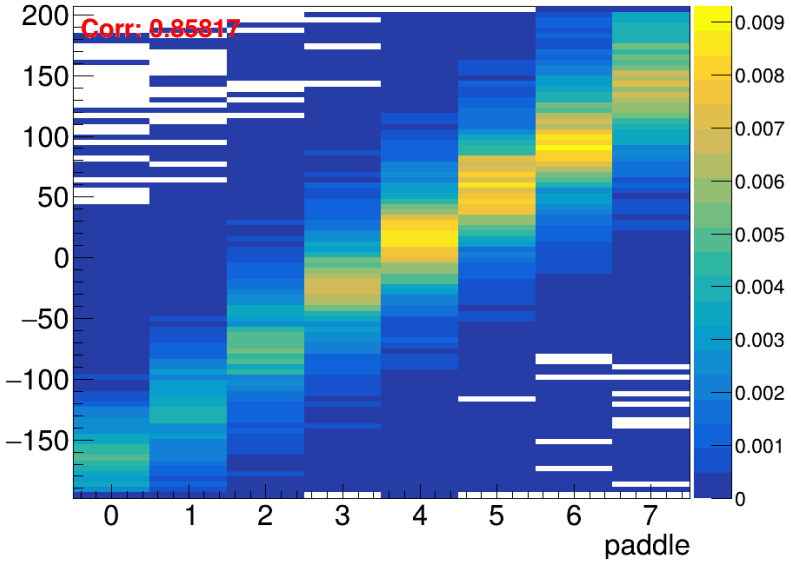
Beam Right: Top GEM X vs Paddle #



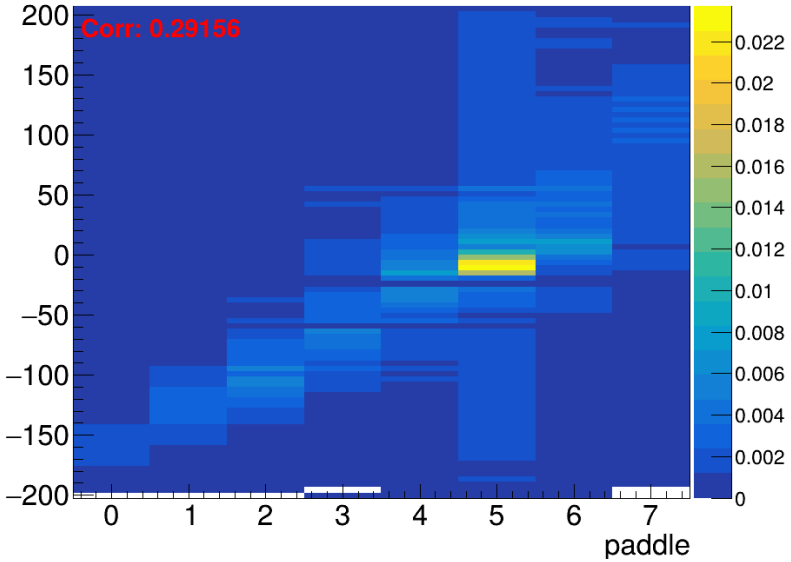
Cosmic Left: Top GEM X vs In-Time Paddle #



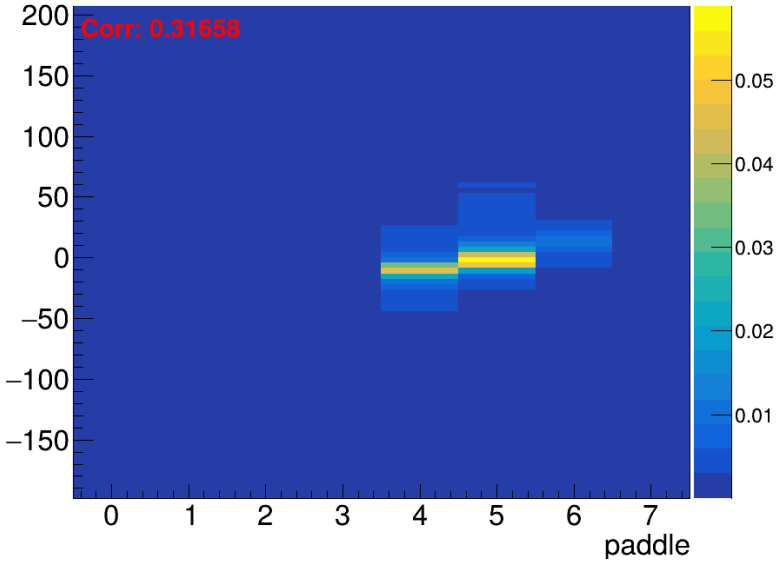
Cosmic Right: Top GEM X vs In-Time Paddle #



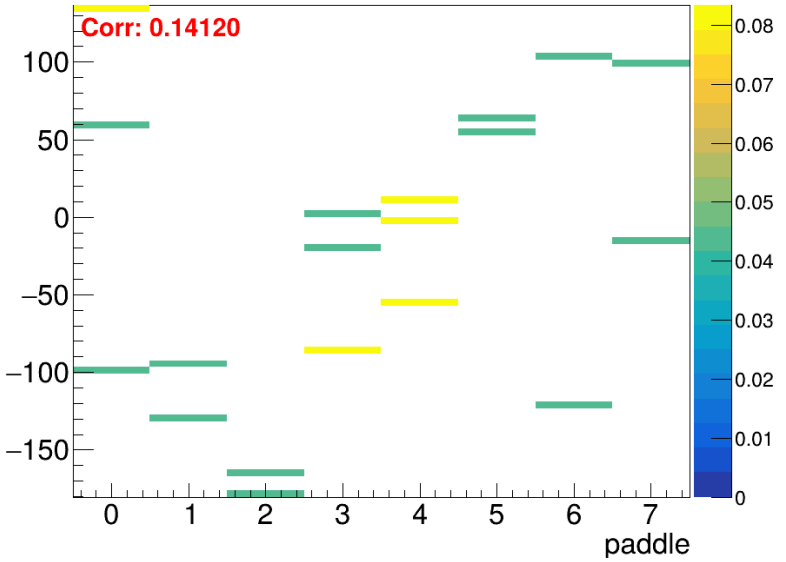
Beam Left: Top GEM X vs In-Time Paddle #



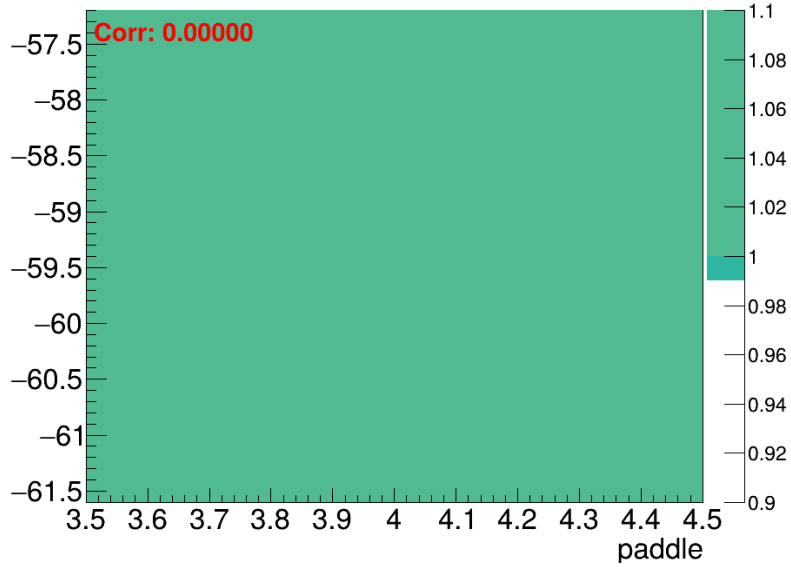
Beam Right: Top GEM X vs In-Time Paddle #



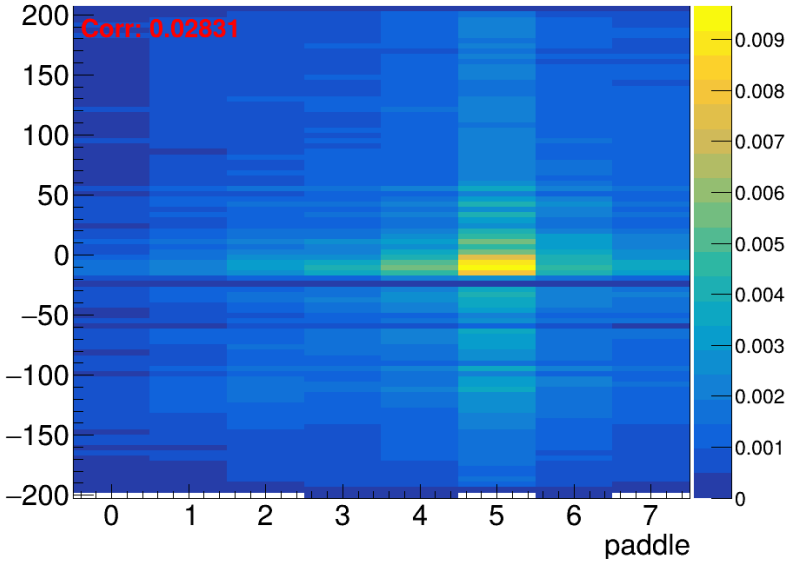
Cosmic Left: Top GEM X vs Out-of-Time Paddle #



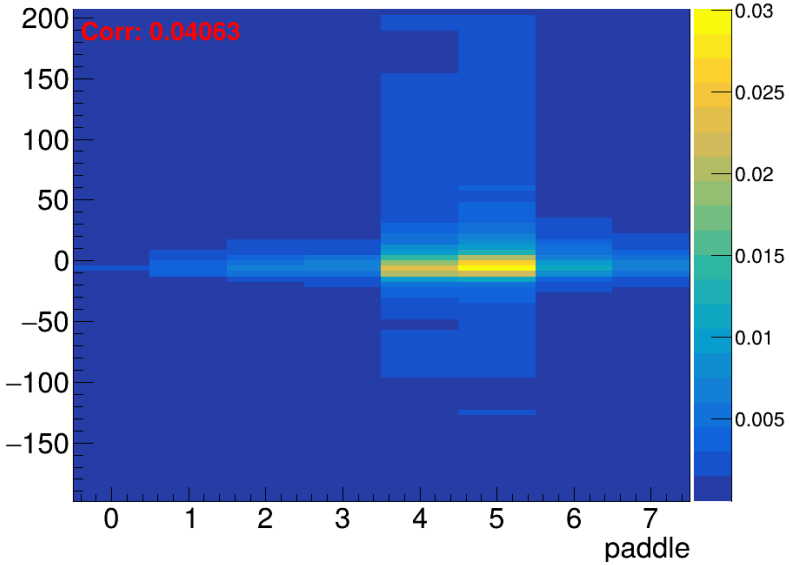
Cosmic Right: Top GEM X vs Out-of-Time Paddle #



Beam Left: Top GEM X vs Out-of-Time Paddle #

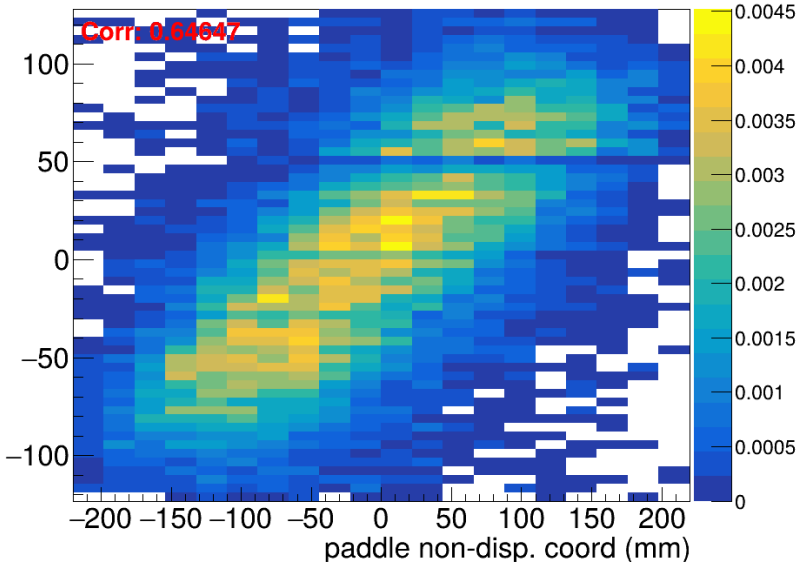


Beam Right: Top GEM X vs Out-of-Time Paddle #

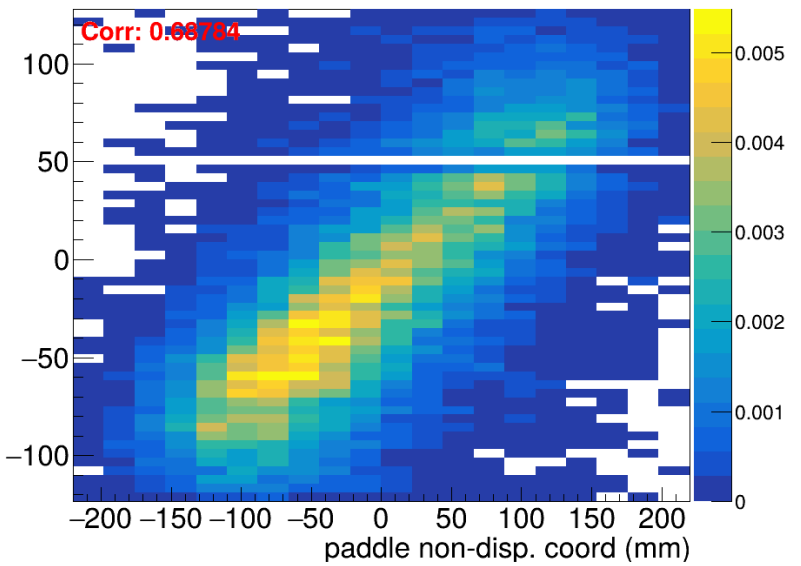


Top GEM vs Trigger Hit Y Correlations

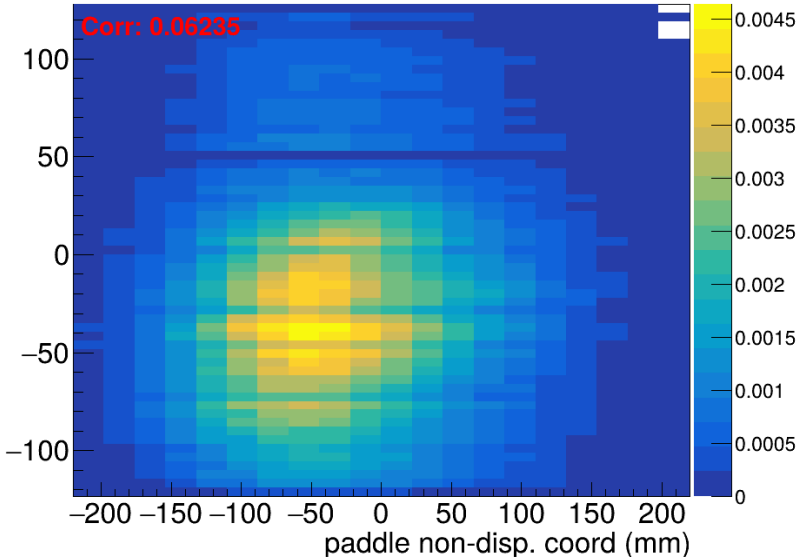
Cosmic Left: Top GEM Y vs Trigger Y



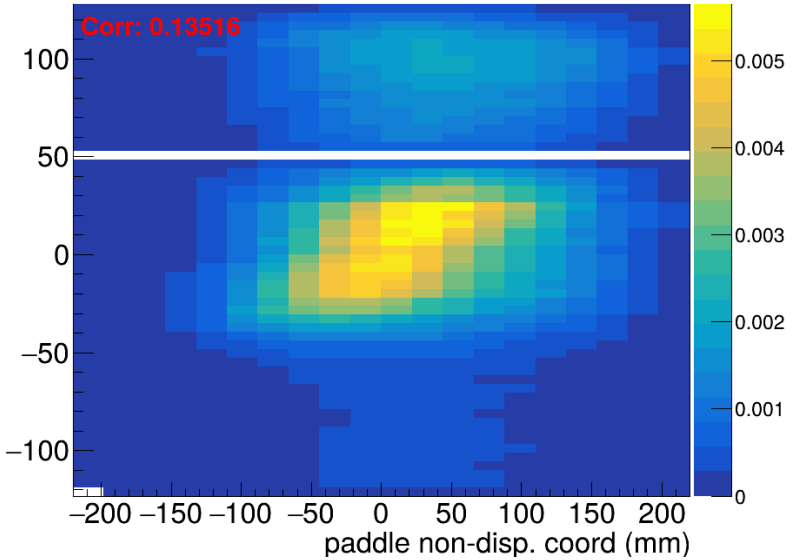
Cosmic Right: Top GEM Y vs Trigger Y



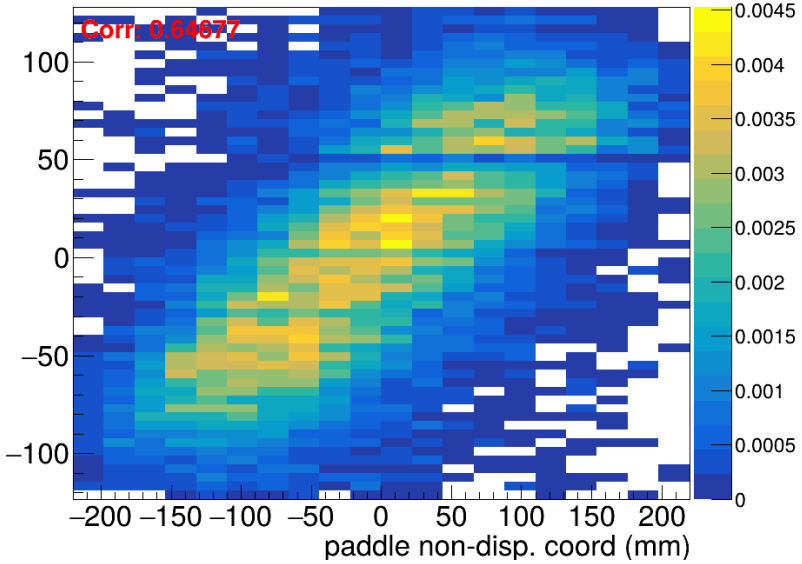
Beam Left: Top GEM Y vs Trigger Y



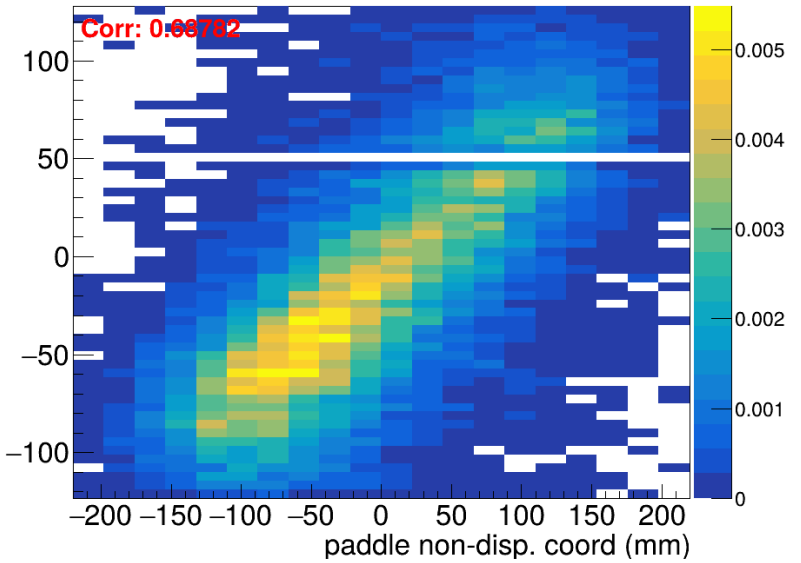
Beam Right: Top GEM Y vs Trigger Y



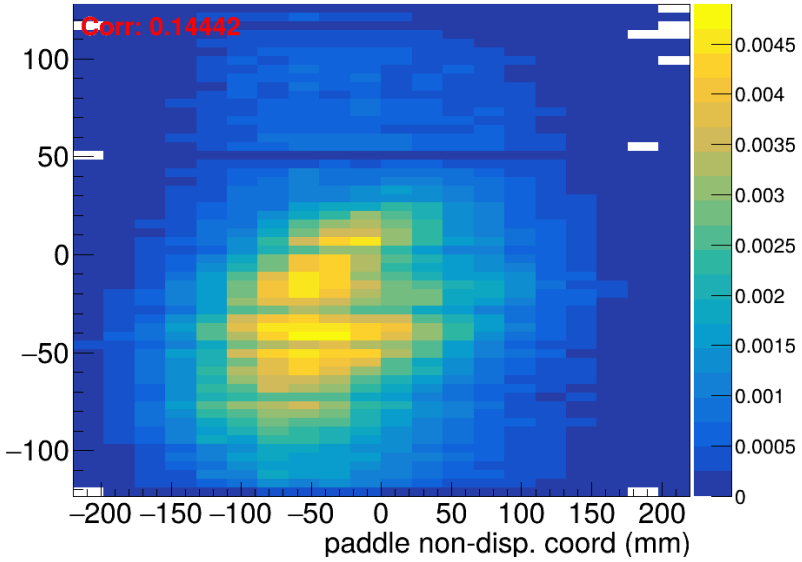
Cosmic Left: Top GEM Y vs In-Time Trigger Y



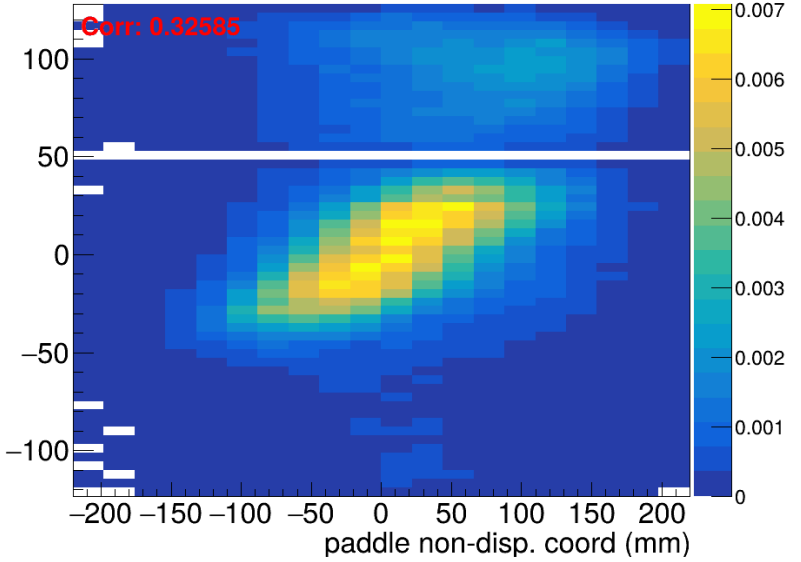
Cosmic Right: Top GEM Y vs In-Time Trigger Y



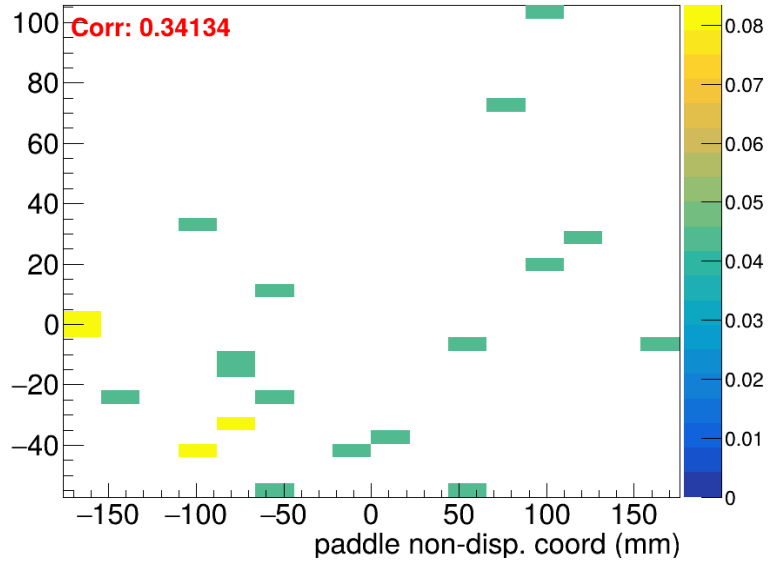
Beam Left: Top GEM Y vs In-Time Trigger Y



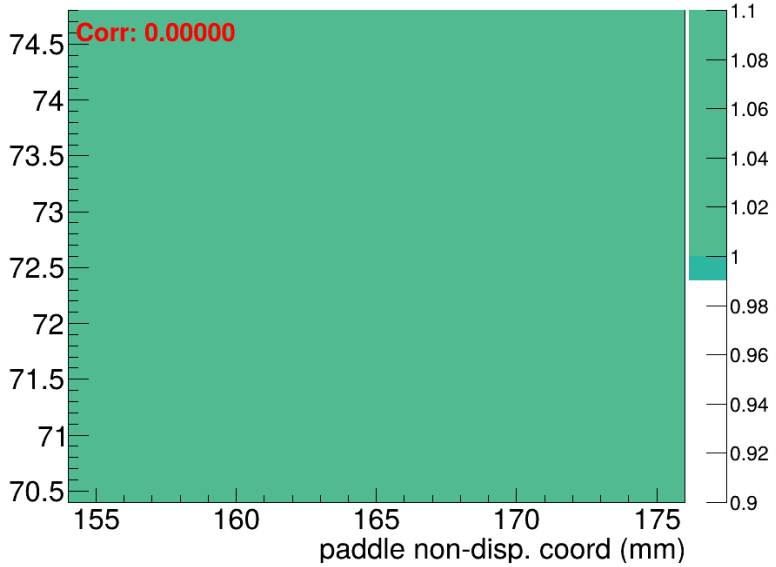
Beam Right: Top GEM Y vs In-Time Trigger Y



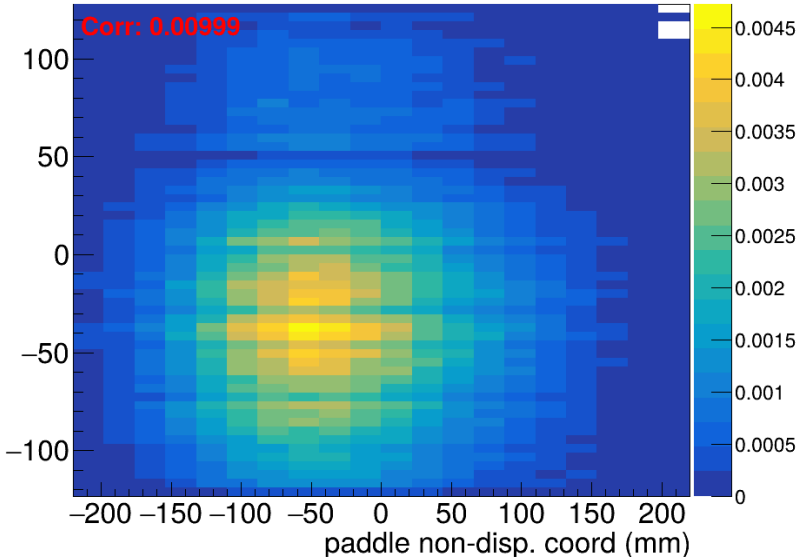
Cosmic Left: Top GEM Y vs Out-of-Time Trigger Y



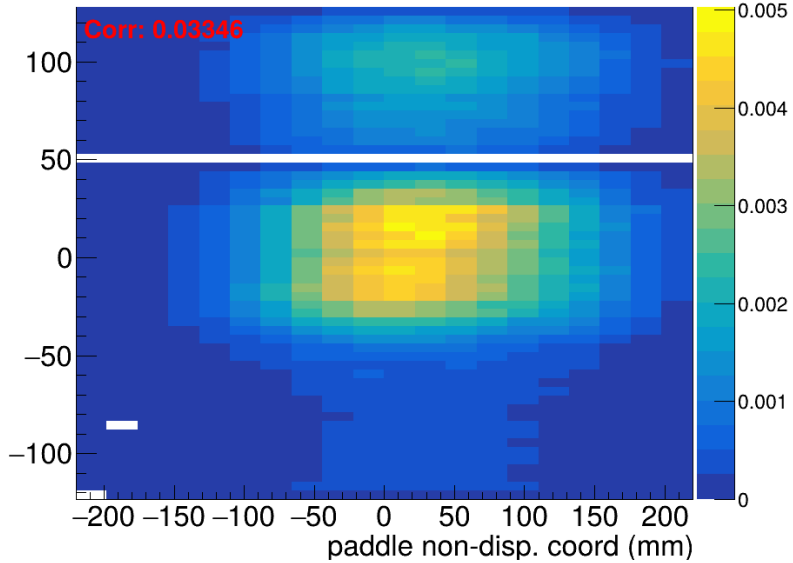
Cosmic Right: Top GEM Y vs Out-of-Time Trigger Y



Beam Left: Top GEM Y vs Out-of-Time Trigger Y



Beam Right: Top GEM Y vs Out-of-Time Trigger Y



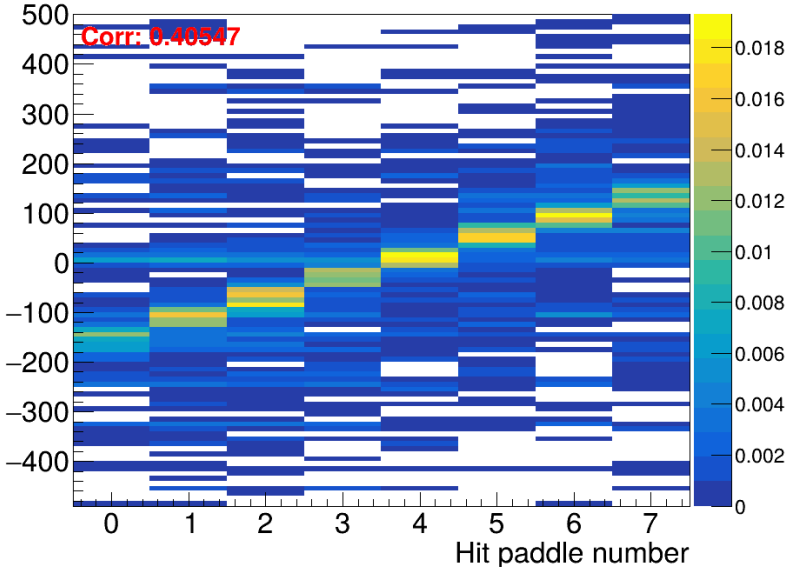
Projected vs Actual Trigger Hit Position Correlations

- Average GEM projection vs trigger hit position correlations
  - Average over left and right

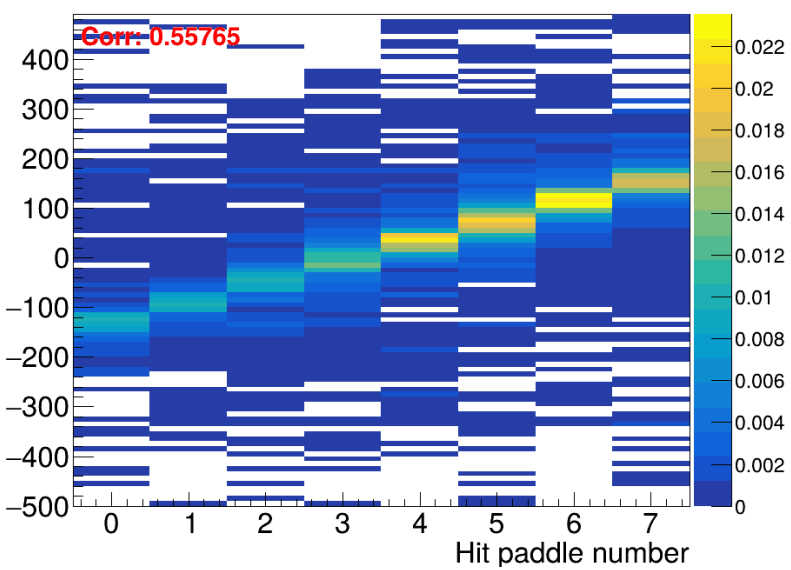
	In Time	Out of Time
Cosmic	0.39	-0.16
Beam	0.13	0.02

Projected vs Actual Trigger Hit X Correlations

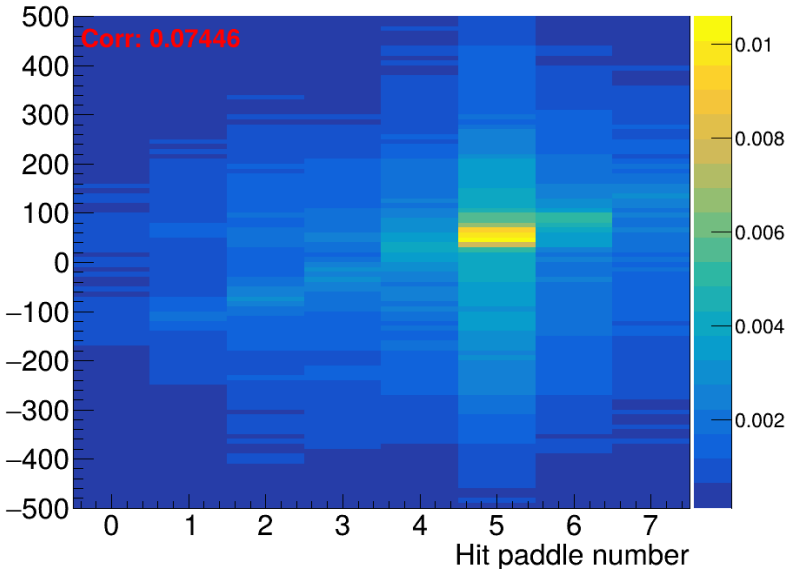
Cosmic Left: Track Proj. X vs Hit Paddle #



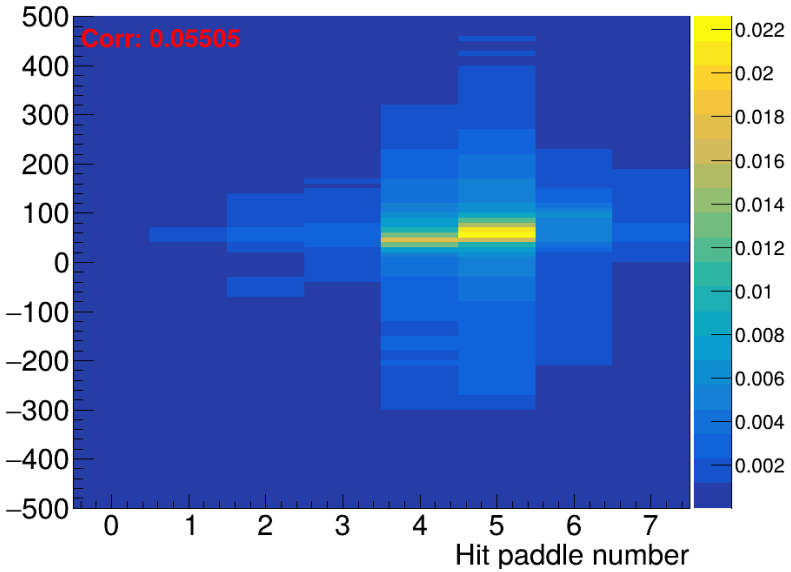
Cosmic Right: Track Proj. X vs Hit Paddle #



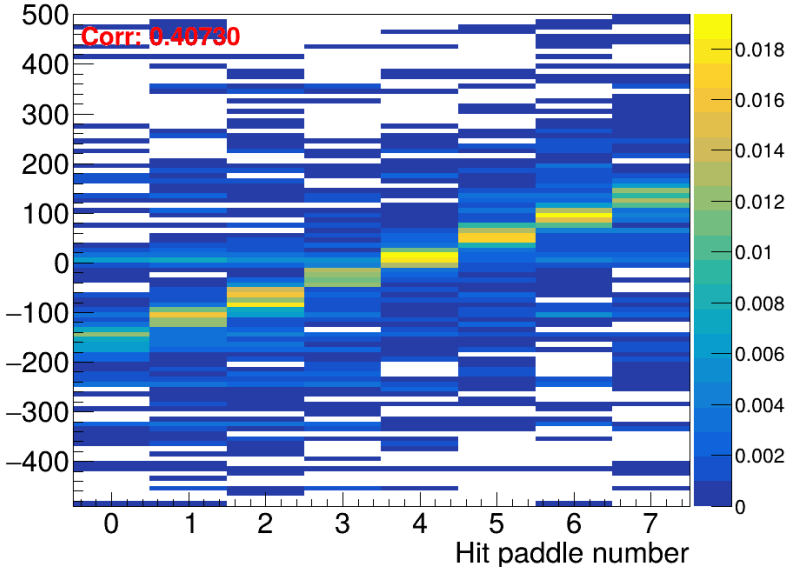
Beam Left: Track Proj. X vs Hit Paddle #



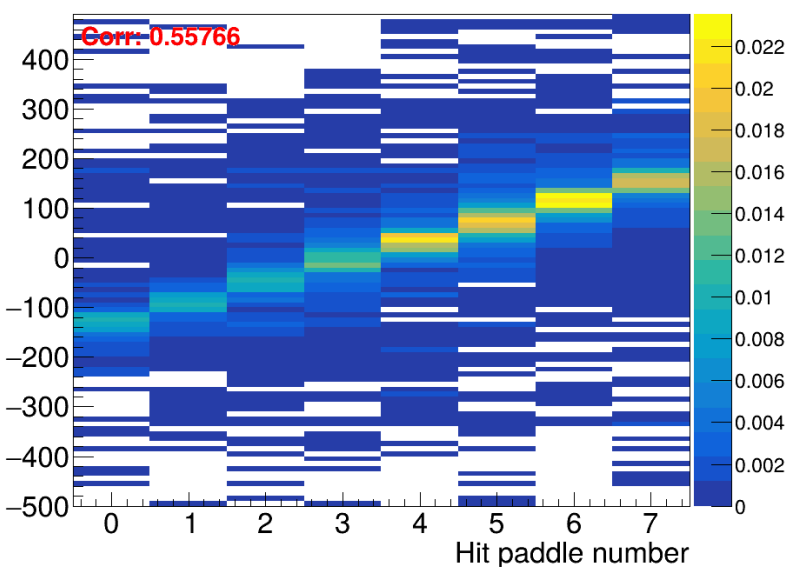
Beam Right: Track Proj. X vs Hit Paddle #



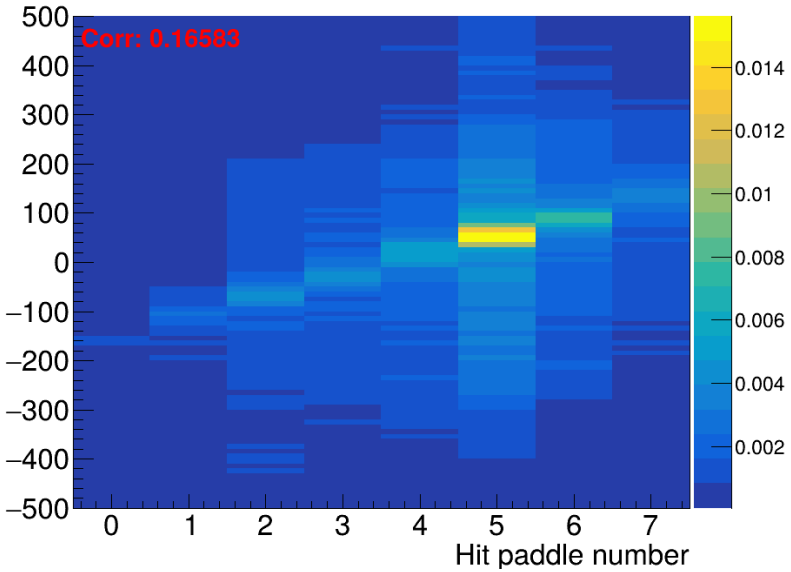
Cosmic Left: Track Proj. X vs In-Time Hit Paddle #



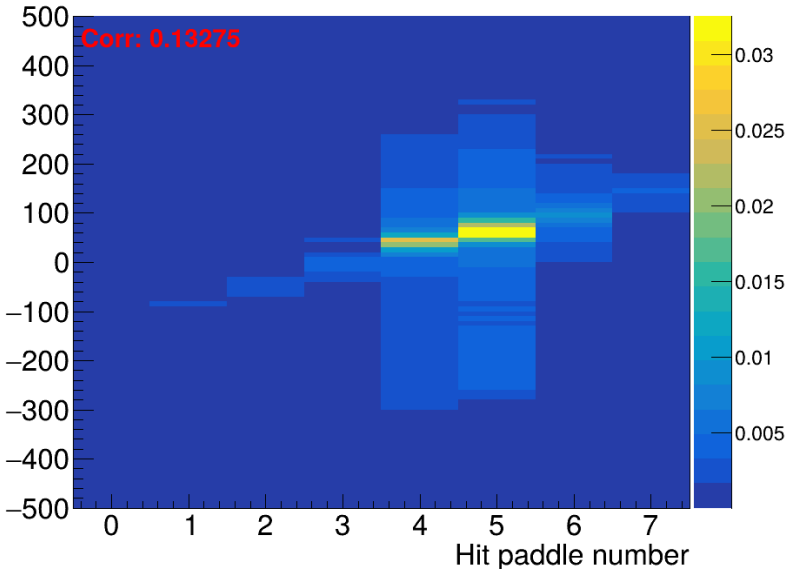
Cosmic Right: Track Proj. X vs In-Time Hit Paddle #



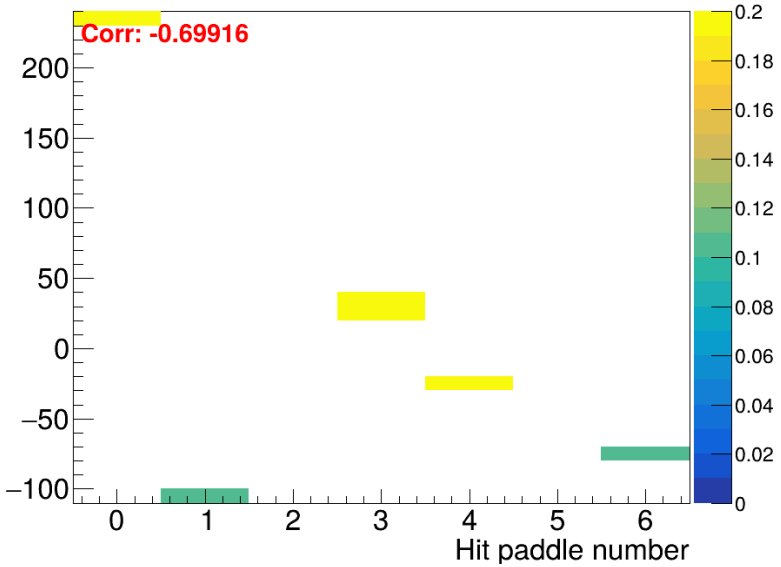
Beam Left: Track Proj. X vs In-Time Hit Paddle #



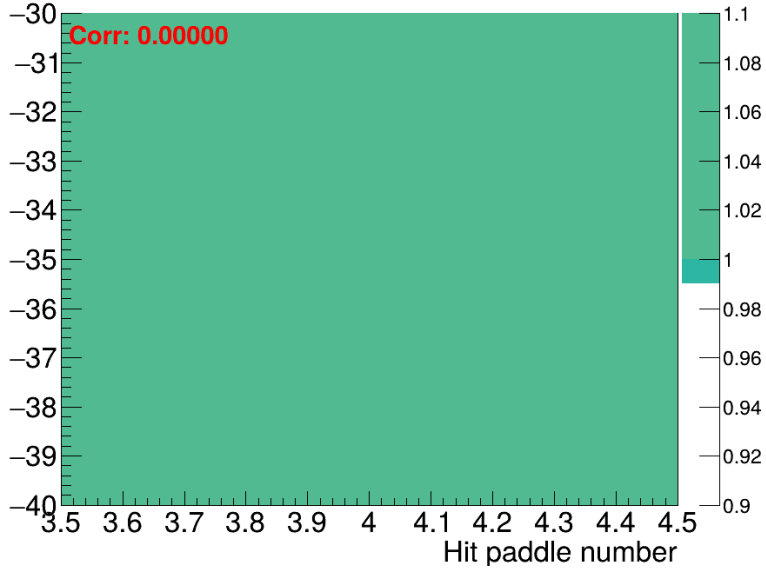
Beam Right: Track Proj. X vs In-Time Hit Paddle #



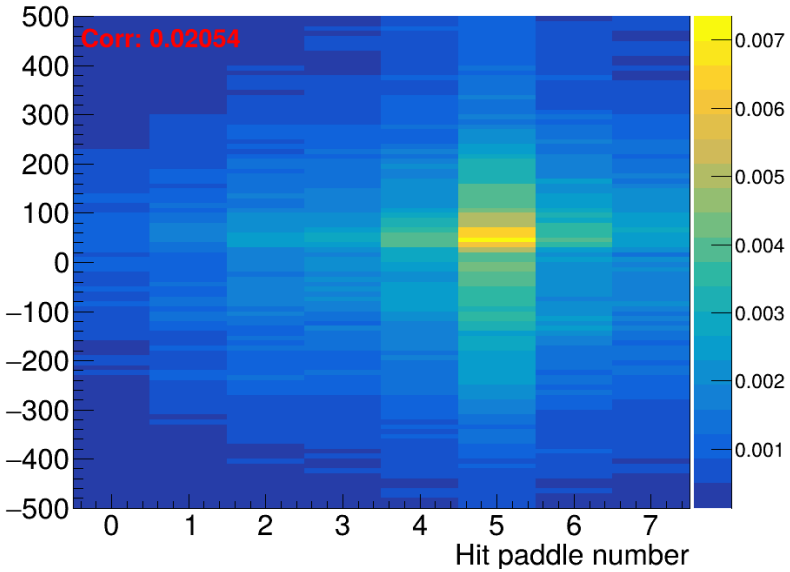
Cosmic Left: Track Proj. X vs Out-of-Time Hit Paddle #



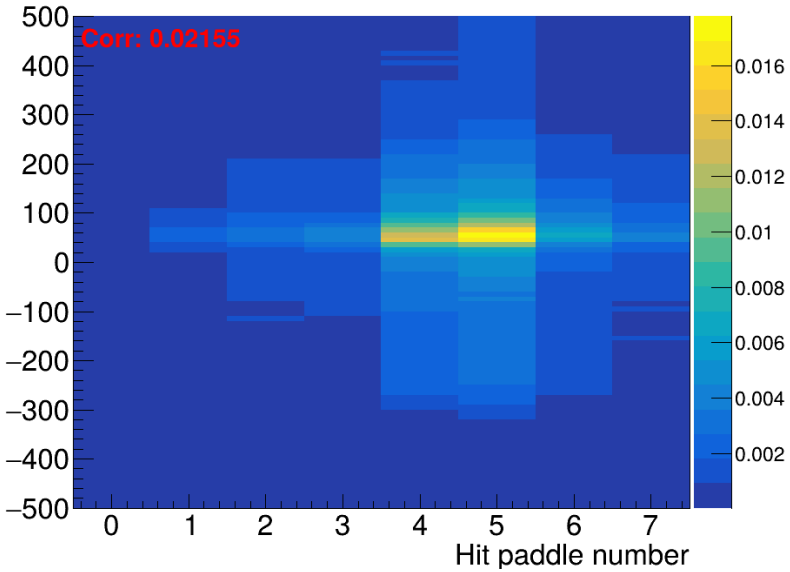
Cosmic Right: Track Proj. X vs Out-of-Time Hit Paddle #



Beam Left: Track Proj. X vs Out-of-Time Hit Paddle #

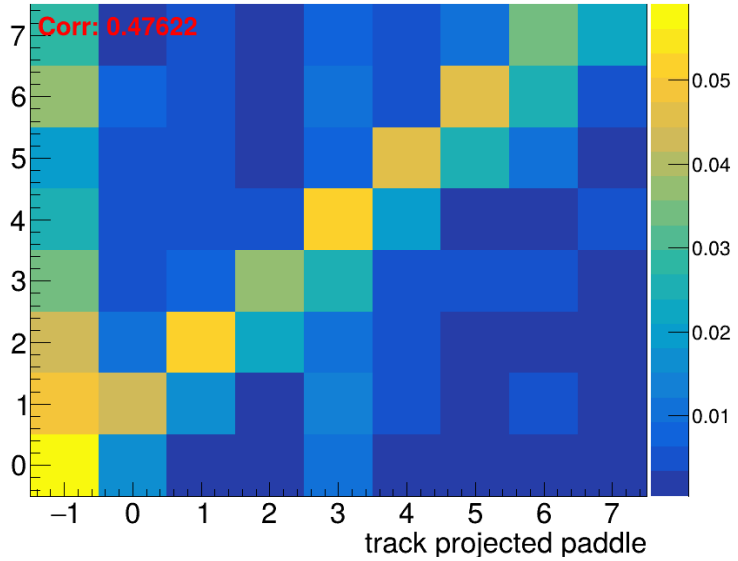


Beam Right: Track Proj. X vs Out-of-Time Hit Paddle #

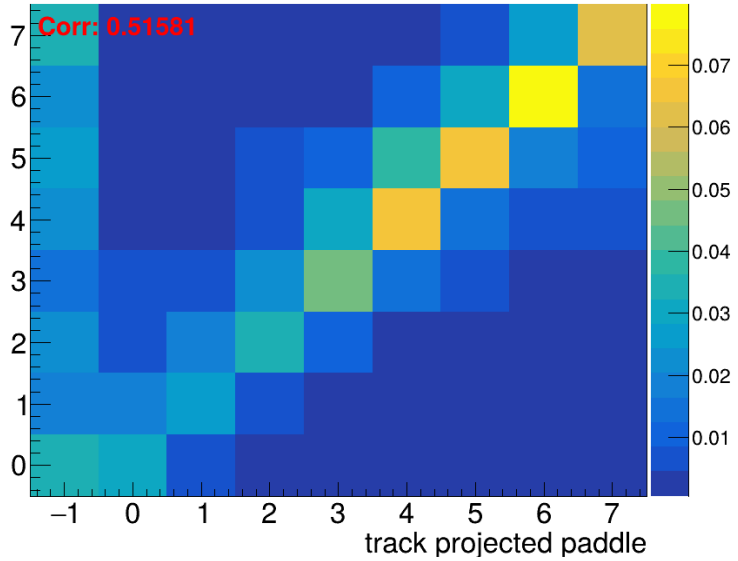


Actual vs Projected Hit Paddle Correlations

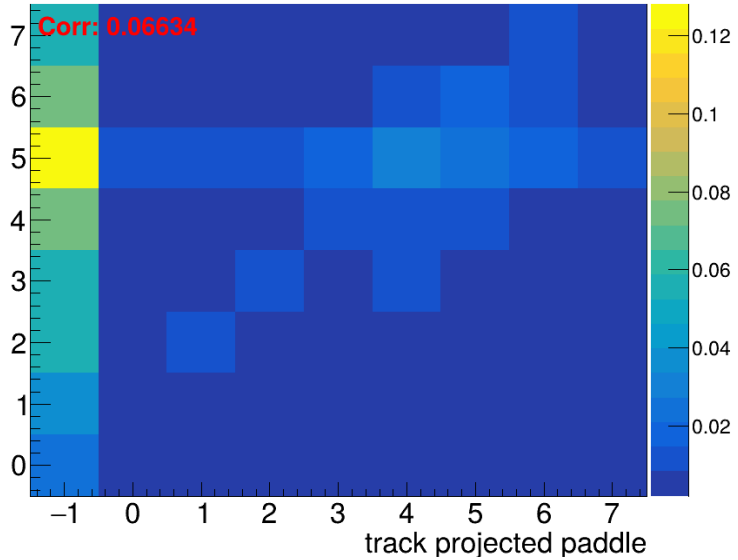
Cosmic Left: Hit Paddle # vs Track Proj. Paddle #



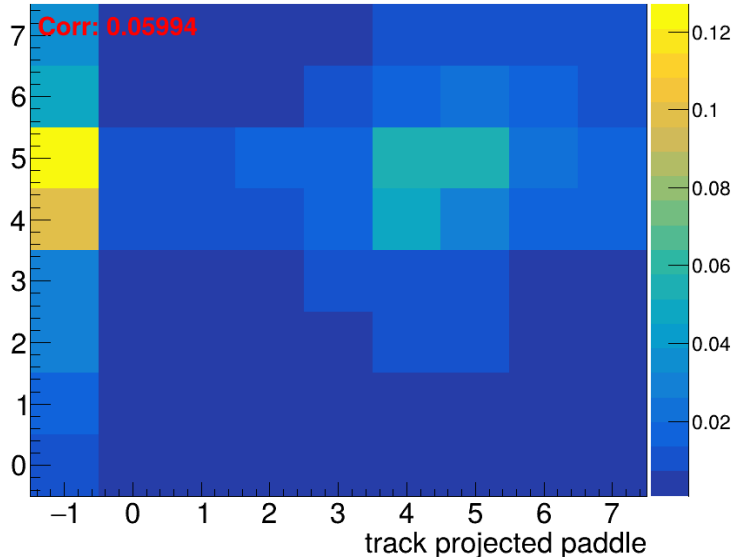
Cosmic Right: Hit Paddle # vs Track Proj. Paddle #



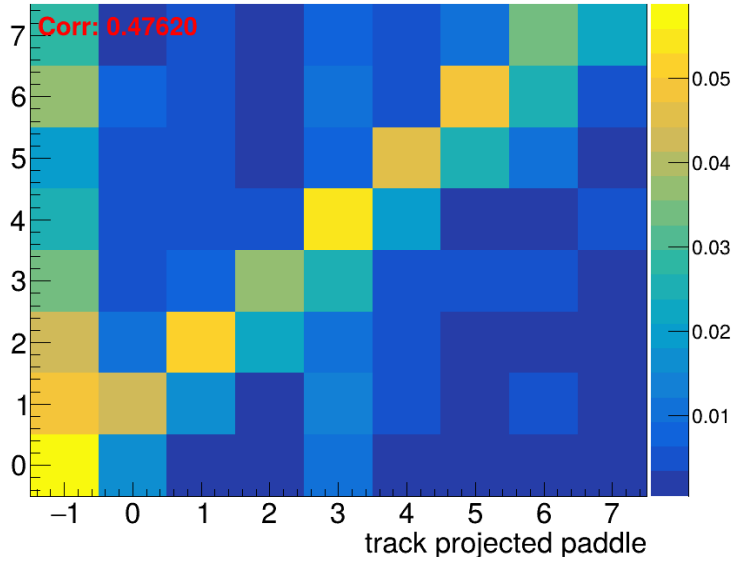
Beam Left: Hit Paddle # vs Track Proj. Paddle #



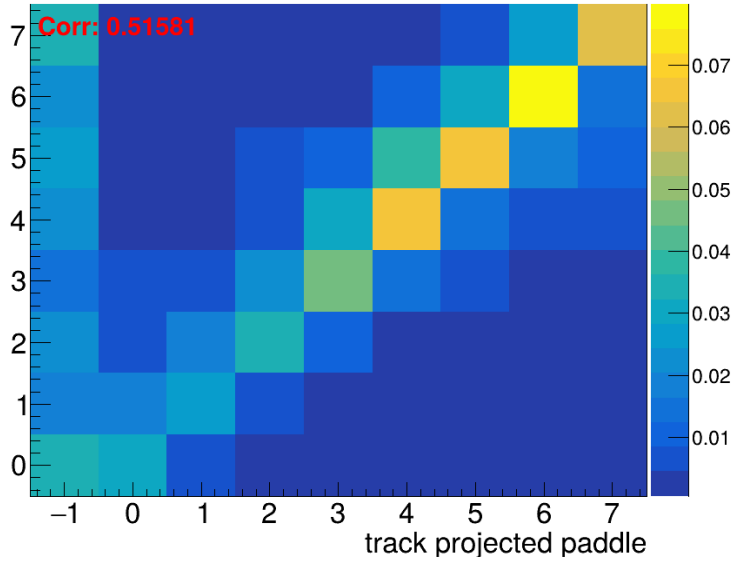
Beam Right: Hit Paddle # vs Track Proj. Paddle #



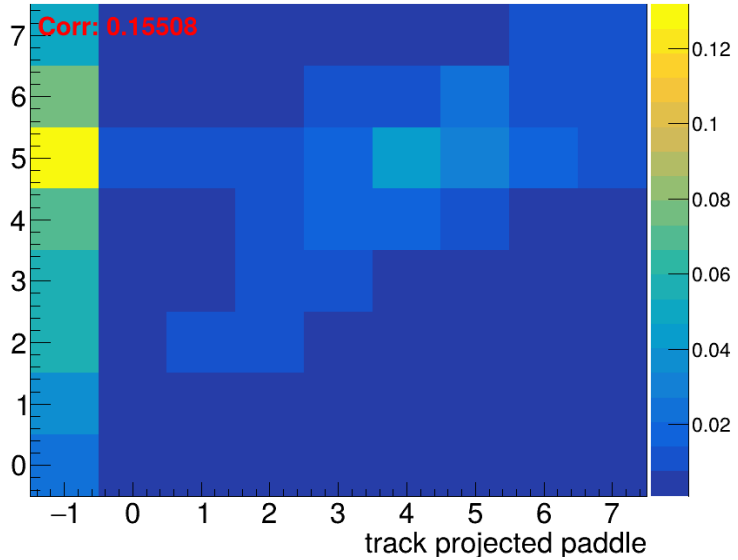
Cosmic Left: In-Time Hit Paddle # vs Track Proj. Paddle #



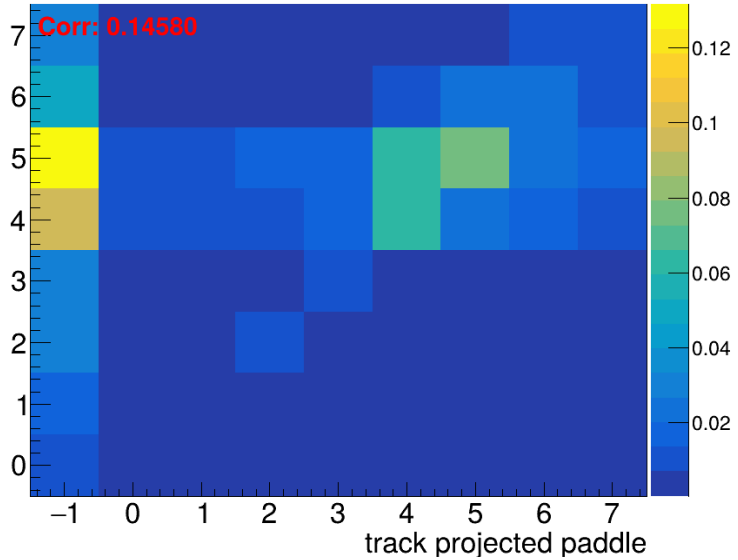
Cosmic Right: In-Time Hit Paddle # vs Track Proj. Paddle #



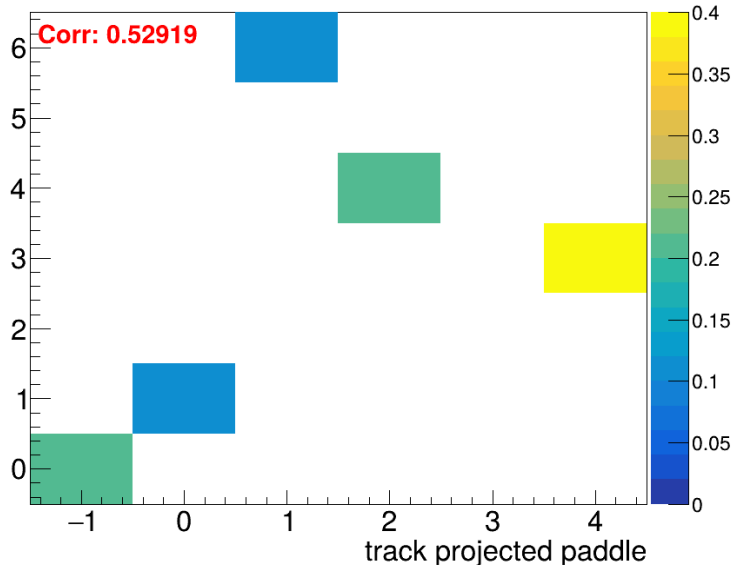
Beam Left: In-Time Hit Paddle # vs Track Proj. Paddle #



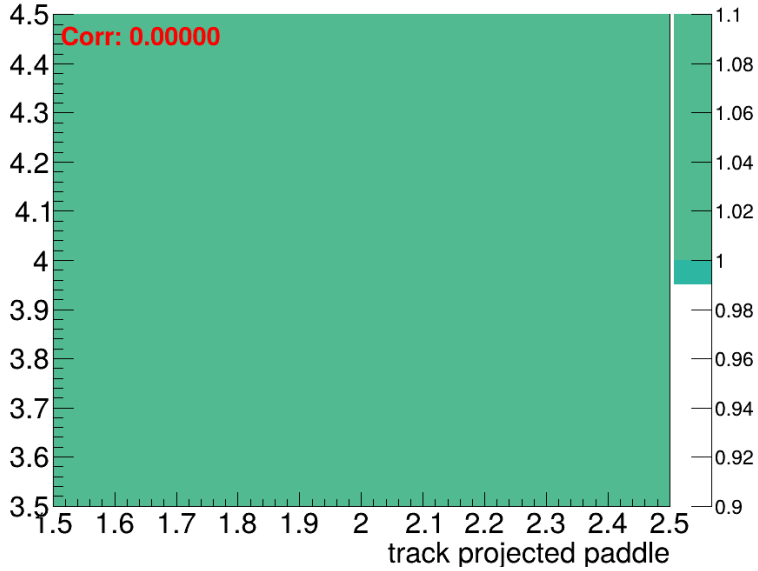
Beam Right: In-Time Hit Paddle # vs Track Proj. Paddle #



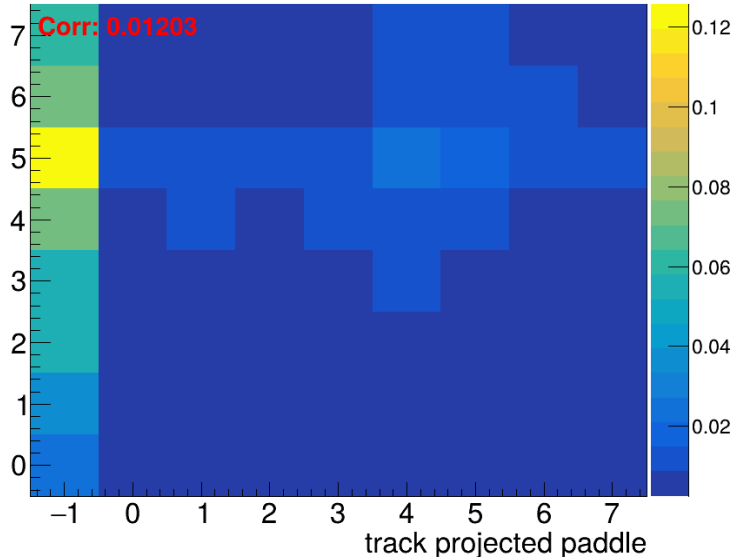
Cosmic Left: Out-of-Time Hit Paddle # vs Track Proj. Paddle #



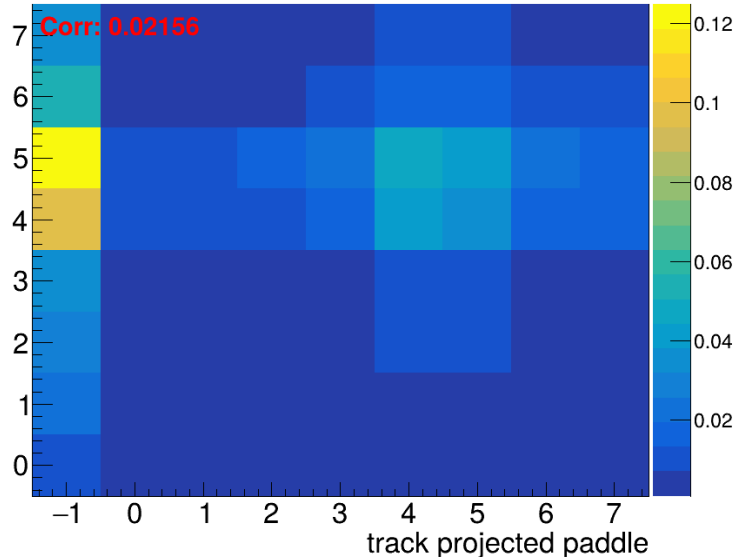
Cosmic Right: Out-of-Time Hit Paddle # vs Track Proj. Paddle #



Beam Left: Out-of-Time Hit Paddle # vs Track Proj. Paddle #

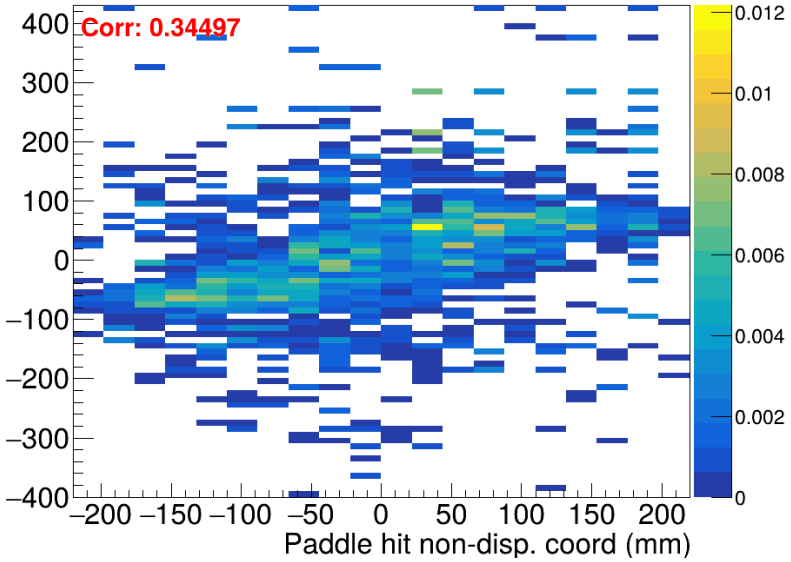


Beam Right: Out-of-Time Hit Paddle # vs Track Proj. Paddle #

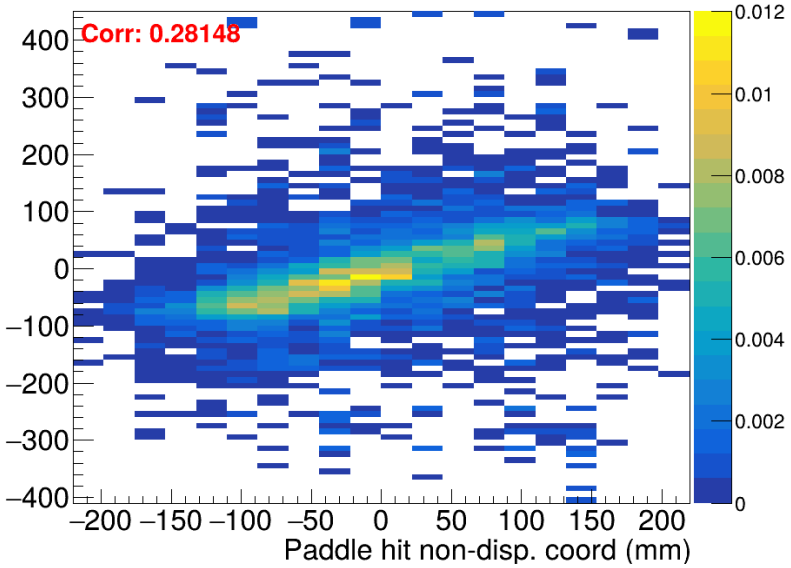


Projected vs Actual Trigger Hit Y Correlations

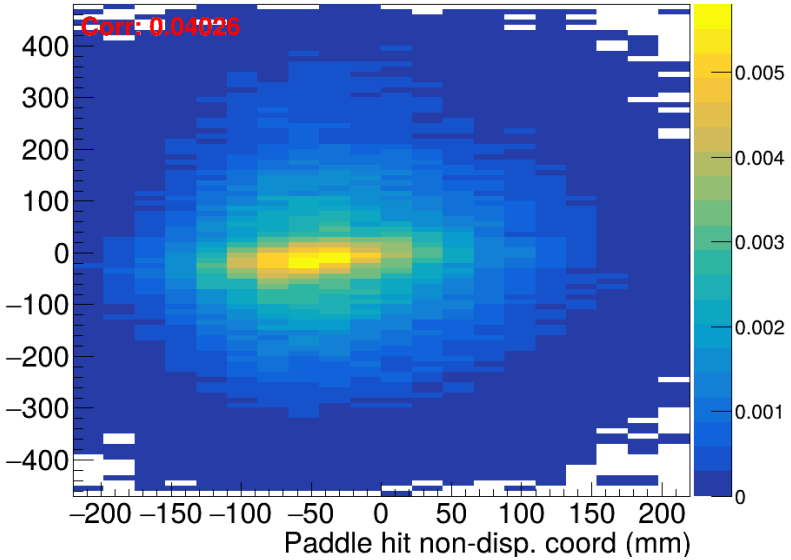
Cosmic Left: Track Proj. Y vs Hit Paddle Y



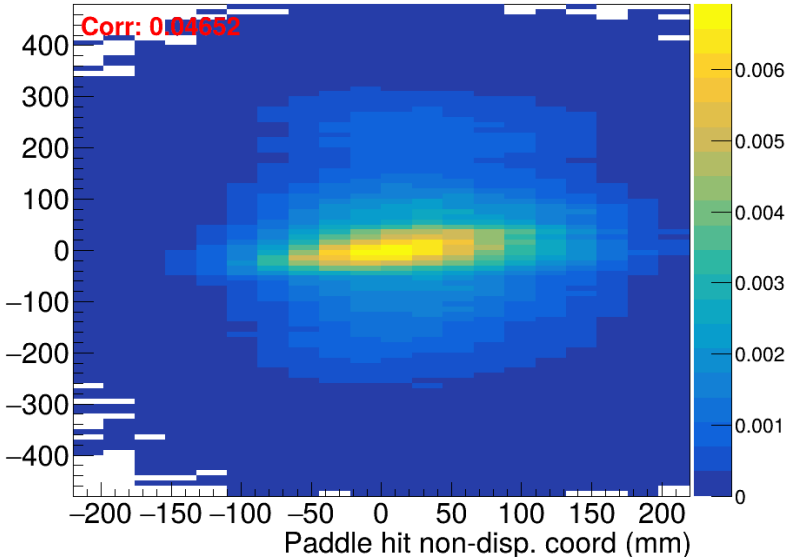
Cosmic Right: Track Proj. Y vs Hit Paddle Y



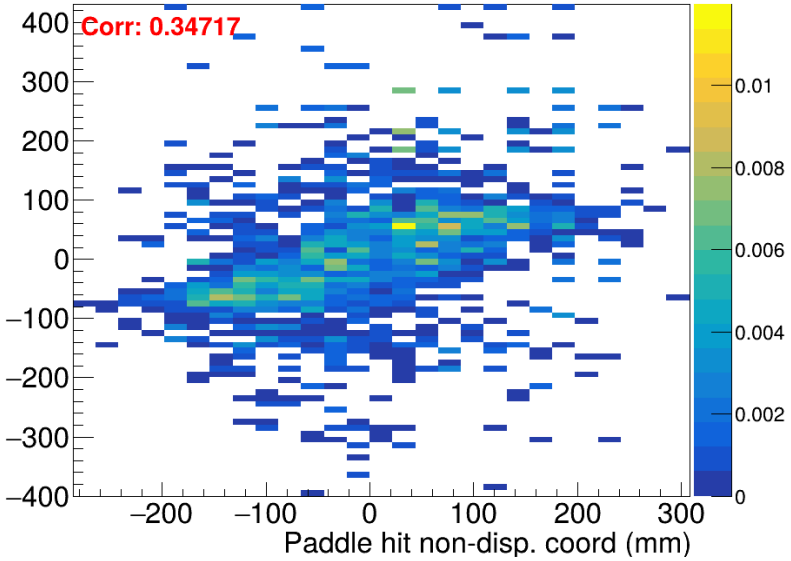
Beam Left: Track Proj. Y vs Hit Paddle Y



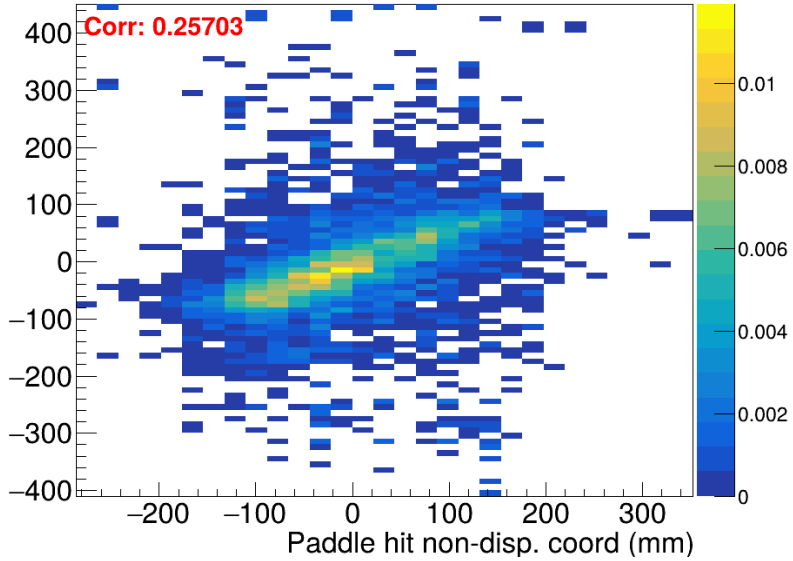
Beam Right: Track Proj. Y vs Hit Paddle Y



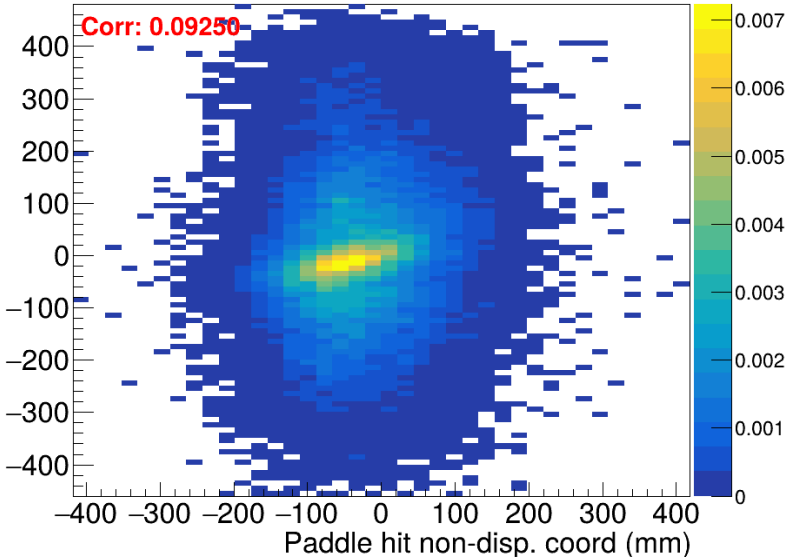
Cosmic Left: Track Proj. Y vs In-Time Hit Paddle Y



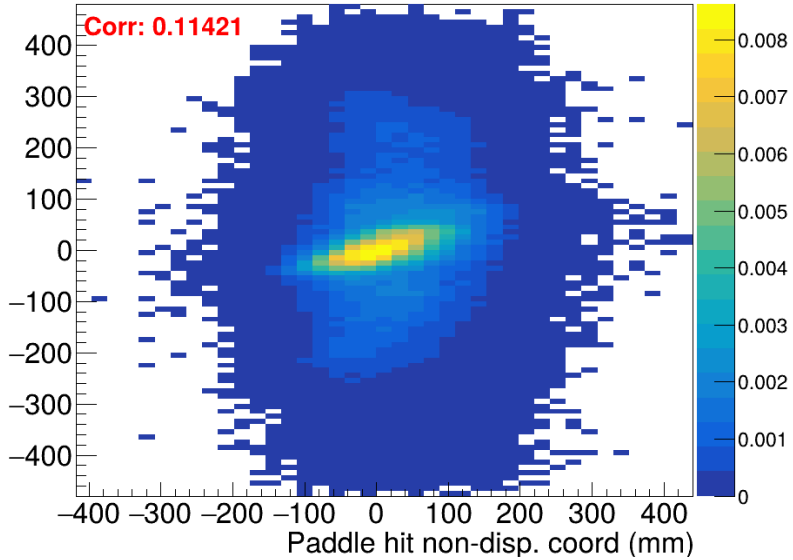
Cosmic Right: Track Proj. Y vs In-Time Hit Paddle Y



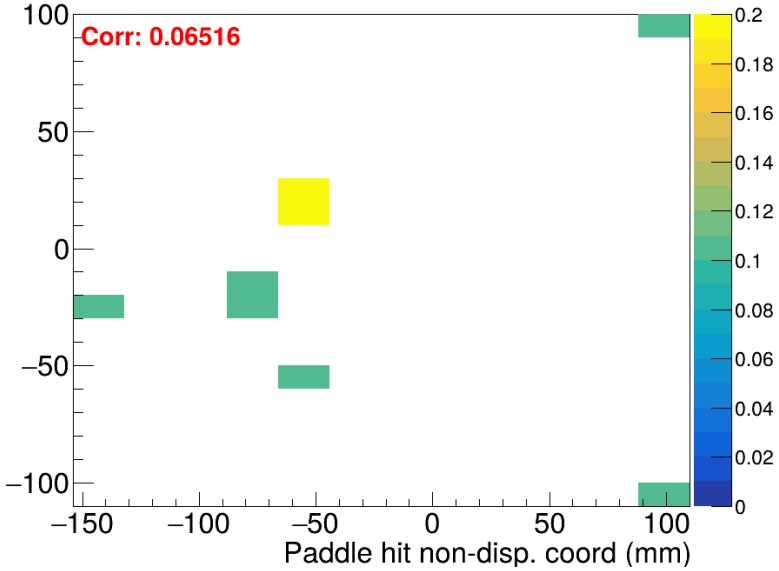
Beam Left: Track Proj. Y vs In-Time Hit Paddle Y



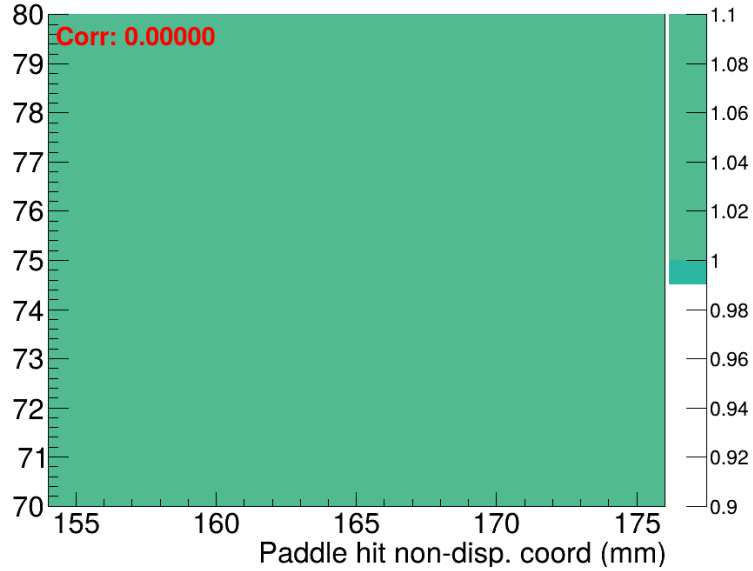
Beam Right: Track Proj. Y vs In-Time Hit Paddle Y



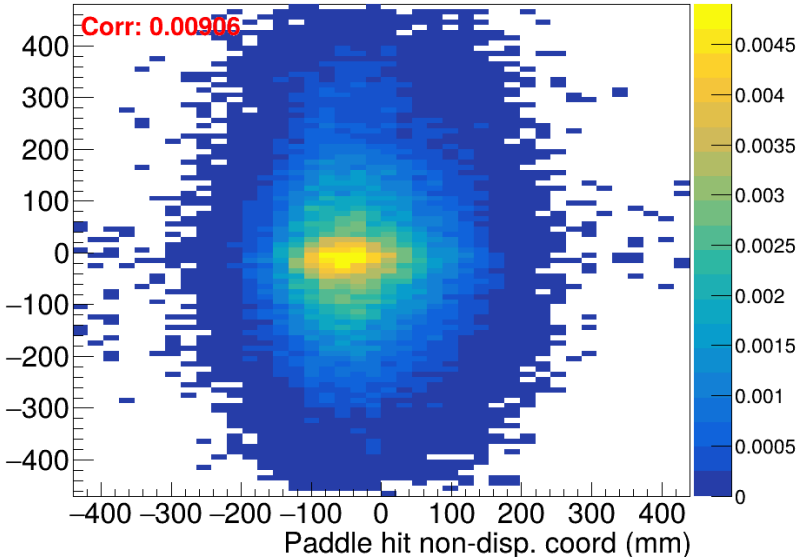
Cosmic Left: Track Proj. Y vs Out-of-Time Hit Paddle Y



Cosmic Right: Track Proj. Y vs Out-of-Time Hit Paddle Y



Beam Left: Track Proj. Y vs Out-of-Time Hit Paddle Y



Beam Right: Track Proj. Y vs Out-of-Time Hit Paddle Y

